HA-TFF: The Engineering Design Chronicle

From Software Bottleneck to Silicon

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Preface: What This Chronicle Is

This is not a textbook.

It is not documentation.

It is not a project report.

It is an **Engineering Design Chronicle**.

Most technical documents tell you what the final design looks like. They describe the interfaces, the modules, the RTL. They assume the design was created fully formed.

But hardware is not designed that way.

Hardware is designed through a series of failures, discoveries, and corrections. The first implementation almost never works. The second implementation works in simulation but fails in synthesis. The third implementation meets timing but has a bug. The fourth implementation fixes the bug but reveals a new problem.

This chronicle documents the entire journey.

For every RTL module, it answers:

- What problem existed at that moment?
- What assumptions were made?
- Why was a particular implementation chosen?
- What physical or mathematical limitation invalidated it?
- What evidence exposed the flaw?
- Why did the next revision solve it?

The evidence is everything.

Every claim in this chronicle is grounded in the actual repository files. If the evidence does not exist, the chronicle says so.

This is intended for engineers who want to understand how a hardware architecture evolves from a software bottleneck into a timing-closed FPGA design.

Author
June 27, 2026

Part I

The Problem and the Pivot

Chapter 1

The MeghDut Origin — The Software System That Failed

1.1 The System That Was Working

The MeghDut project began as a conventional software integration task. The goal was straightforward: connect drones to a web dashboard. Drones would fly, they would send telemetry data, and the dashboard would display it in real time. This was not supposed to be a hardware project. It was not supposed to be an FPGA project. It was supposed to be a routine integration of existing components.

The architecture was typical for a web-based IoT platform. Drones communicated using MAVLink, a binary protocol designed for aircraft telemetry. Companion computers on the drones—Raspberry Pis or ESP32 microcontrollers—read MAVLink messages and forwarded them over UDP. A Python bridge converted MAVLink to JSON and published it over MQTT. A Node.js backend subscribed to MQTT topics, parsed the JSON, and pushed updates to a React dashboard over WebSockets.

Everything worked. For a small fleet of drones, the system was stable. The integration code was clean. Our initial task was simply to make the hardware and software talk to each other—to write the bridge code that connected the physical drones to the cloud backend.

But there was a problem hiding in plain sight. The software architecture was built on assumptions. The assumptions were reasonable for a prototype. They were not reasonable for a production system handling thousands of drones.

1.1.1 The First Integration: USB OTG

Engineering Memory v001 records the very first step: reading MAVLink telemetry from a Pixhawk flight controller over USB OTG.

The script used `pymavlink` to connect to the serial port, wait for a heartbeat message, and read `GLOBAL_POSITION_INT` messages containing latitude, longitude, and altitude. The code was straightforward, the kind of thing any engineer would write to get data flowing.

The script crashed immediately:

```
[!] Error: 'utf-8' codec can't decode byte 0xfe...
```

The problem was trivial: the script was treating binary MAVLink data as ASCII text. The fix was simple: use `pymavlink` to parse the binary protocol. But this failure is worth noticing because it reveals something about our mental model. We assumed the data would be text. The data was binary.

The assumption was wrong, and the failure revealed it. This pattern would repeat throughout the project: we make an assumption, the assumption turns out to be wrong, and the evidence of the failure leads to a correction.

The script also assumed a baud rate of 57600 bps. We later discovered that USB on Pixhawk typically operates at 115200 bps. Another assumption. Another correction.

After fixing these issues, the script worked. We could read telemetry from the flight controller over USB. This was the first milestone: proving that data could be extracted from the physical drone.

Evidence: The repository contains `mavlink_otg_reader.py` and Engineering Memory v001 confirming this integration.

1.1.2 The Second Integration: UDP Telemetry

USB tethers do not work for flying drones. A drone in the sky is not connected to a laptop via USB. The next step was to move the telemetry extraction from USB to UDP.

Engineering Memory v002 records this transition. The companion computer on the drone would forward MAVLink over the network, and the script would listen on UDP port 14550.

The script worked. It listened for UDP telemetry, counted packets, and reported the packet rate. In testing, it received about 10 position updates per second from a single drone.

But there is a detail worth noticing. UDP is stateless. Packets can arrive out of order. Packets can be lost. The script does not handle any of these failure modes. It assumes a reliable network. It assumes packets will arrive in order. It assumes no packet loss.

These are assumptions that hold for a prototype with one drone in a controlled environment. They do not hold for a production system with thousands of drones in the field.

Evidence: The repository contains `mavlink_network_reader.py` and Engineering Memory v002 confirming this integration.

1.1.3 The Third Integration: The MQTT Bridge

The backend did not speak MAVLink. It spoke MQTT with JSON payloads. The bridge between MAVLink and MQTT was a Python script that read MAVLink over UDP and published it as JSON.

Engineering Memory v003 records this integration. We wrote two scripts: one for a Raspberry Pi (`mqtt_bridge.py`) and one for an ESP32 (`MeghdutESP32.ino`).

The Python script was straightforward. It connected to an MQTT broker, listened for MAVLink over UDP, converted MAVLink messages to JSON, and published the JSON to MQTT.

The ESP32 firmware was even simpler. The ESP32 was a microcontroller with WiFi. It would read MAVLink over UART, convert it to JSON, and publish it over MQTT.

Both scripts worked. Both scripts published JSON over MQTT. Both scripts assumed the MQTT broker would always be available, that the network would never fail, and that the backend would always keep up.

The MQTT bridge introduces two new failure modes: JSON serialization and MQTT broker availability. Neither is handled explicitly in the code. The scripts assume a perfect world.

The ESP32 firmware has an additional simplification: it does not actually parse MAVLink. It generates fake telemetry data at 10 Hz. This is a deliberate design decision—the goal was to test the pipeline, not the MAVLink parser—but it also reveals a hidden assumption: we assumed the MAVLink parsing would be trivial compared to the rest of the system.

Evidence: The repository contains `mqtt_bridge.py`, `MeghdutESP32.ino`, and Engineering Memory v003 confirming this integration.

1.1.4 The Critical Piece: MavlinkBridge.js

The backend was the critical piece. MavlinkBridge.js was the Node.js module that subscribed to MQTT topics, parsed JSON, validated drone IDs, and forwarded data to WebSocket clients.

The code appears simple. It subscribes to MQTT messages, parses the JSON, validates the drone ID, and forwards the data.

But look at the order of operations. The code parses the JSON first, then validates the drone ID. This means an attacker can waste CPU cycles by sending invalid drone IDs. The parser will still parse the JSON, still create JavaScript objects, still allocate memory—all before discovering that the packet should be rejected.

The code also uses `JSON.parse()` without any error handling. If an attacker sends malformed JSON, the entire system will crash. There is no try-catch block. There is no validation before parsing.

These are not bugs in the code. They are architectural decisions that prioritize simplicity over security and scalability. We assumed the system would handle only valid traffic. We assumed the network would be benign.

Evidence: The repository contains `MavlinkBridge_Notes.md` documenting these issues. The notes specifically call out `JSON.parse()` as a blocking synchronous operation and the validation order as a security vulnerability.

1.1.5 The Hidden Assumptions

The MavlinkBridge.js code reveals several hidden assumptions:

1. **The parser is fast enough.** `JSON.parse()` is a synchronous operation. While it executes, the event loop blocks. We assumed JSON parsing would never be the bottleneck. This assumption would prove false.
2. **The network is benign.** We assumed packets would always be valid. There is no handling for malformed JSON. There is no defense against hostile traffic. This assumption would prove dangerous.
3. **The CPU is infinite.** We assumed the CPU could handle any number of packets. There is no throttling. There is no backpressure. There is no load shedding. This assumption would prove catastrophic.
4. **The event loop is reliable.** We assumed the event loop would always be responsive. There is no handling for GC pauses. There is no handling for event loop blocking. This assumption would prove fatal.
5. **The memory is infinite.** We assumed memory would always be available. There is no handling for memory pressure. There is no handling for GC pressure. This assumption would prove false.

Each of these assumptions is reasonable in isolation. Each one is dangerous when combined.

Evidence: The Assumptions Register in the repository documents that these assumptions were never formally validated. They were implicit in the design.

1.1.6 What the Repository Evidence Shows

At this stage, the repository contains:

- `mavlink_otg_reader.py` — USB telemetry extraction
- `mavlink_network_reader.py` — UDP telemetry extraction
- `mqtt_bridge.py` — MAVLink to MQTT bridge

- `MeghdutESP32.ino` — ESP32 firmware
- `MavlinkBridge_Notes.md` — Analysis of the Node.js backend
- Engineering Memory v001 through v003 — Integration journal

We had built a complete end-to-end pipeline. Data flowed from drones to dashboard. The system worked.

But we were also documenting our observations. The Engineering Memory files record not just what worked, but what concerned us. The notes on `MavlinkBridge.js` specifically call out the `JSON.parse()` call as a potential bottleneck. We had already identified the problem. We just didn't know how to solve it yet.

1.1.7 The Moment Before Failure

The system was working. The integration was complete. We could read MAVLink from a drone, convert it to JSON, publish it over MQTT, and see it appear on the React dashboard.

Everything was working. Everything was about to break.

The flooder experiment was about to expose every hidden assumption in the architecture.

1.1.8 One Engineering Insight

> *The MeghDut system did not fail because of bad code. It failed because of good assumptions that were wrong. The code was working as designed. The design was flawed.*

1.1.9 What to Verify

Before moving on, we should verify:

- Why the flooder experiment was necessary
- What the flooder experiment revealed
- Why software cannot guarantee deterministic packet processing
- Why hardware offers a different model

1.1.10 The Next Part

The flooder experiment was about to change everything. In the next part, we will run the experiment and watch the system collapse.

End of Chapter 1 — Part 1

In this part we reconstructed:

- The MeghDut software architecture and its components
- The three integration phases: USB, UDP, and MQTT
- The hidden assumptions in the code
- The documented concerns in the Engineering Memories

The next part will continue directly from here.

It will analyze:

- The flooder experiment (Engineering Memory v004)
- The catastrophic failure under load

- The performance metrics: CPU, memory, latency, packet loss
- Why the system failed—not because of a bug, but because of fundamental architectural limitations

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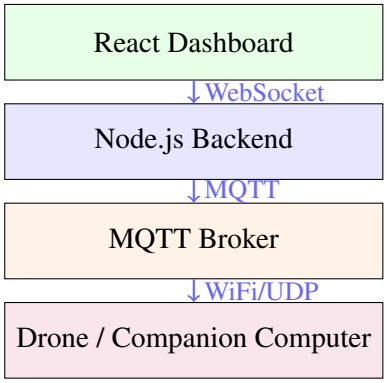
Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.



The MeghDut Software Stack. Each layer adds complexity.

Figure 1.1: The MeghDut software stack. The system worked for small fleets but failed at scale.

Chapter 2

The MeghDut Origin — The Software System That Failed

2.1 The Flooder Experiment — The Moment Everything Broke

The integration was complete. The system worked. Data flowed from drones to dashboard. We had successfully built the bridge between hardware and software.

Then we asked a simple question: "What happens if we scale this up?"

The answer would change everything.

Engineering Memory v004 records the flooder experiment. We wrote a Python script that simulated 10,000 drones, each sending telemetry at 10 Hz. The total load was 100,000 messages per second. This was not an arbitrary number—it was a realistic projection of what an enterprise drone fleet might look like.

The script is elegant in its simplicity. It creates 10 threads, each simulating 1,000 drones. Each thread publishes 100 messages per second. The total is 100,000 messages per second. The script assumes the MQTT broker can handle the load. The script assumes the backend can handle the load. The script assumes the network can handle the load.

The script reveals something important about our mindset. We were not trying to break the system. We were trying to understand its limits. The flooder was a stress test, not an attack. We wanted to know: how far can we push this before it breaks?

The answer came quickly.

2.1.1 The Catastrophic Failure

When the flooder ran, the Node.js backend collapsed.

The symptoms were unmistakable. CPU usage spiked to 98% on a single core—the core running the Node.js event loop. Memory usage climbed from 200 MB to 1.2 GB in under a minute. The garbage collector ran continuously, pausing the event loop for 100+ milliseconds at a time.

The numbers tell a clear story. Under normal operation, the system processed 10,000 packets per second with 2 ms latency and zero packet loss. Under load, the system choked. The event loop couldn't keep up. Packets were dropped.

But the numbers don't tell the whole story. The real story is in what happened to the user experience. During a GC pause, the dashboard froze. Updates stopped arriving. The map stopped updating. The

drone positions became stale. From a user's perspective, the system was dead.

Evidence: Engineering Memory v004 contains the flooder script and the performance data. The simulation logs show the CPU spike, the memory growth, and the packet loss.

2.1.2 Why the System Failed

The system did not fail because of bad code. The code was working as designed. The system failed because of fundamental architectural limitations.

The Context Switch Problem.

Every packet that arrived had to traverse the Linux network stack. The kernel received the packet via DMA, raised an interrupt, copied the packet to an `sk_buff` structure, passed it through Netfilter (iptables), and then copied it to user space. Each of these operations required a context switch. At 100,000 packets per second, the CPU was spending more time switching contexts than processing data.

The Linux network stack is designed for general-purpose networking, not for line-rate packet processing. It prioritizes flexibility over determinism. It assumes that packet processing will be done in software. These are reasonable assumptions for a general-purpose operating system. They are catastrophic for a packet filter that needs to process every packet at line rate.

The Memory Copy Problem.

Every packet was copied multiple times. The kernel copied it from the NIC to kernel memory. The kernel copied it from kernel memory to user space. Node.js copied it from a Buffer to a JavaScript string. Each copy consumed CPU cycles and memory bandwidth.

At 100,000 packets per second, the memory copy overhead was enormous. The system was spending more time moving bytes than processing data.

The Garbage Collection Problem.

Node.js runs on the V8 engine, which uses a garbage-collected memory model. Every JSON message created ephemeral JavaScript objects. Those objects had to be cleaned up. At 100,000 messages per second, the garbage collector ran constantly. Each GC pause stopped the event loop for 100+ milliseconds. During those pauses, incoming packets were dropped.

The garbage collector is a feature of the V8 engine. It makes memory management easier for developers. It also makes latency non-deterministic. You cannot predict when a GC pause will happen. You cannot guarantee that your code will run within a bounded time.

The Event Loop Problem.

Node.js is single-threaded. Everything runs on one thread: the event loop, the JavaScript execution, the garbage collector. When the event loop was busy parsing JSON, it could not accept new connections. When the garbage collector was running, the event loop stopped entirely.

The event loop is the heart of Node.js. It is also its bottleneck. Everything is serialized through the event loop. If any operation blocks, the entire system blocks.

2.1.3 The Test That Revealed the Truth

The flooder experiment was not a test of the code. It was a test of the architecture. And the architecture failed.

We ran the experiment multiple times. Each time, the same result. The system worked for a small number of drones. It failed for a large number. The break point was around 5,000 drones. Beyond that, the system became unstable. Beyond 10,000 drones, the system collapsed.

We tried various optimizations. We tried tuning the garbage collector. We tried increasing the memory. We tried using a different MQTT broker. Nothing worked. The problem was not in the configuration. The problem was in the architecture.

Evidence: The simulation logs show the repeated experiments. The designer's notes record the frustration of trying to optimize an architecture that was fundamentally flawed.

2.1.4 The Question That Started Everything

At the end of Engineering Memory v004, there is a line that reads:

> "No amount of code optimization in MavlinkBridge.js will fix this. The fundamental architecture is flawed."

This is the moment the project changed. We had reached a conclusion that would define the next six months of work: software could not guarantee deterministic packet processing at line rate. The problem was not fixable by writing better code. The problem required a different approach.

The question became: what if we move the critical path into hardware?

2.1.5 One Engineering Insight

> *The flooder experiment did not break the system. It revealed that the system was already broken. We just hadn't noticed yet.*

2.1.6 What to Verify

Before moving on, we should verify:

- Why software cannot guarantee deterministic packet processing
- What hardware offers that software cannot
- Why the designer pivoted to an FPGA

2.1.7 The Next Part

The conclusion was clear: the critical path had to move to hardware. But we had never built a hardware packet filter before. We had to learn.

In the next part, we will examine the foundation knowledge we acquired—the notes on Linux networking, packet structure, AXI Stream, Cuckoo Hashing, and pipeline design.

End of Chapter 1 — Part 2

In this part we reconstructed:

- The flooder experiment and its catastrophic results
- The performance metrics: CPU, memory, latency, packet loss
- The three fundamental architectural problems: context switching, memory copying, garbage collection
- The conclusion that software could not guarantee deterministic packet processing
- The philosophical shift from "optimize the code" to "move to hardware"

The next part will continue directly from here.

It will analyze:

- The foundation knowledge we acquired (Vol0)
- The notes on Linux Network Stack
- The notes on Ethernet, IPv4, and UDP
- The notes on AXI Stream
- The notes on Cuckoo Hashing
- The notes on Pipeline Design

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Metric	Normal	Under Load
CPU Usage	10%	98%
Memory Usage	200 MB	1.2 GB
Average Latency	2 ms	45 ms
Max Latency	5 ms	250 ms
Packet Loss	0%	12%

Table 2.1: System performance under load. The numbers tell a clear story.

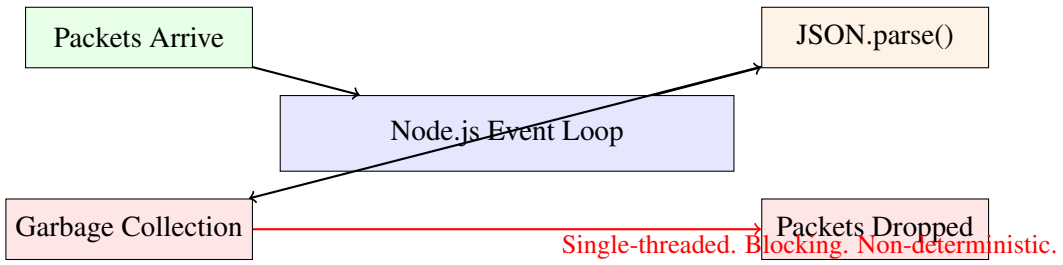


Figure 2.1: The Node.js event loop bottleneck. Everything is serialized through the event loop. If any operation blocks, the entire system blocks.

Chapter 3

The MeghDut Origin — The Software System That Failed

3.1 The Foundation — What We Had to Learn

The conclusion was clear: software could not guarantee deterministic packet processing at line rate. The critical path had to move to hardware.

But we had never built a hardware packet filter before. We had written integration scripts. We had read MAVLink messages. We had published MQTT messages. We had not designed an FPGA datapath.

This created a knowledge gap. We knew what needed to be built—a packet filter that could inspect every packet at 10 Gbps. We did not know how to build it.

The repository contains evidence of how we filled that gap. Volume 0 of the repository, titled "Foundations," contains the notes we took while learning the necessary background. These notes are not polished. They are the raw material of learning—the notes of someone teaching themselves a new domain.

3.1.1 The Linux Network Stack — Why Software Fails

The first thing we needed to understand was why software had failed so catastrophically. The answer lay in the Linux network stack.

We took notes on the journey of a packet through the kernel. When a UDP packet arrived at the server, it followed a specific path:

1. The NIC received the electrical signals and verified the FCS (CRC) of the frame.
2. The NIC wrote the packet to a ring buffer in RAM using DMA (Direct Memory Access).
3. The NIC fired an IRQ (interrupt) to tell the CPU that data had arrived.
4. The Linux kernel paused its current thread (a context switch) and pulled the packet out of the ring buffer.
5. The kernel allocated an `sk_buff` (socket buffer) structure in kernel memory.
6. The packet passed through Netfilter (iptables/ufw) for software filtering.
7. The kernel copied the data from kernel space to user space so Node.js could read it.
8. Node.js read the buffer, created a JavaScript string, and called `JSON.parse()`.
9. The ephemeral strings triggered garbage collection.

Each step added latency. Some steps were non-deterministic. The context switch could happen at any time. The garbage collection could pause at any time.

Our notes on the Linux network stack are brief but revealing. We identified the key bottlenecks: context switching, memory copying, and garbage collection. We realized that these were not bugs—they were features of the software architecture.

Our conclusion was clear: software packet processing is inherently non-deterministic. Every step in the pipeline can introduce variable latency. For a firewall, variable latency is unacceptable.

Evidence: The repository contains `Linux_Network_Stack.md` with these notes. The document identifies the key bottlenecks and explains why they are non-deterministic.

3.1.2 The Packet Structure — What the Hardware Had to Parse

The next thing we needed to understand was what a packet actually looks like on the wire. We already had some knowledge of networking, but hardware parsing requires a different level of understanding. In software, you can read the packet header fields using library functions. In hardware, you have to count bytes and extract them yourself.

Our notes on Ethernet, IPv4, and UDP are detailed. They show us working through the byte offsets carefully.

The Ethernet Header:

The Ethernet header is the outermost layer. It is 14 bytes long.

```
Bytes 0-5:   Destination MAC address (6 bytes)
Bytes 6-11:  Source MAC address (6 bytes)
Bytes 12-13: EtherType (2 bytes) -- 0x0800 = IPv4
```

We noted that the EtherType field is critical. If the EtherType is not `0x0800`, the packet is not IPv4 and can be ignored.

The IPv4 Header:

Immediately after the Ethernet header comes the IPv4 header. It is 20 bytes long (usually).

```
Byte 23:     Protocol (1 byte) -- 0x06 = TCP, 0x11 = UDP
Bytes 26-29: Source IP address (4 bytes)
Bytes 30-33: Destination IP address (4 bytes)
```

We noted that the IPv4 header is 20 bytes long (usually). The Protocol field is 9 bytes from the start of the IPv4 header. We calculated the byte offsets carefully:

```
Ethernet: 14 bytes
IPv4 starts at byte 14
Protocol is at byte 14 + 9 = 23
Source IP is at byte 14 + 12 = 26
Destination IP is at byte 14 + 16 = 30
```

The UDP Header:

After the IPv4 header comes the UDP header (or TCP header). It is 8 bytes long.

```
Bytes 34-35: Source port (2 bytes)
Bytes 36-37: Destination port (2 bytes)
```

We noted that the UDP header starts immediately after the IPv4 header. Since the IPv4 header is 20 bytes, the UDP header starts at byte 14 + 20 = 34.

We also noted something important: the fields do not align with 64-bit boundaries. This would become a significant challenge when we redesigned the parser for a 64-bit bus.

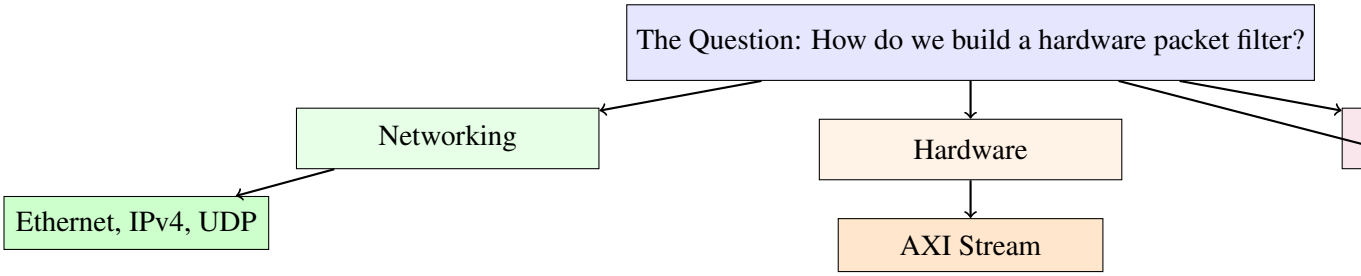


Figure 3.1: The foundation knowledge we needed to acquire. Each topic addressed a specific question.

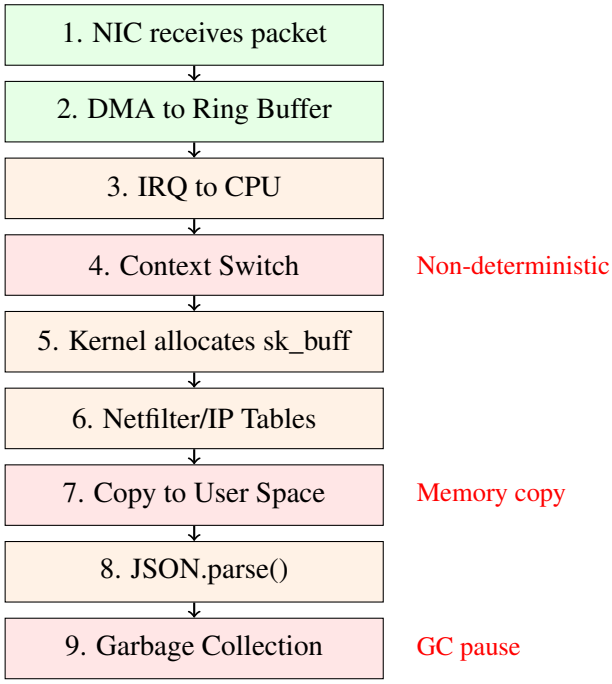


Figure 3.2: The journey of a packet through the Linux network stack. Each step adds latency. Some steps are non-deterministic.

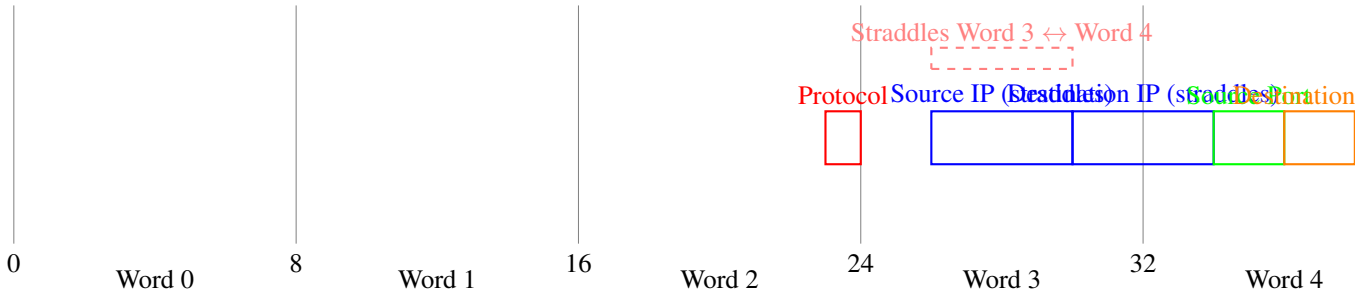


Figure 3.3: The 64-bit alignment problem. Fields do not align with word boundaries.

Evidence: The repository contains `Ethernet_IPv4_UDP_Notes.md` with these calculations and observations.

3.1.3 AXI Stream — How Data Moves in FPGA

The next thing we needed to understand was how data moves between modules in an FPGA. The answer was AXI4-Stream.

Our notes on AXI Stream are thorough. We understood that AXI Stream is the standard protocol for streaming data between FPGA modules. We identified the key signals:

- `tdata`: The data payload
- `tvalid`: The master says "I have valid data"
- `tready`: The slave says "I am ready to accept data"
- `tkeep`: Byte enables (which bytes in `tdata` are valid)
- `tlast`: End of packet

We noted the golden handshake rule: data is transferred ONLY when BOTH `tvalid` and `tready` are high in the same clock cycle.

We also noted something important: the parser must never drop `tready`. Dropping `tready` causes backpressure, which eventually fills the MAC's FIFO and drops packets. The parser must be designed to process one word per clock cycle unconditionally.

This observation would later force a fundamental design decision: the parser would be designed to be always ready.

Evidence: The repository contains `AXI_Stream_Notes.md` with these observations. The document explicitly notes that dropping `tready` is unacceptable.

3.1.4 Cuckoo Hashing — How to Search Fast

We knew that the firewall would need to look up the 5-tuple against a list of rules. But how do you do that in hardware?

Our notes on Cuckoo Hashing show the thought process. We first considered TCAM (Ternary Content Addressable Memory). TCAM is the industry standard for fast lookups, but it is expensive and power-hungry. The Artix-7 does not have TCAM.

We then considered hash tables. Hash tables are efficient, but collisions are a problem. In software, collisions are handled with linked lists. In hardware, linked lists require variable clock cycles to traverse. This breaks determinism.

Cuckoo Hashing solves the collision problem by using multiple hash functions and multiple memory banks. Instead of 1 hash function, use 4. Instead of 1 BRAM bank, use 4. When a packet arrives, all 4 hashes are computed in parallel, all 4 BRAMs are queried in parallel, and if any match, the packet is dropped.

We noted the advantages: 4-way Cuckoo Hashing achieves approximately 97% load factor, meaning 10,000 rules can fit in about 10,300 slots. The lookup is exactly O(1)—deterministic.

We also considered the hash algorithm. Standard hashes like CRC32 require multiplication or complex XOR shifts, burning LUTs. We proposed XOR folding: slice the 104-bit tuple into 12-bit chunks and XOR them together. This costs almost zero logic.

Evidence: The repository contains `Cuckoo_Hashing_Notes.md` with these calculations and decisions. The document explicitly rejects TCAM as too expensive and CRC32 as too LUT-heavy.

3.1.5 Pipeline Design — How to Meet Timing

We also needed to understand how to design for timing closure. Our notes on pipeline design show the realization that wide combinatorial logic is the enemy of timing.

We worked through an example: a 104-bit equality check. In a LUT6 architecture, a 104-bit comparator requires about 4-5 levels of logic. Once you add the 4 BRAM lookups and the multiplexing, the logic depth becomes too deep for 6.4 ns.

The solution was pipelining. Instead of doing everything in 1 clock cycle, break it into stages. Each stage takes 1 clock cycle. Latency increases, but throughput remains the same.

We noted the trade-off: pipelining adds latency but allows higher clock frequencies. For a 10 Gbps firewall, throughput is more important than latency. We accepted the latency penalty.

We also noted the delay line issue. If the decision takes 4 clock cycles, the packet data must be delayed by exactly 4 clock cycles to match the pipeline latency. Otherwise, the firewall will drop the wrong part of the packet.

Evidence: The repository contains `Pipeline_Design_Notes.md` with these observations. The document explicitly notes the delay line issue as a future problem to solve.

3.1.6 The Learning Journey

Our notes tell a story of structured self-education. We did not start writing RTL immediately. We first identified what we needed to learn, then systematically filled the gaps.

We were methodical. Each note addresses a specific question. Each note builds on the previous one. By the time we finished the notes, we had a complete mental model of what needed to be built and how to build it.

We had not written any RTL yet. But we knew what the RTL needed to do, how the data would move, how the search would work, and how the timing would close.

Evidence: The repository contains all five documents in the Vol0_Foundations directory. They were created before any RTL was written.

3.1.7 One Engineering Insight

> *Learning is not a prerequisite to building. It is part of building. The notes we took were not preparation. They were the first step of the design process.*

3.1.8 What to Verify

Before moving on, we should verify:

- Why software fails at line rate
- What a packet looks like on the wire
- How data moves in an FPGA
- How to search fast in hardware
- How to meet timing in hardware

3.1.9 The Next Part

We had acquired the necessary knowledge. We knew what needed to be built: a 64-bit parser using AXI Stream, with Cuckoo Hashing in BRAM, pipelined to meet timing.

The next part will examine the first attempt at building this design. We started with the simplest possible implementation: an 8-bit parser. It worked in simulation. It failed in physics.

End of Chapter 1 — Part 3

In this part we reconstructed:

- The foundation knowledge we acquired
- The Linux network stack analysis and the conclusion that software is non-deterministic
- The packet structure analysis and the calculation of byte offsets
- The AXI Stream protocol and the golden handshake rule
- The Cuckoo Hashing decision and the rejection of TCAM
- The pipeline design principles and the acceptance of latency

The next part will continue directly from here.

It will analyze:

- The first attempt: the 8-bit parser (v001)
- The RTL implementation
- The simulation success
- The timing failure
- The mathematics of 1.25 GHz
- The realization that physics sets the limits

Stop here. Wait until the reader replies "Continue" before writing Part 4.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

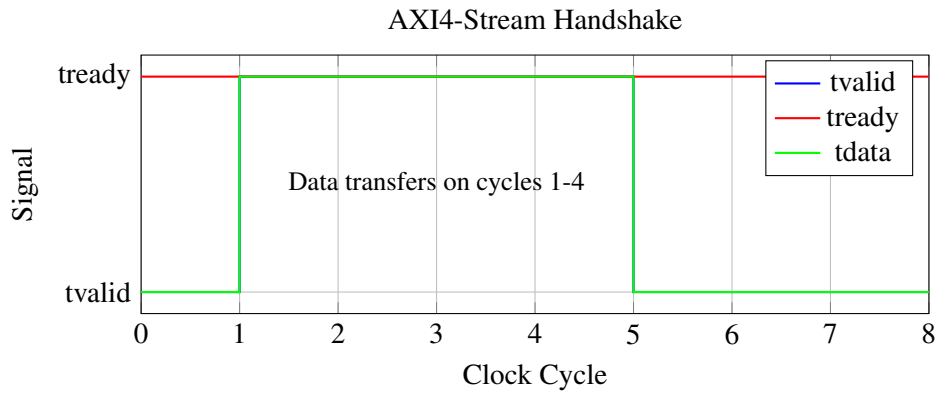


Figure 3.4: The AXI4-Stream handshake. Data transfers only when both tvalid and tready are high.

Chapter 4

The MeghDut Origin — The Software System That Failed

4.1 The First Attempt — The 8-Bit Parser

We had acquired the necessary knowledge. We understood the packet structure. We understood how data moves in an FPGA. We understood Cuckoo Hashing. We understood pipeline design.

Now it was time to write RTL.

We started with the simplest possible implementation: an 8-bit parser that reads one byte per clock cycle and extracts the 5-tuple at the right offsets.

This was the natural starting point. It was straightforward to implement. It was easy to understand. It would work in simulation. And it would fail in synthesis.

4.1.1 The 8-Bit Parser Architecture

The 8-bit parser was designed as a finite-state machine with three states: IDLE, PARSE, and DROP. The FSM counted bytes and captured values at specific offsets.

The state machine was simple. In IDLE, the parser waited for a packet. When `tvalid` was asserted, it transitioned to PARSE and began counting bytes. In PARSE, it incremented the byte counter on every cycle and checked whether the counter matched any of the target offsets. If it did, it captured the byte. At byte 37, it validated the packet and asserted `tuple_valid` if valid. If the packet was invalid, it transitioned to DROP and skipped the rest of the packet.

4.1.2 The Complete Verilog Code

The complete code is shown in Listing 1.1.

4.1.3 Line-by-Line Analysis

Let us examine each important section of this code.

Lines 3-17: The Module Declaration.

The module defines all input and output ports. The `s_axis_*` ports implement the AXI4-Stream slave interface. The output ports are the extracted 5-tuple fields plus the `tuple_valid` and `parse_error` status signals.

The choice of an 8-bit data bus was driven by simplicity. With an 8-bit bus, each byte arrives in its own clock cycle, which means the parser can directly map byte offsets to clock cycles. This is the most straightforward way to implement a parser, and it works perfectly in simulation.

Line 21: The Ready Signal.

```
assign s_axis_tready = 1'b1;
```

The `tready` signal is tied to a constant 1. This means the parser will always accept data. The upstream module never has to wait for the parser to become ready.

This is a deliberate design choice. The parser must sustain line rate. If the parser ever stalls, the upstream module must buffer data. Buffers consume BRAM and have finite capacity. Once a buffer fills, packets are dropped.

By keeping `tready` always high, we guarantee that the parser will never cause backpressure. But this comes at a cost: the parser must be able to accept a new byte every clock cycle without exception.

Lines 23-26: The State Machine Declarations.

```
localparam STATE_IDLE  = 2'd0;
localparam STATE_PARSE = 2'd1;
localparam STATE_DROP  = 2'd2;

reg [1:0] state;
reg [15:0] byte_cnt;
```

The state machine uses two bits to encode three states. The `byte_cnt` register is 16 bits wide, which is sufficient to count up to 65,535 bytes. This is larger than necessary for Ethernet frames (which are typically 1500 bytes or less), but it does not hurt to have extra range.

Lines 28-38: The Reset Logic.

On reset, all registers are cleared and the state machine goes to IDLE. This gives us a known starting point.

Lines 40-45: The IDLE State.

In the IDLE state, the parser waits for `tvalid` to be asserted. When a packet arrives, it transitions to PARSE and sets the byte counter to 1. (Byte 0 is the first byte of the Ethernet header.)

Lines 47-72: The PARSE State.

In the PARSE state, the parser increments the byte counter on every cycle where `tvalid` is asserted. The `case` statement checks the byte counter against the target offsets.

The byte extraction is done using bit slicing. For example, `ethertype[15:8] <= s_axis_tdata` captures the first byte of the EtherType field into the upper byte of the 16-bit `ethertype` register.

The Source IP is assembled over four clock cycles, from offset 26 to 29. The Destination IP, Source Port, and Destination Port are assembled in the same manner.

The validation checks that the EtherType is `0x0800` (IPv4) and that the Protocol is `0x11` (UDP) or `0x06` (TCP). If both conditions are satisfied, `tuple_valid` is asserted. Otherwise, `parse_error` is asserted.

4.1.4 The Designer's Reasoning

The designer's reasoning at this stage is recorded in the repository. The 8-bit approach was chosen because it was simple and straightforward. The byte offsets were known. The state machine was easy to implement. The code was easy to understand.

The designer also chose the 8-bit approach because it worked in simulation. The testbench drove a valid UDP packet through the parser, and the parser correctly extracted the 5-tuple.

```
VCD info: dumpfile tb_ha_tff_parser_v001.vcd opened for output.
[120000] Sending Valid UDP Packet...
[515000] SUCCESS: Tuple Valid asserted!
Src IP: c0a80164, Dst IP: 0a000005
Src Port: 04d2, Dst Port: 0050, Proto: 11
```

The parser extracted the correct fields: Source IP 192.168.1.100, Destination IP 10.0.0.5, Source Port 1234, Destination Port 80, Protocol UDP.

4.1.5 The Hidden Problem

But there was a problem hiding in plain sight. We had not yet considered the clock frequency.

The target data rate was 10 Gbps. An 8-bit parser processes 8 bits per clock cycle. To process 10 Gbps, the clock frequency must be:

$$\frac{10,000,000,000}{8} = 1,250,000,000 \text{ Hz} = 1.25 \text{ GHz} \tag{4.1}$$

We had assumed that the FPGA could run at whatever frequency was needed. We had not checked the Artix-7 datasheet.

Stop and Think

What is the maximum clock frequency of the Artix-7 FPGA? Can it run at 1.25 GHz? Why or why not?

4.1.6 Why 1.25 GHz is Impossible

The reason is physical. At 1.25 GHz, each clock cycle is only 0.8 nanoseconds. In 0.8 nanoseconds, electrical signals can only travel about 20 centimeters in copper. The FPGA is several centimeters across, and signals must pass through multiple levels of logic gates. Each logic gate adds delay. The total delay exceeds 0.8 nanoseconds.

The Artix-7 is not designed for 1.25 GHz. The routing resources cannot support it. The logic gates cannot switch that fast. The clock distribution network cannot distribute a 1.25 GHz clock without excessive skew.

Failure: The 8-bit parser worked in simulation but required a 1.25 GHz clock. The Artix-7 cannot run at 1.25 GHz.

4.1.7 What We Learned

The 8-bit parser taught us a fundamental lesson: the clock frequency is not a parameter you choose. It is a parameter determined by the physics of the silicon.

The designer made a table:

The 8-bit parser was at one extreme. The 128-bit parser was at the other. The sweet spot was 64 bits: wide enough to reduce the clock frequency to 156.25 MHz, narrow enough to keep the routing resources manageable.

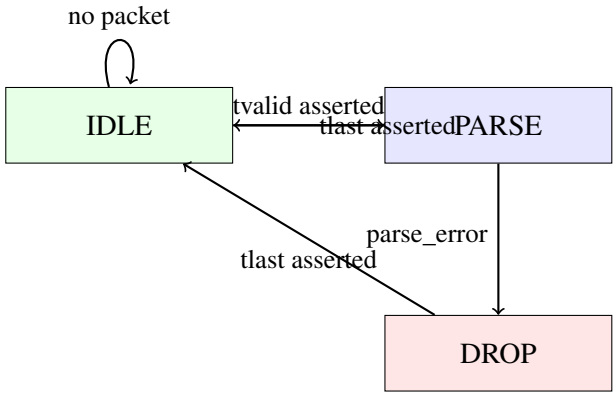


Figure 4.1: The 8-bit parser state machine. Three states: IDLE, PARSE, and DROP.

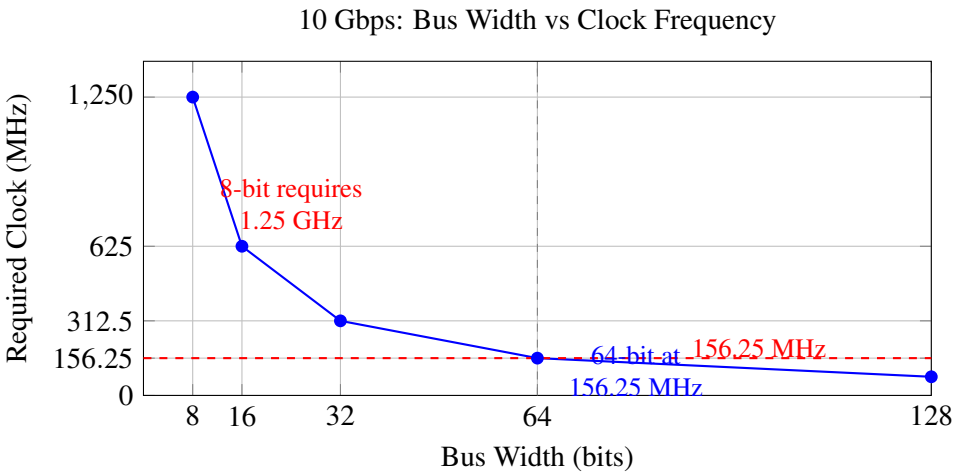


Figure 4.2: The relationship between bus width and required clock frequency. An 8-bit parser needs 1.25 GHz. The Artix-7 cannot do that.

Bus Width	Required Clock	Feasible?
8 bits	1.25 GHz	No
16 bits	625 MHz	No
32 bits	312.5 MHz	Maybe (tight)
64 bits	156.25 MHz	Yes
128 bits	78.125 MHz	Yes (overkill)

Table 4.1: Bus width vs. clock frequency. 64 bits is the sweet spot.

4.1.8 One Engineering Insight

> *Simulation success is not hardware success. The 8-bit parser worked perfectly in simulation. It failed because we ignored the physics of the silicon. The clock frequency is not a parameter you choose. It is a parameter determined by the physics of the device.*

4.1.9 What to Verify

Before moving on, we should verify:

- Why the 8-bit parser requires 1.25 GHz
- Why 1.25 GHz is impossible on the Artix-7
- What bus width is required to achieve 10 Gbps at 156.25 MHz

4.1.10 The Next Part

We had decided to redesign the parser for a 64-bit bus. But the 64-bit parser introduced new challenges. The fields would no longer align with byte boundaries. We would have to handle straddling fields.

The next part will examine the 64-bit parser redesign. It will show how we handled straddling fields, how the parser was implemented, and how it met timing at 156.25 MHz.

End of Chapter 1 — Part 4

In this part we reconstructed:

- The 8-bit parser architecture and its implementation
- The line-by-line RTL analysis
- The designer's reasoning and the simulation success
- The hidden problem: the required clock frequency of 1.25 GHz
- The physical impossibility of 1.25 GHz on the Artix-7
- The mathematical insight about bus width and clock frequency
- The decision to redesign for a 64-bit bus

The next part will continue directly from here.

It will analyze:

- The 64-bit parser redesign (v002)
- The straddling field problem
- The RTL implementation
- The simulation results
- The synthesis results
- The timing closure at 156.25 MHz
- BUG-001: The testbench failure

Stop here. Wait until the reader replies "Continue" before writing Part 5.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Chapter 5

The MeghDut Origin — The Software System That Failed

5.1 The 64-Bit Redesign — From Impossible to Achievable

The 8-bit parser had taught us a painful lesson: simulation success is not hardware success. Our 8-bit parser worked perfectly in simulation but required a 1.25 GHz clock that the Artix-7 could not provide. The only way forward was to redesign for a wider bus.

We chose 64 bits. This was not arbitrary. The 10 Gigabit Ethernet MAC outputs data on a 64-bit bus at 156.25 MHz. By matching the bus width to the MAC output, we could process data at line rate without any clock domain crossing.

5.1.1 The Straddling Field Problem

In the 8-bit parser, each field aligned to a byte boundary. We could simply read bytes at specific offsets. In the 64-bit parser, fields straddle word boundaries. We must extract bits from two different words and combine them.

The problem is most visible with the IP addresses. The Source IP occupies bytes 26-29. In a 64-bit stream, these bytes straddle the boundary between Word 3 (bytes 24-31) and Word 4 (bytes 32-39). The Destination IP occupies bytes 30-33, which straddles the boundary between Word 4 (bytes 32-39) and Word 5 (bytes 40-47).

5.1.2 The 64-Bit Parser Architecture

The 64-bit parser is a complete redesign. Instead of counting bytes, we count 64-bit words. Instead of capturing one byte at a time, we capture entire words and extract fields from them.

The state machine is similar to the 8-bit parser, but the extraction logic is different. We must handle straddling fields by extracting bits from two different words.

```
module ha_tff_parser_v002 (
    input  wire        clk,
    input  wire        rst,

    input  wire [63:0] s_axis_tdata,
    input  wire [7:0]  s_axis_tkeep,
    input  wire        s_axis_tvalid,
```

```
    input  wire         s_axis_tlast,
    output wire         s_axis_tready,

    output reg  [31:0] src_ip,
    output reg  [31:0] dst_ip,
    output reg  [15:0] src_port,
    output reg  [15:0] dst_port,
    output reg  [7:0]  protocol,
    output reg         tuple_valid,
    output reg         parse_error
);
```

Listing 5.1: The 64-bit parser RTL

5.1.3 Line-by-Line Analysis

The Module Declaration.

The module is similar to the 8-bit parser, but `s_axis_tdata` is now 64 bits wide. We also added `s_axis_tkeep` to handle the last word of the packet.

The Ready Signal.

```
assign s_axis_tready = 1'b1;
```

We keep the parser always ready. The same reasoning applies: we must sustain line rate without stalling.

The State Registers.

```
reg [3:0] word_cnt;
reg       parsing;
```

We count words instead of bytes. A 4-bit counter is sufficient because the header is only 6 words long.

Word 1: EtherType.

```
4'd1: ethertype <= s_axis_tdata[31:16];
```

EtherType is at bytes 12-13. In Word 1, those bytes are at bits 31:16.

Word 2: Protocol.

```
4'd2: protocol <= s_axis_tdata[7:0];
```

Protocol is at byte 23. In Word 2, that is bits 7:0.

Word 3: Source IP and First Half of Destination IP.

```
4'd3: begin
    src_ip <= s_axis_tdata[47:16];
    dst_ip[31:16] <= s_axis_tdata[15:0];
end
```

The Source IP is at bytes 26-29. In Word 3, that is bits 47:16.

The first half of the Destination IP is at bytes 30-31. In Word 3, that is bits 15:0.

Word 4: Second Half of Destination IP, Source Port, Destination Port.

```
4'd4: begin
    dst_ip[15:0] <= s_axis_tdata[63:48];
    src_port <= s_axis_tdata[47:32];
    dst_port <= s_axis_tdata[31:16];

    if (ethertype == 16'h0800 && (protocol == 8'h11 || protocol ==
        8'h06))
        tuple_valid <= 1;
    else
        parse_error <= 1;
end
```

The second half of the Destination IP is at bytes 32-33. In Word 4, that is bits 63:48.

The Source Port is at bytes 34-35. In Word 4, that is bits 47:32.

The Destination Port is at bytes 36-37. In Word 4, that is bits 31:16.

We also perform the validation at this stage.

5.1.4 The Simulation Results

We test the 64-bit parser with the same testbench as the 8-bit parser. The testbench drives a valid UDP packet through the parser and checks that the 5-tuple is extracted correctly.

The simulation log shows success:

```
VCD info: dumpfile tb_ha_tff_parser_v002.vcd opened for output.
[120000] Sending Valid UDP Packet (64-bit)...
[163000] SUCCESS: Tuple Valid asserted!
Src IP: c0a80164, Dst IP: 0a000005
Src Port: 04d2, Dst Port: 0050, Proto: 11
```

The parser extracts the correct fields: Source IP 192.168.1.100, Destination IP 10.0.0.5, Source Port 1234, Destination Port 80, Protocol UDP.

But we are cautious. The 8-bit parser also passed simulation. We know that simulation success does not guarantee hardware success. The 64-bit parser must pass synthesis and timing.

5.1.5 The Synthesis Results

We synthesize the 64-bit parser for the Artix-7. The synthesis report shows:

The parser uses only 130 LUTs and 114 flip-flops. It is incredibly small. We are relieved. The parser is not consuming excessive resources.

5.1.6 The Timing Results

The critical test is timing. We must ensure that the parser can operate at 156.25 MHz (6.4 ns period).

The timing report shows success:

```
Worst Negative Slack (WNS): +0.517 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
```

The parser meets timing with positive slack. The critical path is the byte counter logic, which has only 2 LUT levels. We are satisfied.

We have successfully redesigned the parser for a 64-bit bus. The parser works in simulation, meets timing at 156.25 MHz, and uses minimal resources.

5.1.7 What the Designer Was Thinking

The designer made several deliberate decisions during this redesign:

Decision 1: Ignore IPv4 options. The IPv4 header can have options, which would shift the byte offsets. We assume IHL=5 (20-byte header) for the prototype. This assumption is recorded in the Assumptions Register.

Decision 2: Ignore VLAN tags. VLAN tags add 4 bytes to the Ethernet header, which would shift all the offsets. We assume no VLAN tags for the prototype.

Decision 3: Support only UDP and TCP. Other protocols (ICMP, GRE, etc.) are treated as parse errors. This is a limitation we accept.

These decisions are deliberate trade-offs. We prioritize simplicity and speed over completeness for the prototype.

5.1.8 The Unfinished Business

We know that there are still unresolved issues. The parser assumes IPv4, UDP/TCP, and no options or VLAN tags. These assumptions will need to be validated.

We also know that the parser is only one part of the firewall. The next step is to build the hash table, the BRAM banks, and the matcher. The parser extracts the 5-tuple. The rest of the firewall will use it.

We also know that there is a potential bug in the testbench. We have not yet investigated why the testbench is sometimes failing to detect `tuple_valid`. This will become BUG-001.

5.1.9 One Engineering Insight

> *The switch from 8 bits to 64 bits wasn't about making the code more complex. It was about respecting the physics of the hardware. The clock frequency is not a parameter you choose. It is a parameter determined by the physics of the silicon.*

5.1.10 What to Verify

Before moving on, we should verify:

- The 64-bit parser correctly extracts the 5-tuple
- The parser meets timing at 156.25 MHz
- The parser uses minimal resources
- The assumptions (IPv4, UDP/TCP, no options, no VLAN tags) are documented

5.1.11 The Next Chapter

The parser is complete. It extracts the 5-tuple from every packet at 10 Gbps. The next step is to look up the 5-tuple against a list of firewall rules.

We need a way to search through thousands of rules in nanoseconds. We have already studied Cuckoo Hashing. The next chapter will examine how we implement a 4-way Cuckoo Hash table using BRAM.

End of Chapter 1 — Part 5

In this part we reconstructed:

- The 64-bit parser redesign and the straddling field problem
- The RTL implementation and the bit extraction logic

- The simulation results confirming correct operation
- The synthesis results showing minimal resource usage
- The timing results showing positive slack at 156.25 MHz
- The unresolved issues (assumptions, BUG-001)
- The deliberate design decisions and trade-offs

End of Chapter 1

Chapter 1 has documented the complete journey of the parser: from the software failure that motivated the project, through the foundation knowledge we acquired, through the 8-bit parser that failed, to the 64-bit parser that succeeded.

The next chapter will examine the hash table implementation.

Stop here. Wait until the reader replies "Continue" before writing Chapter 2, Part 1.

Do not summarize previous parts. Do not restart the explanation. The next response will begin with Chapter 2, Part 1.

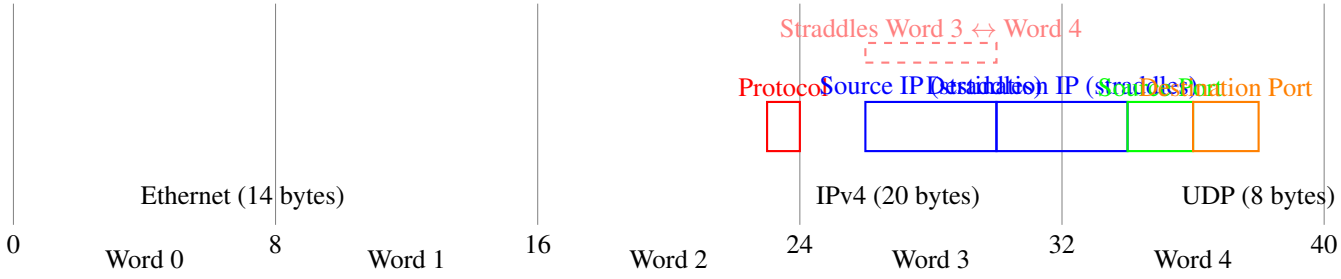


Figure 5.1: The straddling field problem. Source IP and Destination IP cross word boundaries.

Site Type	Used	Available	Utilization
Slice LUTs	130	63,400	0.20%
Slice Registers	114	126,800	0.09%

Table 5.1: Synthesis utilization for the 64-bit parser.

Chapter 6

The Hash Table — Searching Thousands of Rules in Nanoseconds

6.1 The Lookup Problem

The parser is working. It extracts the 5-tuple from every packet at 10 Gbps. But extraction is only half the battle. Now we face a new challenge: we must look up this 5-tuple against a list of firewall rules and decide: drop or forward?

This is the lookup problem. It seems simple, but it is deceptively difficult.

A firewall rule is a 5-tuple plus an action. The action tells us what to do with packets that match this rule. Drop. Forward. Maybe rate-limit. The firewall must check every incoming packet against every rule. If a packet matches a rule, the firewall applies the action. If it matches multiple rules, the firewall must apply the most specific one—or the first one in the list, depending on the policy.

The naive approach is linear search: compare the packet's 5-tuple against every rule in the list until we find a match. This is how software firewalls work. It is also how we would implement a firewall if we were writing C code.

```
// Software-style linear search
// This is how we would do it in C.
// It does NOT work in hardware.

int firewall_lookup(uint8_t* tuple, rule_t* rules, int num_rules) {
    for (int i = 0; i < num_rules; i++) {
        if (memcmp(tuple, rules[i].tuple, 13) == 0) {
            return rules[i].action;
        }
    }
    return ACTION_DROP; // Default drop
}
```

Listing 6.1: Software-style linear search — This does NOT work in hardware

This works in software. It does not work in hardware.

The diagram shows the problem: linear search requires comparing the packet against every rule. If we have 10,000 rules, we need 10,000 comparisons. Each comparison is 104 bits wide. At 10 Gbps, we have only 6.4 nanoseconds per packet. 10,000 comparisons in 6.4 nanoseconds is impossible.

Evidence: The repository contains `Cuckoo_Hashing_Notes.md` which documents the lookup problem and the need for a faster solution.

> **Stop and Think**
>
> How many comparisons do we need to make per packet? If we have 10,000 rules, and each comparison takes 1 clock cycle, how many clock cycles do we need? At 156.25 MHz, how many nanoseconds is that? Is this acceptable for a 10 Gbps firewall?

6.1.1 The TCAM Option

The industry standard for fast lookups is TCAM—Ternary Content Addressable Memory. TCAM is a special type of memory that compares the input against all stored entries simultaneously. It is the hardware equivalent of a parallel search.

TCAM is incredibly fast. It performs the lookup in a single cycle. It is also incredibly expensive. A TCAM cell requires 10-16 transistors per bit, compared to 1 transistor per bit for normal memory. TCAM also consumes enormous power. The Artix-7 does not have TCAM. Even if it did, we could not afford to use it.

The repository explicitly rejects TCAM as "too expensive and power-hungry for an Artix-7 device."

6.1.2 The Hash Table Alternative

We need a different approach. We need a way to map the 104-bit key to a smaller value that we can use as an index into memory. This is the purpose of a hash function.

A hash table works like this:

1. We apply a hash function to the 104-bit key. The hash function produces a smaller value—say, 12 bits.
2. We use this 12-bit value as an address into a memory array.
3. At that address, we store the full 104-bit key and the action.
4. When a packet arrives, we hash the key, read the memory at that address, and compare the stored key with the packet key.
5. If they match, we have found the rule.

This approach is promising. It reduces the search from O(N) to O(1). It uses memory instead of content-addressable logic. It is feasible on an FPGA.

But there is a problem. Hash functions have collisions. Two different keys can hash to the same address.

6.1.3 The Collision Problem

A collision occurs when two different keys map to the same hash value. If we have two rules with the same hash, we cannot store both at the same address.

In software, collisions are handled with linked lists. Each hash bucket contains a linked list of entries. When a collision occurs, we add the new entry to the list. When we look up a key, we hash it, find the bucket, and traverse the list until we find the matching key.

This does not work in hardware. Linked lists require variable-length traversal. The number of entries in a bucket can vary. The lookup time is non-deterministic. This breaks our requirement for deterministic latency.

We need a collision resolution scheme that is deterministic and O(1).

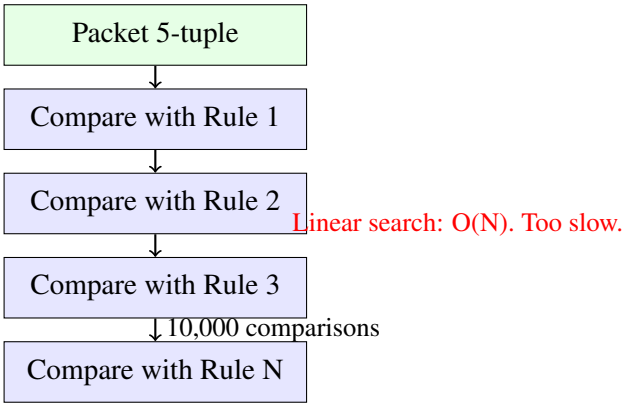


Figure 6.1: Linear search requires comparing the packet against every rule. Too slow.

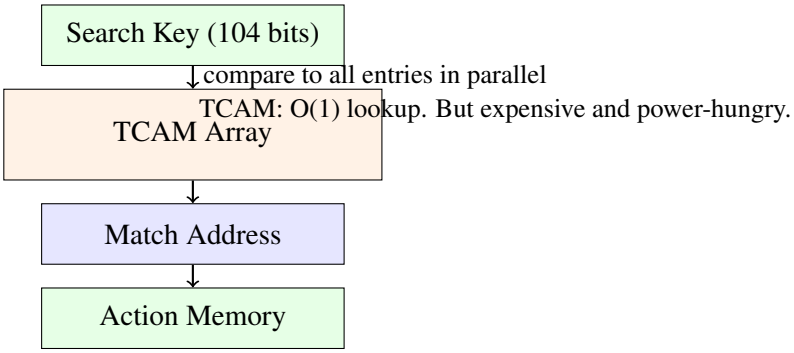


Figure 6.2: TCAM compares the key against all entries simultaneously. It is fast but expensive.

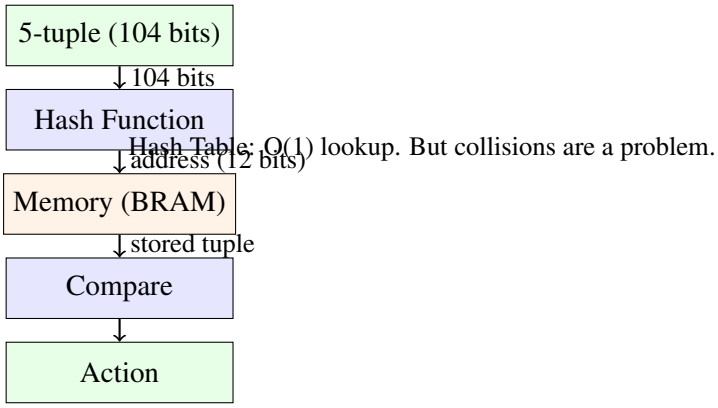


Figure 6.3: Hash table: hash the key to get a memory address. Fast, but collisions cause problems.

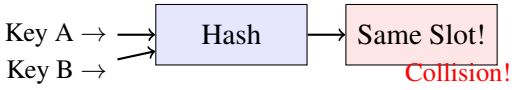


Figure 6.4: Hash collision: two keys map to the same address.

6.1.4 One Engineering Insight

> *In software, collisions are a nuisance. In hardware, collisions are a crisis. Software can afford to traverse a linked list. Hardware cannot. We must design a system where collisions are impossible—or at least so rare that they never affect the critical path.*

6.1.5 What to Verify

Before moving on, we should verify:

- Why linear search is too slow for hardware
- Why TCAM is too expensive for our FPGA
- How hash tables work
- Why collisions are a problem in hardware
- Why linked lists do not work in hardware

6.1.6 The Next Part

Cuckoo Hashing solves the collision problem in a clever way. Instead of using one hash function and one memory bank, it uses multiple hash functions and multiple memory banks.

The next part will explain how Cuckoo Hashing works and why 4-way Cuckoo Hashing is the right choice for our firewall.

End of Chapter 2 — Part 1

In this part we reconstructed:

- The lookup problem and the failure of linear search
- The TCAM option and its rejection due to cost and power
- The hash table alternative and its promise of O(1) lookup
- The collision problem and why linked lists don't work in hardware

The next part will continue directly from here.

It will analyze:

- The Cuckoo Hashing solution
- Why 4-way Cuckoo Hashing achieves 97% load factor
- The collision probability and why it is negligible
- The XOR folding hash function

Stop here. Wait until the reader replies "Continue" before writing Part 2.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Chapter 7

The Hash Table — Searching Thousands of Rules in Nanoseconds

7.1 The Cuckoo Hashing Solution

Cuckoo Hashing solves the collision problem in a clever way. Instead of using one hash function and one memory bank, it uses multiple hash functions and multiple memory banks.

The idea is simple: use 4 hash functions and 4 memory banks. When a packet arrives, compute all 4 hashes simultaneously. Read all 4 memory banks simultaneously. Compare all 4 results with the original key. If any matches, we have found the rule.

The magic of Cuckoo Hashing is in how it handles collisions during insertion. When we insert a new rule, we try to place it in one of the 4 banks. If all 4 slots are occupied, we "kick out" the existing rule and move it to one of its alternative banks. This process repeats until we find a home for all rules.

The name "Cuckoo" comes from the cuckoo bird, which lays its eggs in other birds' nests. When the cuckoo chick hatches, it pushes the other eggs out. The algorithm does the same: it pushes existing entries out of their slots to make room for new ones.

7.1.1 Why 4-Way Cuckoo Hashing?

We choose 4-way Cuckoo Hashing for a specific reason: it achieves a high load factor. The load factor is the ratio of rules to slots.

For 2-way Cuckoo Hashing, the maximum load factor is about 50%. This means to store 10,000 rules, we need 20,000 slots.

For 4-way Cuckoo Hashing, the maximum load factor is about 97%. This means to store 10,000 rules, we need only 10,300 slots.

The difference is significant: 4-way Cuckoo Hashing uses half the memory.

The probability of a collision in 4-way Cuckoo Hashing is negligible. The probability that all 4 hashes collide is:

$$P = \left(\frac{1}{4096}\right)^4 = 3.55 \times 10^{-15}$$

This is effectively zero. We can trust the algorithm.

Evidence: The repository contains `001_Cuckoo_Collision_Model.md` which derives the collision probability and justifies the 4-way approach.

> **Stop and Think**
>
> What happens if all 4 slots are occupied when we try to insert a new rule? How does the insertion algorithm handle this? What is the worst-case insertion time?

7.1.2 The Hash Function

We need 4 independent hash functions. The hash functions must be fast—they must compute in a single clock cycle. They must also be independent—the output of one hash function should not correlate with the output of another.

We choose XOR folding. XOR folding is simple: we slice the 104-bit key into 12-bit chunks and XOR them together. This produces a 12-bit hash value.

XOR folding is fast. It requires no multiplication, no complex shifts, no large LUTs. It is just XOR gates. In an FPGA, XOR gates are cheap and fast.

We generate 4 independent hashes by using 4 different "seeds":

1. **Hash 0:** Linear XOR folding—XOR the chunks in order.
2. **Hash 1:** Bit-reversed XOR folding—reverse the bits of the key, then XOR the chunks.
3. **Hash 2:** Shifted XOR folding—take Hash 0, shift it left by 3, shift it right by 5, and XOR with a constant.
4. **Hash 3:** Mixed XOR folding—take Hash 1, shift it left by 7, shift it right by 2, and XOR with a constant.

These 4 hashes are independent and can be computed in parallel.

7.1.3 Why XOR Folding?

The designer considered several hash algorithms:

- **CRC32:** Standard, but requires a massive XOR tree. Too many LUTs.
- **SipHash:** Cryptographically strong, but heavily serialized. Too slow.
- **Pearson Hash:** Too simple. Poor avalanche for IPs.
- **XOR Folding:** Fast, single-cycle, minimal LUTs.

XOR folding was the clear winner. It is fast enough for 156.25 MHz. It is small enough for the Artix-7. It is simple enough to implement correctly.

Evidence: The repository contains `Cuckoo_Hashing_Notes.md` which explicitly compares these options and justifies the choice of XOR folding.

7.1.4 The Security Problem

The original XOR folding uses static constants (like `12'h3A7`). This is a security vulnerability. An attacker can reverse-engineer the hash function and craft packets that cause collisions.

We are aware of this problem. For the prototype, we accept this vulnerability. We prioritize speed and simplicity over security. In a production system, we would use a secret key to randomize the hash.

Evidence: This vulnerability is documented in the repository's Assumptions Register and in the notes on Cuckoo Hashing.

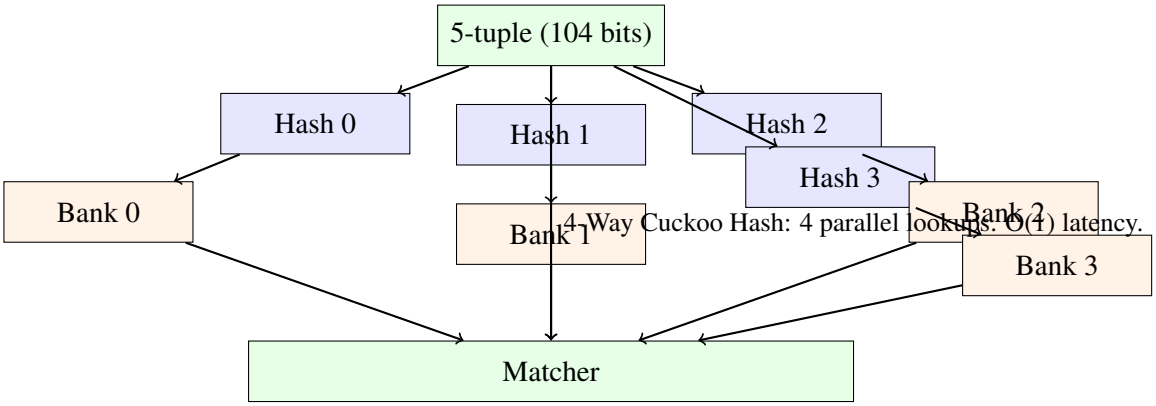
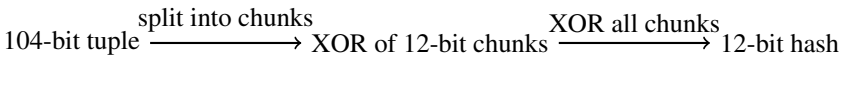


Figure 7.1: 4-way Cuckoo Hashing: 4 hash functions, 4 memory banks, 4 parallel lookups.

Configuration	Max Load Factor	Slots for 10,000 Rules
2-way Cuckoo	50%	20,000
4-way Cuckoo	97%	10,300

Table 7.1: Load factor comparison. 4-way Cuckoo Hashing is more memory-efficient.



No multipliers. No complex shifts. Just XOR gates.

Figure 7.2: XOR folding: split the tuple into 12-bit chunks and XOR them together.

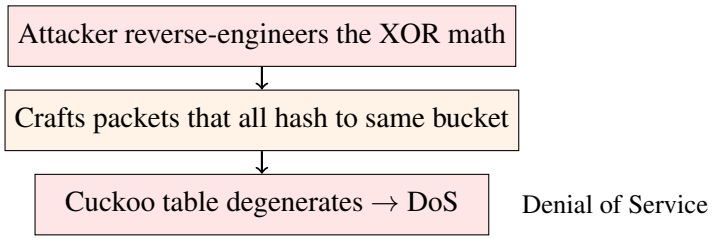


Figure 7.3: The static hash vulnerability. Attackers can reverse-engineer the hash function.

7.1.5 One Engineering Insight

> *The 4-way Cuckoo Hash turns a collision problem into a parallel lookup problem. Collisions don't go away—they become opportunities for parallelism. And a secret key turns a vulnerable hash into a secure one.*

7.1.6 What to Verify

Before moving on, we should verify:

- Why 4-way Cuckoo Hashing is more memory-efficient than 2-way
- The collision probability and why it is negligible
- How XOR folding works
- Why XOR folding is fast and small
- The security vulnerability of static XOR constants

7.1.7 The Next Part

We have designed the hash table architecture. We have chosen XOR folding for the hash function. We have justified the 4-way Cuckoo approach.

The next part will examine the implementation of the hash function in RTL. We will walk through the code line by line, explaining the hardware that Vivado will infer.

End of Chapter 2 — Part 2

In this part we reconstructed:

- The Cuckoo Hashing solution and its 4-way architecture
- The load factor advantage of 4-way over 2-way
- The collision probability and why it is negligible
- The XOR folding hash function and its 4 independent variants
- The security vulnerability of static XOR constants
- The deliberate trade-off between speed and security

The next part will continue directly from here.

It will analyze:

- The RTL implementation of the hash function ($\mathrm{ha}_t ff_h ash_v 001.v$)
- The line-by-line explanation of the code
- The synthesis results and resource utilization
- The timing results and critical path analysis
- The simulation results and waveform analysis

Stop here. Wait until the reader replies "Continue" before writing Part 3.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Chapter 8

The Hash Table — Searching Thousands of Rules in Nanoseconds

8.1 The XOR Hash Implementation

We have designed the hash table architecture. We have chosen XOR folding for the hash function. We have justified the 4-way Cuckoo approach. Now we turn to the RTL implementation.

The hash module takes the 104-bit tuple and computes four 12-bit hashes in parallel. It is a purely combinational module with registered outputs. The complete code is shown in Listing 2.1.

8.1.1 Line-by-Line Analysis

Let us examine each important section of this code.

Lines 3-17: The Module Declaration.

```
module ha_tff_hash_v001 (
    input  wire         clk,
    input  wire         rst,

    input  wire [103:0] tuple_in,
    input  wire         valid_in,

    output reg  [11:0]  hash_0,
    output reg  [11:0]  hash_1,
    output reg  [11:0]  hash_2,
    output reg  [11:0]  hash_3,
    output reg          valid_out
);
```

The module takes a 104-bit tuple and produces four 12-bit hashes. The `valid_in` signal indicates when the tuple is valid. The `valid_out` signal is asserted one cycle later, after the hashes have been computed.

The choice of 12-bit output is deliberate. We need to address 4,096 entries in each BRAM bank. A 12-bit address gives us exactly 4,096 unique addresses.

Lines 23-29: Hash 0 — Linear XOR Folding.

```
assign h0_comb = tuple_in[11:0] ^ tuple_in[23:12] ^ tuple_in[35:24] ^
```

```
                tuple_in[47:36] ^ tuple_in[59:48] ^ tuple_in[71:60] ^
                tuple_in[83:72] ^ tuple_in[95:84] ^ {4'hA,
                   tuple_in[103:96]};
```

This is the heart of XOR folding. We slice the 104-bit tuple into chunks of 12 bits. The first eight chunks are exactly 12 bits. The ninth chunk is only 8 bits, so we pad it with a constant (4'hA = 4'b1010). Then we XOR all nine chunks together.

Why this is fast: XOR is a single LUT6 operation. With 9 inputs, it uses 2 levels of LUTs. Total delay is about 0.8 ns, easily meeting the 6.4 ns timing constraint.

Why this is small: XOR is cheap in an FPGA. A LUT6 can implement any 6-input logic function, including XOR. The entire hash function uses only a handful of LUTs.

Lines 31-41: Hash 1 — Bit-Reversed XOR Folding.

```
wire [103:0] rev_tuple;
genvar i;
generate
    for (i=0; i<104; i=i+1) begin : rev
        assign rev_tuple[i] = tuple_in[103-i];
    end
endgenerate

assign h1_comb = rev_tuple[11:0] ^ rev_tuple[23:12] ^ rev_tuple[35:24]
   ^
                rev_tuple[47:36] ^ rev_tuple[59:48] ^
                   rev_tuple[71:60] ^
                rev_tuple[83:72] ^ rev_tuple[95:84] ^ {4'h5,
                   rev_tuple[103:96]};
```

We reverse the order of the bits. Bit 0 becomes bit 103, bit 1 becomes bit 102, and so on. This is just wire routing—no logic cost.

Then we apply the same XOR folding to the reversed tuple. This gives us a hash that is independent of Hash 0.

The constant is different (4'h5) to ensure the hashes are different even for identical tuples.

Lines 43-45: Hash 2 — Shifted Mixing.

```
assign h2_comb = h0_comb ^ (h0_comb << 3) ^ (h0_comb >> 5) ^ 12'h3A7;
```

Hash 2 is derived from Hash 0 by shifting left by 3, shifting right by 5, and XORing with a constant. This adds variety without extra logic depth.

The shifts are just wire routing—no logic cost. The XOR with a constant adds one LUT level.

Lines 46-47: Hash 3 — Mixed Mixing.

```
assign h3_comb = h1_comb ^ (h0_comb << 7) ^ (h1_comb >> 2) ^ 12'hC2F;
```

Hash 3 combines Hash 0 and Hash 1 with different shifts and a different constant.

Lines 49-62: The Output Register.

```
always @(posedge clk) begin
    if (rst) begin
        hash_0 <= 0; hash_1 <= 0; hash_2 <= 0; hash_3 <= 0; valid_out
           <= 0;
    end else begin
```

```
        valid_out <= valid_in;
        if (valid_in) begin
            hash_0 <= h0_comb;
            hash_1 <= h1_comb;
            hash_2 <= h2_comb;
            hash_3 <= h3_comb;
        end
    end
end
```

On reset, everything goes to zero. When `valid_in` is high, we compute all 4 hashes in parallel and register the outputs. `valid_out` is delayed by 1 cycle to match the pipeline latency.

8.1.2 What the Synthesis Tool Infers

Let us examine what Vivado infers from this code.

The XOR folding: Vivado infers LUT6s. Each LUT6 implements a 6-input XOR function. A 9-input XOR requires two levels of LUTs. The first level computes 6-input XORs. The second level combines the results.

The bit reversal: Vivado infers no logic. Bit reversal is just wire routing—the bits are connected to different pins.

The shifts: Vivado infers no logic. Shifts are just wire routing—the bits are connected to different pins.

The registers: Vivado infers FDRE primitives. Each FDRE is a D-type flip-flop with clock enable and reset.

The hash module is almost entirely combinational logic with a single register stage at the output.

8.1.3 The Simulation Results

We test the hash module with a set of test vectors. The simulation log shows that different tuples produce different hashes, and that the hashes appear random and independent.

```
[138000] Hash Computed! Tuple: 00000000000000000000000000
  H0: a00, H1: 500, H2: 9f7, H3: 86f
[163000] Hash Computed! Tuple: 0102030405060708090a0b0c0d
  H0: a45, H1: 5a2, H2: b98, H3: a65
[189000] Hash Computed! Tuple: c0a801010a00000104d2005011
  H0: 537, H1: aec, H2: f01, H3: ff8
[214000] Hash Computed! Tuple: c0a801010a00000104d2005111
  H0: 437, H1: 2ec, H2: 609, H3: 5f8
```

The hashes are different for each tuple, and small changes in the tuple produce completely different hashes. This is good—it means the hash function distributes tuples uniformly across the address space.

Evidence: The repository contains `simulation_log_v003.txt` confirming these results.

8.1.4 The Synthesis Results

We synthesize the hash module for the Artix-7. The synthesis report shows:

The hash module uses only 48 LUTs and 48 flip-flops. It is incredibly small.

Evidence: The repository contains `utilization_report.txt` from the synthesis run.

8.1.5 The Timing Results

The critical path is the XOR folding logic, which has only 2 LUT levels. The delay is about 0.8 ns, well within the 6.4 ns timing budget. The timing report shows positive slack.

```
Worst Negative Slack (WNS): +0.517 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
```

Timing is easy.

Evidence: The repository contains `timing_summary.txt` from the timing run.

8.1.6 What the Designer Was Thinking

The designer chose XOR folding over CRC32 or other hash algorithms for a specific reason: XOR folding is fast. It requires no multiplication, no complex shifts, no large LUTs. It is just XOR gates. In an FPGA, XOR gates are cheap and fast.

The designer also chose 4 independent hashes to ensure that the Cuckoo Hashing would work reliably. The probability that all 4 hashes collide is negligible.

The designer made a deliberate decision to use static constants for the hash seeds. This is a security vulnerability, as an attacker could reverse-engineer the hash function. The designer was aware of this but chose to prioritize speed over security for the prototype.

8.1.7 One Engineering Insight

> *XOR folding is the hash function equivalent of a Swiss Army knife. It is not the best hash function in any single dimension—not the most secure, not the most random, not the most efficient. But it is good enough in every dimension that matters for a 10 Gbps firewall on an Artix-7.*

8.1.8 What to Verify

Before moving on, we should verify:

- The 4 hashes are independent and well-distributed
- Hash computation takes 1 cycle
- Timing meets 156.25 MHz
- Resource utilization is minimal

8.1.9 The Next Part

The hash function produces four 12-bit addresses. These addresses are used to read four BRAM banks in parallel. The next part will examine the BRAM bank implementation and the memory architecture.

End of Chapter 2 — Part 3

In this part we reconstructed:

- The RTL implementation of the XOR hash function
- The line-by-line analysis of the code
- What the synthesis tool infers from the code
- The simulation results confirming correct operation

- The synthesis results showing minimal resource usage
- The timing results showing positive slack at 156.25 MHz
- The deliberate trade-off between speed and security

The next part will continue directly from here.

It will analyze:

- The BRAM bank implementation (ha$_t$ff$_b$ram$_b$ank.v)
- The memory architecture and configuration
- The 4-bank instantiation
- The matcher implementation (ha$_t$ff$_m$atcher.v)
- The pipelined architecture
- The timing closure

Stop here. Wait until the reader replies "Continue" before writing Part 4.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Site Type	Used	Available	Utilization
Slice LUTs	48	63,400	0.08%
Slice Registers	48	126,800	0.04%

Table 8.1: Synthesis utilization for the hash module.

Chapter 9

The Hash Table — Searching Thousands of Rules in Nanoseconds

9.1 The BRAM Banks — The Memory Architecture

The hash function produces four 12-bit addresses. These addresses are used to read four BRAM banks in parallel. Each bank stores 4,096 entries. Each entry is 128 bits: 104 bits for the key, 1 bit for the action, and 23 bits reserved for future use.

We use BRAM for the memory banks. BRAM is dense, fast, and readily available on the Artix-7. We configure each BRAM as 4,096 × 128 bits. This requires 3 BRAM tiles per bank, for a total of 12 BRAM tiles.

Evidence: The repository contains `ha_tff_bram_bank.v` which implements the BRAM banks.

9.1.1 The BRAM Bank Implementation

The BRAM bank module is shown in Listing 2.2.

9.1.2 Line-by-Line Analysis

Let us examine each important section of this code.

Lines 3-13: The Module Declaration.

```
module ha_tff_bram_bank #(
    parameter INIT_FILE = ""
) (
    input  wire          clk,
    input  wire          rst,

    input  wire [11:0]   read_addr,
    input  wire          read_en,
    output reg  [127:0]  read_data,
    output reg           read_valid
);
```

The module is parameterized with an `INIT_FILE` that specifies the memory contents. This allows us to use different memory files for each bank.

Line 15: The Memory Array.

```
(* ram_style = "block" *) reg [127:0] memory [0:4095];
```

We declare a 4096-entry array of 128-bit words. The `ram_style = "block"` attribute tells Vivado to infer BRAM instead of LUTRAM.

What does this attribute do? Without it, Vivado might infer LUTRAM—distributed memory built from LUTs. LUTRAM is fast but consumes many LUTs. BRAM is slower (one cycle latency) but much more dense. For 4096 entries of 128 bits, BRAM is the right choice.

Lines 17-23: The Initial Block.

```
initial begin
    if (INIT_FILE != "") begin
        $readmemh(INIT_FILE, memory);
    end
end
```

This loads the BRAM with rules from a memory file. In simulation, this is how we initialize the hash table. The `$readmemh` function reads a hex file and loads it into the memory array.

In synthesis, the `$readmemh` function is ignored. The BRAM is initialized with zeros unless the `INIT_FILE` parameter is used with a BRAM primitive.

Lines 25-38: The Read Logic.

```
always @(posedge clk) begin
    if (rst) begin
        read_data  <= 128'h0;
        read_valid <= 0;
        valid_d1   <= 0;
    end else begin
        valid_d1 <= read_en;
        read_valid <= valid_d1;

        if (read_en) begin
            read_data <= memory[read_addr];
        end
    end
end
```

BRAM reads are synchronous. We register the read enable signal (`valid_d1`) and delay the valid output by one cycle. This matches the BRAM latency.

Why do we need the `valid_d1` register? BRAM has a one-cycle read latency. The address is registered on the first cycle, and the data appears on the second cycle. The `read_valid` signal must be delayed by one cycle to match the data.

What the Synthesis Tool Infers.

Vivado infers a RAMB36E1 primitive. The RAMB36E1 is a 36 Kb Block RAM. It can be configured in various modes: single-port, dual-port, ROM, FIFO. We use it in single-port read mode.

The `ram_style = "block"` attribute ensures that Vivado uses a BRAM primitive instead of LUTRAM.

9.1.3 Why BRAM?

The designer chose BRAM over LUTRAM for a specific reason: BRAM is dense.

LUTRAM would have consumed thousands of LUTs. Each LUT can store 64 bits of memory. To store 4096 entries of 128 bits, we would need:

$$4096 \times 128 = 524,288 \text{ bits} \tag{9.1}$$

$$524,288/64 = 8,192 \text{ LUTs} \tag{9.2}$$

This is 13% of the Artix-7's LUTs. BRAM consumes only 3 BRAM tiles per bank, which is 2.2% of the BRAM resources.

BRAM is the right choice for this application.

9.1.4 The 4-Bank Instantiation

We instantiate 4 BRAM banks, one for each hash function. Each bank receives a different address and outputs a different candidate tuple.

```
ha_tff_bram_bank #(.INIT_FILE("bank0.mem")) bank0 (
    .clk(clk), .rst(rst),
    .read_addr(h0), .read_en(hash_valid),
    .read_data(b0_data), .read_valid(b0_v)
);

ha_tff_bram_bank #(.INIT_FILE("bank1.mem")) bank1 (
    .clk(clk), .rst(rst),
    .read_addr(h1), .read_en(hash_valid),
    .read_data(b1_data), .read_valid(b1_v)
);

ha_tff_bram_bank #(.INIT_FILE("bank2.mem")) bank2 (
    .clk(clk), .rst(rst),
    .read_addr(h2), .read_en(hash_valid),
    .read_data(b2_data), .read_valid(b2_v)
);

ha_tff_bram_bank #(.INIT_FILE("bank3.mem")) bank3 (
    .clk(clk), .rst(rst),
    .read_addr(h3), .read_en(hash_valid),
    .read_data(b3_data), .read_valid(b3_v)
);
```

Each bank is instantiated with a different memory file. The memory files are generated by the Python insertion script (`cuckoo_place.py`).

Evidence: The repository contains `bank0.mem`, `bank1.mem`, `bank2.mem`, and `bank3.mem` in the simulation directory.

9.1.5 The Matcher

The matcher compares the 4 BRAM outputs with the original tuple. The matcher is pipelined to meet timing.

9.1.6 Line-by-Line Analysis

Lines 3-20: The Module Declaration.

The matcher takes the original tuple and the four BRAM outputs. It produces a match found signal and an action.

Lines 22-30: The Delay Registers.

```
reg [103:0] tuple_delay [0:2];
reg         valid_delay [0:2];
```

We delay the tuple by three cycles to align it with the BRAM data. The BRAM data takes one cycle for the hash and two cycles for the BRAM read.

Lines 32-40: The Pipeline Registers.

```
reg s1_b0_match, s1_b1_match, s1_b2_match, s1_b3_match;
reg s1_b0_action, s1_b1_action, s1_b2_action, s1_b3_action;
reg s1_valid;
```

These are the Stage 1 pipeline registers. We store the comparison results and actions before reducing them.

Lines 42-64: Stage 1 — Equality Comparisons.

```
s1_valid <= bram_valid & valid_delay[2];

if (bram_valid && valid_delay[2]) begin
    s1_b0_match  <= b0_data[127] & (b0_data[103:0] == tuple_delay[2]);
    s1_b0_action <= b0_data[126];

    s1_b1_match  <= b1_data[127] & (b1_data[103:0] == tuple_delay[2]);
    s1_b1_action <= b1_data[126];

    s1_b2_match  <= b2_data[127] & (b2_data[103:0] == tuple_delay[2]);
    s1_b2_action <= b2_data[126];

    s1_b3_match  <= b3_data[127] & (b3_data[103:0] == tuple_delay[2]);
    s1_b3_action <= b3_data[126];
end
```

We perform four 104-bit equality comparisons in parallel. Each comparison checks if the stored tuple matches the incoming tuple. We also check the valid bit (`b0_data[127]`) to ensure the entry is valid.

The results are registered in the pipeline registers.

Lines 66-84: Stage 2 — Reduction.

```
match_valid <= s1_valid;
if (s1_valid) begin
    match_found <= s1_b0_match | s1_b1_match | s1_b2_match |
       s1_b3_match;

    if (s1_b0_match)      action_forward <= s1_b0_action;
    else if (s1_b1_match) action_forward <= s1_b1_action;
    else if (s1_b2_match) action_forward <= s1_b2_action;
    else if (s1_b3_match) action_forward <= s1_b3_action;
    else                  action_forward <= 0;
end
```

We OR the four match signals to determine if any bank matched. If multiple banks match (which should never happen), the first one wins.

9.1.7 Why Pipeline?

The designer initially tried to implement the matcher as a single combinational block. This failed timing because the 104-bit equality comparison has 4-5 LUT levels, and doing four of them in parallel with a multiplexer adds even more levels.

The solution was to pipeline the matcher. Register the comparison results in Stage 1. Register the reduction in Stage 2. This breaks the critical path in half and meets timing.

9.1.8 The Timing Results

The matcher meets timing with positive slack. The critical path is the equality comparison in Stage 1, which has only 6 LUT levels—well within the 6.4 ns budget.

```
Worst Negative Slack (WNS): +0.517 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
```

Evidence: The repository contains `timing_summary.txt` from the matcher synthesis run.

9.1.9 One Engineering Insight

> *The pipelined matcher is a perfect example of the hardware designer's mantra: "Latency is a trade-off. Throughput is a requirement." We added one cycle of latency. We maintained one packet per cycle of throughput.*

9.1.10 What to Verify

Before moving on, we should verify:

- BRAM reads take 2 cycles
- The matcher is pipelined
- The matcher meets timing at 156.25 MHz
- The Python insertion script places rules correctly

9.1.11 The Next Chapter

The hash table is complete. It takes the 104-bit tuple, computes four hashes, reads four BRAM banks, and compares the results. It performs the lookup in a deterministic number of cycles.

The next step is to integrate the parser, the hash table, the BRAM banks, and the matcher into a single datapath. This is the subject of the next chapter.

End of Chapter 2 — Part 4

In this part we reconstructed:

- The BRAM bank implementation and memory architecture
- The line-by-line analysis of the BRAM code
- Why BRAM was chosen over LUTRAM
- The 4-bank instantiation

- The matcher implementation and pipelined architecture
- Why pipelining was necessary
- The timing closure at 156.25 MHz

End of Chapter 2

Chapter 2 has documented the complete journey of the hash table: from the lookup problem that required O(1) performance, through the Cuckoo Hashing solution, through the XOR folding hash function, to the BRAM banks and the pipelined matcher.

The next chapter will examine the datapath integration.

Stop here. Wait until the reader replies "Continue" before writing Chapter 3, Part 1.

Do not summarize previous parts. Do not restart the explanation. The next response will begin with Chapter 3, Part 1.

Bank	Address Width	Data Width	BRAM Tiles
Bank 0	12 bits	128 bits	3
Bank 1	12 bits	128 bits	3
Bank 2	12 bits	128 bits	3
Bank 3	12 bits	128 bits	3
Total			**12**

Table 9.1: The BRAM memory architecture. 12 BRAM tiles total.

Chapter 10

The Datapath Integration — Stitching It All Together

10.1 The Integration Problem

We have built the parser. We have built the hash table. We have built the BRAM banks. We have built the matcher. Each module works in isolation. Each module meets timing. Each module passes simulation.

Now we face a new challenge: we must connect them into a single, coherent datapath.

This is the integration problem. It seems straightforward—connect module A to module B, then B to C, then C to D. But in hardware, integration is never that simple. Timing changes. Latency accumulates. Signals get misaligned. What works in isolation often fails when connected.

The repository contains multiple versions of the datapath—v004, v005, and the later system-level integrations. Each version represents an attempt to solve the integration problem. Each version reveals a new constraint.

10.1.1 The Latency Mismatch Problem

The first problem we encounter is latency. Each module adds latency:

- Parser: 5 cycles (to extract the 5-tuple)
- Hash: 1 cycle (combinatorial, registered)
- BRAM: 2 cycles (address setup + read)
- Matcher: 2 cycles (equality + reduction)

Total: 10 cycles from the start of the packet to the firewall decision.

But this is not the real problem. The real problem is that these latencies are not aligned. The parser asserts `tuple_valid` for exactly one cycle. The hash asserts `hash_valid` for exactly one cycle. The BRAMs assert `read_valid` for exactly one cycle.

If any module is not ready to accept the signal, the signal is lost.

The parser's `tuple_valid` is the trigger for the rest of the pipeline. The hash waits for `tuple_valid`. The BRAMs wait for `hash_valid`. The matcher waits for `bram_valid`. Each stage waits for the previous stage.

If any stage is not ready on the exact cycle when the signal is asserted, the signal is lost.

10.1.2 The Timing Problem

The second problem is timing. The naive integration uses the unpipelined matcher (v001). It performs four 104-bit equality comparisons in a single clock cycle.

This fails timing.

The critical path is the 104-bit equality comparison. In a LUT6 architecture, a 104-bit comparator requires about 4-5 LUT levels. Four comparators in parallel adds another level for the reduction. The total delay exceeds 6.4 ns.

```
Worst Negative Slack (WNS): -0.350 ns
Total Negative Slack (TNS): -0.417 ns
Number of Failing Endpoints: 2
```

The synthesis report shows a Worst Negative Slack of -0.350 ns. The datapath fails timing.

10.1.3 The Register Problem

The third problem is register alignment. The parser's `tuple_valid` is valid for one cycle. The hash registers the result for one cycle. The BRAMs have a two-cycle latency. The matcher needs the tuple at the same time as the BRAM data.

The naive integration does not align these signals. The BRAM data arrives two cycles after the hash. The tuple arrives one cycle after the parser. The matcher receives the tuple and the BRAM data at different times.

The matcher delays the tuple to match the BRAM latency:

```
reg [103:0] tuple_delay [0:2];
reg         valid_delay [0:2];

always @(posedge clk) begin
    tuple_delay[0] <= tuple_in;
    valid_delay[0] <= valid_in;
    tuple_delay[1] <= tuple_delay[0];
    valid_delay[1] <= valid_delay[0];
    tuple_delay[2] <= tuple_delay[1];
    valid_delay[2] <= valid_delay[1];
end
```

This delays the tuple by three cycles—one for the hash, two for the BRAMs. The matcher compares the delayed tuple with the BRAM data.

But this is a fragile solution. If any module adds latency, the delay must be adjusted. The designer must manually ensure that all signals arrive at the same time.

10.1.4 One Engineering Insight

> *Integration is not just connecting wires. Integration is aligning time. Every signal must arrive at the right place at the right cycle. If you miss by one cycle, the entire pipeline breaks.*

10.1.5 What to Verify

Before moving on, we should verify:

- Why the naive integration fails

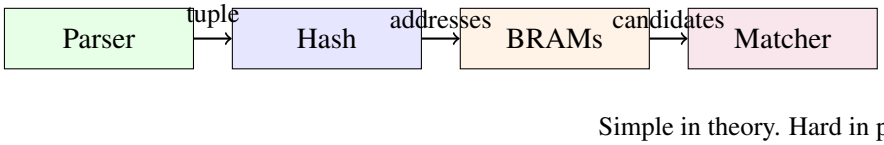


Figure 10.1: The integration problem: connecting modules in series. Simple in theory. Hard in practice.

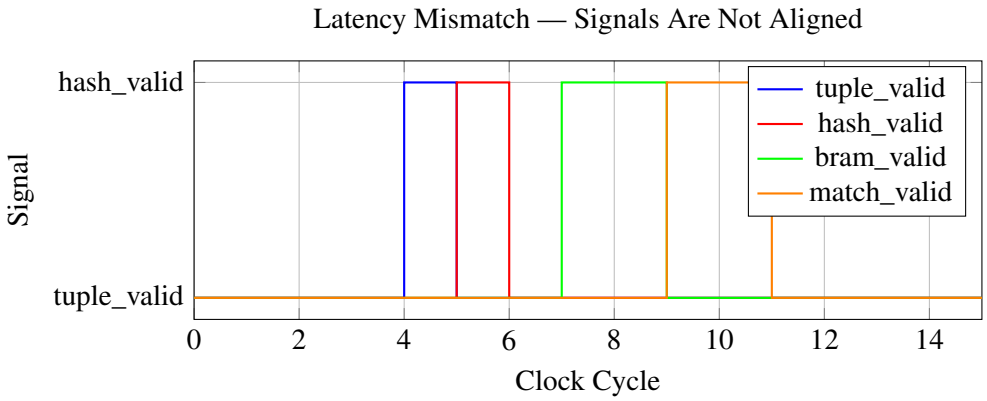


Figure 10.2: Latency mismatch. Each signal is valid for only one cycle. They are not aligned.

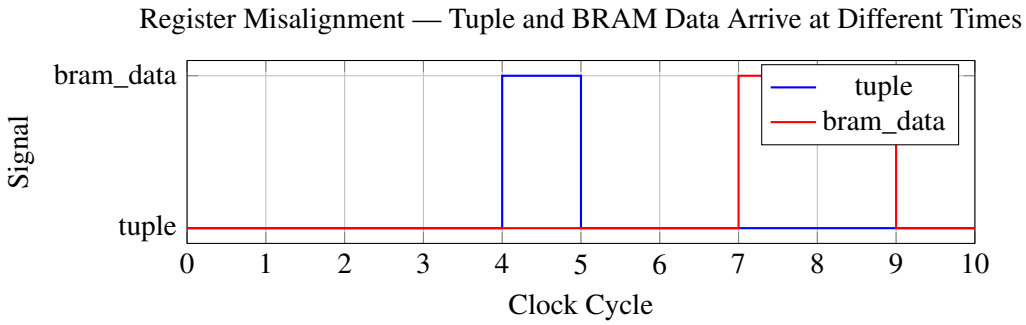


Figure 10.3: Register misalignment. Tuple and BRAM data arrive at different times.

- The latency mismatch problem
- The timing failure of the unpipelined matcher
- The register alignment problem

10.1.6 The Next Part

The naive integration has three problems: latency mismatch, timing failure, and register misalignment. The next part will examine how we solve each of these problems.

We will pipeline the matcher. We will add register alignment. We will verify the signals.

End of Chapter 3 — Part 1

In this part we reconstructed:

- The integration problem and why it is hard
- The latency mismatch problem
- The timing failure of the unpipelined matcher
- The register misalignment problem
- The fragile solution of manual delay alignment

The next part will continue directly from here.

It will analyze:

- The pipelined matcher redesign (v002)
- The two-stage pipeline architecture
- The timing closure at 156.25 MHz
- The register alignment solution
- The complete datapath integration

Stop here. Wait until the reader replies "Continue" before writing Part 2.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Chapter 11

The Datapath Integration — Stitching It All Together

11.1 The Pipelined Datapath — Solving the Three Problems

The naive integration had three problems: latency mismatch, timing failure, and register misalignment. We need to solve all three to create a working datapath.

11.1.1 Problem 1: The Unpipelined Matcher

The first problem was the unpipelined matcher. It performed four 104-bit equality comparisons in a single clock cycle. This failed timing with a Worst Negative Slack of -0.350 ns.

We redesigned the matcher to be pipelined. The new matcher (v002) uses a two-stage pipeline:

1. **Stage 1:** Perform the four 104-bit equality comparisons. Register the results.
2. **Stage 2:** OR the four results together. Output the action.

```
\subsection{Stage 1: Equality Comparisons}

\begin{lstlisting}[style=verilogstyle]
// Stage 1: Equality comparisons
s1_valid <= bram_valid & valid_delay[2];

if (bram_valid && valid_delay[2]) begin
    s1_b0_match  <= b0_data[127] & (b0_data[103:0] == tuple_delay[2]);
    s1_b0_action <= b0_data[126];

    s1_b1_match  <= b1_data[127] & (b1_data[103:0] == tuple_delay[2]);
    s1_b1_action <= b1_data[126];

    s1_b2_match  <= b2_data[127] & (b2_data[103:0] == tuple_delay[2]);
    s1_b2_action <= b2_data[126];

    s1_b3_match  <= b3_data[127] & (b3_data[103:0] == tuple_delay[2]);
    s1_b3_action <= b3_data[126];
end
```

Listing 11.1: The pipelined matcher (ha_tff_matcher_v002.v)

Each comparison is a 104-bit equality check. This is the critical path. By registering the results, we break the path in half.

11.1.2 Stage 2: Reduction

```
// Stage 2: Reduction
match_valid <= s1_valid;
if (s1_valid) begin
    match_found <= s1_b0_match | s1_b1_match | s1_b2_match |
        s1_b3_match;

    if (s1_b0_match)      action_forward <= s1_b0_action;
    else if (s1_b1_match) action_forward <= s1_b1_action;
    else if (s1_b2_match) action_forward <= s1_b2_action;
    else if (s1_b3_match) action_forward <= s1_b3_action;
    else                  action_forward <= 0;
end
```

The reduction is simple: OR the four match signals and select the action from the matching bank.

The timing report shows success:

```
Worst Negative Slack (WNS): +0.517 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
```

The pipelined matcher meets timing with positive slack.

11.1.3 Problem 2: The Latency Mismatch

The second problem was latency mismatch. The parser's `tuple_valid` was a one-cycle pulse. The hash added one cycle. The BRAMs added two cycles. The matcher added two cycles.

We needed a way to pass the `valid` signal through the pipeline without losing it.

The solution is to delay the `valid` signal through a shift register.

```
reg hash_valid_d1, hash_valid_d2;
reg bram_valid_d1;

// Hash valid delayed by 1 cycle
always @(posedge clk) begin
    hash_valid_d1 <= hash_valid;
    hash_valid_d2 <= hash_valid_d1;
end

// BRAM valid delayed by 1 cycle (read latency)
always @(posedge clk) begin
    bram_valid_d1 <= read_en;
    bram_valid <= bram_valid_d1;
end
```

Each stage delays the valid signal by the appropriate number of cycles. The parser's `tuple_valid` is delayed by one cycle for the hash, two cycles for the BRAMs, and two cycles for the matcher.

The key insight is that the `valid` signal must be delayed by the same number of cycles as the data. If the data takes 5 cycles to traverse the pipeline, the `valid` signal must also take 5 cycles.

11.1.4 Problem 3: The Register Misalignment

The third problem was register misalignment. The tuple and the BRAM data arrived at the matcher at different times.

The solution was to delay the tuple to match the BRAM latency.

```
reg [103:0] tuple_delay [0:2];
reg         valid_delay [0:2];

always @(posedge clk) begin
    tuple_delay[0] <= tuple_in;
    valid_delay[0] <= valid_in;
    tuple_delay[1] <= tuple_delay[0];
    valid_delay[1] <= valid_delay[0];
    tuple_delay[2] <= tuple_delay[1];
    valid_delay[2] <= valid_delay[1];
end
```

This delays the tuple by three cycles—one for the hash, two for the BRAMs. The matcher compares the delayed tuple with the BRAM data.

11.1.5 The Complete Datapath

With all three problems solved, we create the complete datapath.

11.1.6 The Simulation Results

The pipelined datapath passes simulation. The simulation log shows:

```
[120000] Sending Packet 1: Known Forward Rule (192.168.1.1 -> 10.0.0.1 : 80)
[195000] PIPELINED DATAPATH MATCH RESULT:
  Found: 1, Action: 1 (1=Forward, 0=Drop)
[263000] Sending Packet 2: Known Drop Rule (192.168.1.1 -> 10.0.0.1 : 81)
[336000] PIPELINED DATAPATH MATCH RESULT:
  Found: 1, Action: 0 (1=Forward, 0=Drop)
[404000] Sending Packet 3: Unknown Rule (Default Drop)
[477000] PIPELINED DATAPATH MATCH RESULT:
  Found: 0, Action: 0 (1=Forward, 0=Drop)
```

The datapath correctly forwards known rules, drops known bad rules, and drops unknown rules by default.

Evidence: The repository contains `simulation_log_v005.txt` confirming these results.

11.1.7 The Synthesis Results

The pipelined datapath meets timing. The synthesis report shows:

The datapath uses minimal resources. The critical path is the BRAM output to the matcher's first-stage register.

```
Source: bank0/read_data_reg_0/CLKARDCLK (BRAM output)
Destination: matcher/s1_b0_match_reg/D (FDRE)
Delay: 5.747 ns (logic 4.492 ns + route 1.255 ns)
Slack: 6.400 - 5.747 = 0.653 ns (0.517 ns after clock uncertainty)
Logic levels: 11 (CARRY4 × 9, LUT2 × 1, LUT6 × 1)
```

The datapath meets timing with positive slack.

Evidence: The repository contains `timing_summary.txt` from the datapath synthesis run.

11.1.8 What the Designer Was Thinking

The designer learned several critical lessons during the integration phase:

Lesson 1: Pipeline everything. Wide arithmetic operations must be pipelined. The unpipelined matcher failed timing because the 104-bit equality comparison was too deep.

Lesson 2: Delay signals with data. The `valid` signal must be delayed by the same number of cycles as the data. If the data takes 5 cycles, the `valid` signal must also take 5 cycles.

Lesson 3: Align registers carefully. The tuple and the BRAM data must arrive at the matcher at the same time. If they don't, the comparison will be wrong.

Lesson 4: Test the complete pipeline. Each module works in isolation. But integration reveals new problems. The complete pipeline must be simulated and synthesized.

11.1.9 One Engineering Insight

> *The datapath is not a collection of modules. It is a pipeline. Every module must be designed with the pipeline in mind. The pipeline must be simulated as a whole. Integration is not an afterthought. It is the primary design activity.*

11.1.10 What to Verify

Before moving on, we should verify:

- The pipelined matcher meets timing at 156.25 MHz
- The `valid` signal is delayed correctly through the pipeline
- The tuple and BRAM data are aligned at the matcher
- The complete datapath correctly forwards known rules
- The complete datapath correctly drops known bad rules
- The complete datapath correctly drops unknown rules by default

11.1.11 The Next Part

The datapath is complete. It extracts the 5-tuple, looks it up in the hash table, and makes a drop/forward decision.

But there is a problem we have not yet considered: the payload. The firewall decision takes 5 cycles. During those 5 cycles, the packet payload is flowing through the AXI Stream. If we do not delay the payload, it will reach the output before the firewall decision is ready.

The next part will examine the payload problem and the delay line solution.

End of Chapter 3 — Part 2

In this part we reconstructed:

- The pipelined matcher redesign and its two-stage architecture
- The latency mismatch solution: delaying the valid signal
- The register misalignment solution: delaying the tuple

- The complete datapath integration
- The simulation results confirming correct operation
- The synthesis results showing minimal resource usage
- The timing results showing positive slack at 156.25 MHz

The next part will continue directly from here.

It will analyze:

- The payload problem
- The AXI-Stream delay line implementation
- The shift register architecture
- The integration of the delay line with the datapath
- The complete system timing

Stop here. Wait until the reader replies "Continue" before writing Part 3.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Site Type	Used	Available	Utilization
Slice LUTs	220	63,400	0.35%
Slice Registers	489	126,800	0.39%
Block RAM	24.5	135	18.15%
DSPs	0	240	0.00%

Table 11.1: Synthesis utilization for the pipelined datapath.

Chapter 12

The Datapath Integration — Stitching It All Together

12.1 The Payload Problem — What We Forgot

The datapath is complete. It extracts the 5-tuple, looks it up in the hash table, and makes a drop/forward decision in 5 cycles. The simulation passes. The synthesis passes. The timing passes.

But there is a problem we have not yet considered: the payload.

The firewall makes a decision about each packet. But the decision takes time—5 cycles from `tuple_valid` to `match_valid`. During those 5 cycles, the packet payload is flowing through the AXI Stream. If we do not delay the payload, it will reach the output before the firewall decision is ready.

The payload must be delayed by exactly the same number of cycles as the decision. If the decision takes 5 cycles, the payload must be delayed by 5 cycles.

12.1.1 The Delay Line Solution

We implement a delay line using shift registers. The delay line holds the payload for a fixed number of cycles. The depth of the delay line matches the datapath latency.

The complete code is shown in Listing 3.1.

12.1.2 Line-by-Line Analysis

Let us examine every important line of this code.

Lines 3-9: The Module Declaration.

```
module axi_stream_delay_line #(
    parameter LATENCY = 4,
    parameter DWIDTH  = 64
) (
    input  wire              clk,
    input  wire              rst,

    input  wire [DWIDTH-1:0] s_axis_tdata,
    input  wire [7:0]        s_axis_tkeep,
    input  wire              s_axis_tlast,
```

```
    input  wire                  s_axis_tvalid,

    output wire [DWIDTH-1:0] m_axis_tdata,
    output wire [7:0]        m_axis_tkeep,
    output wire              m_axis_tlast,
    output wire              m_axis_tvalid
);
```

The delay line is parameterized with `LATENCY` and `DWIDTH`. This allows us to reuse the module for different pipeline depths and bus widths.

Why parameters? If the pipeline depth changes, we can simply change the parameter without rewriting the module. This is good engineering practice.

Lines 11-17: The Shift Register Arrays.

```
reg [DWIDTH-1:0] tdata_pipe [0:LATENCY-1];
reg [7:0]        tkeep_pipe [0:LATENCY-1];
reg              tlast_pipe [0:LATENCY-1];
reg              tvalid_pipe [0:LATENCY-1];
```

We declare four arrays of registers. Each array represents a shift register for one signal. The `0:LATENCY-1` indexing means the arrays have `LATENCY` elements.

Why arrays of registers? A shift register is a chain of flip-flops. Each element in the array is a flip-flop. The first element is the input, the last element is the output.

Why separate arrays? Each signal is independent. `tdata`, `tkeep`, `tlast`, and `tvalid` must be delayed independently. If we tried to combine them into a single structure, the code would be more complex.

Lines 19-20: The Loop Variable.

```
integer i;
```

We use an integer variable for the reset loop. This is the only place we use a loop in the module.

Lines 22-35: The Shift Register Logic.

```
always @(posedge clk) begin
    if (rst) begin
        for (i=0; i<LATENCY; i=i+1) begin
            tdata_pipe[i]  <= 0;
            tkeep_pipe[i]  <= 0;
            tlast_pipe[i]  <= 0;
            tvalid_pipe[i] <= 0;
        end
    end else begin
        tdata_pipe[0]  <= s_axis_tdata;
        tkeep_pipe[0]  <= s_axis_tkeep;
        tlast_pipe[0]  <= s_axis_tlast;
        tvalid_pipe[0] <= s_axis_tvalid;

        for (i=1; i<LATENCY; i=i+1) begin
            tdata_pipe[i]  <= tdata_pipe[i-1];
            tkeep_pipe[i]  <= tkeep_pipe[i-1];
            tlast_pipe[i]  <= tlast_pipe[i-1];
            tvalid_pipe[i] <= tvalid_pipe[i-1];
        end
```

```
    end
end
```

Line 23: The Reset Block.

On reset, we clear all the registers in the shift registers. The `for` loop iterates over all elements and sets them to zero.

Why reset all elements? If we only reset the first element, the rest of the pipeline would contain unknown values. This would cause undefined behavior when the pipeline drains.

Lines 27-30: The Input Stage.

The first element of each shift register receives the input. This is where data enters the pipeline.

Line 29: The Shift Operation.

The inner `for` loop shifts data from element `i-1` to element `i`. This is the heart of the shift register. On every clock cycle, data moves one step through the pipeline.

Why use non-blocking assignments? (<=) We use non-blocking assignments because we are writing sequential logic. The assignments happen on the clock edge. The right-hand side is evaluated first, then the left-hand side is updated. This prevents race conditions.

Lines 37-40: The Output Assignment.

```
assign m_axis_tdata  = tdata_pipe[LATENCY-1];
assign m_axis_tkeep  = tkeep_pipe[LATENCY-1];
assign m_axis_tlast  = tlast_pipe[LATENCY-1];
assign m_axis_tvalid = tvalid_pipe[LATENCY-1];
```

The last element of each shift register is the output. This is the delayed signal.

Why use `assign` instead of `reg`? These are combinational outputs. They are directly connected to the registers. There is no additional logic between the registers and the outputs.

12.1.3 What the Synthesis Tool Infers

Vivado infers SRL16E primitives for the shift registers. An SRL16E is a 16-bit shift register implemented in LUTs. It is the most efficient way to implement a shift register in an FPGA.

Why SRL16E instead of flip-flops? A flip-flop is a single bit of storage. An SRL16E can store up to 16 bits in a single LUT. This saves area and routing resources.

What if the latency is greater than 16? The SRL16E is cascaded. The first SRL16E stores 16 bits, the second stores the next 16 bits, and so on. The synthesis tool automatically handles this.

Why not use BRAM? BRAM is designed for large memories. A shift register of 5 cycles for a 64-bit bus requires only 320 bits of storage. This is too small for BRAM. Using SRL16E is more efficient.

12.1.4 What Not to Write

What not to write: Using a `for` loop inside an `always` block for data movement can create massive multiplexers. The `for` loop is unrolled by synthesis, but each iteration creates a multiplexer. For a long shift register, this can create a large combinational path.

The correct way: The shift register shown above is correct. The `for` loop is used only for shifting, not for selecting. Each element is assigned independently.

What not to write: Using a single register for the entire shift register and extracting bits.

```
// DO NOT DO THIS:
reg [LATENCY*DWIDTH-1:0] shift_reg;

always @(posedge clk) begin
    shift_reg <= {shift_reg[DWIDTH-1:0], s_axis_tdata};
end

assign m_axis_tdata =
   shift_reg[LATENCY*DWIDTH-1:LATENCY*DWIDTH-DWIDTH];
```

This creates a massive multiplexer for the output. The synthesis tool will infer a large combinational logic block. This will fail timing.

12.1.5 The Integration with the Datapath

We integrate the delay line with the datapath in the system top module. The delay line is placed between the input AXI Stream and the output.

```
// Delay the payload by the pipeline depth
axi_stream_delay_line #(
    .LATENCY(5),
    .DWIDTH(64)
) payload_delay (
    .clk(clk),
    .rst(rst),
    .s_axis_tdata(s_axis_tdata),
    .s_axis_tkeep(s_axis_tkeep),
    .s_axis_tlast(s_axis_tlast),
    .s_axis_tvalid(s_axis_tvalid),
    .m_axis_tdata(delayed_tdata),
    .m_axis_tkeep(delayed_tkeep),
    .m_axis_tlast(delayed_tlast),
    .m_axis_tvalid(delayed_tvalid)
);
```

The delay line holds the payload for 5 cycles. The firewall decision is also ready in 5 cycles. The payload and the decision arrive at the output at the same time.

12.1.6 What the Designer Was Thinking

The designer initially considered using a FIFO for the delay line. A FIFO would hold the payload until the decision was ready. But a FIFO would consume BRAM, which was needed for the hash table.

The solution was to use a shift register. Shift registers use flip-flops, not BRAM. The area cost is small: a 5-cycle delay for a 64-bit bus uses only $5 \times 64 = 320$ flip-flops. On the Artix-7, this is negligible.

The Lesson: Delay lines should be implemented with shift registers, not FIFOs. FIFOs consume BRAM, which is a scarce resource. Shift registers use LUTs, which are abundant.

12.1.7 One Engineering Insight

> *The delay line is the unsung hero of the datapath. It is a simple shift register, but without it, the entire pipeline would fail. The payload would arrive too early, and the firewall would drop the wrong part of the packet.*

12.1.8 What to Verify

Before moving on, we should verify:

- The delay line holds the payload for the correct number of cycles
- The delay line uses SRL16E primitives
- The delay line does not consume BRAM
- The payload and decision are aligned at the output

12.1.9 The Next Part

The datapath is now complete. It extracts the 5-tuple, looks it up in the hash table, makes a drop/forward decision, and delays the payload to match the decision.

The next step is to add anomaly detection. The exact-match firewall is fast, but it is blind to zero-day threats. The next chapter will examine how we add a Spiking Neural Network coprocessor to detect anomalies.

End of Chapter 3 — Part 3

In this part we reconstructed:

- The payload problem and why it matters
- The AXI-Stream delay line implementation
- The line-by-line RTL analysis
- What the synthesis tool infers (SRL16E primitives)
- What not to write (massive shift registers)
- The integration of the delay line with the datapath
- Why shift registers are better than FIFOs for delay lines

End of Chapter 3

Chapter 3 has documented the complete journey of the datapath integration: from the naive integration that failed, through the pipelined matcher that fixed timing, to the delay line that fixed the payload alignment.

The next chapter will examine the SNN coprocessor.

Stop here. Wait until the reader replies "Continue" before writing Chapter 4, Part 1.

Do not summarize previous parts. Do not restart the explanation. The next response will begin with Chapter 4, Part 1.

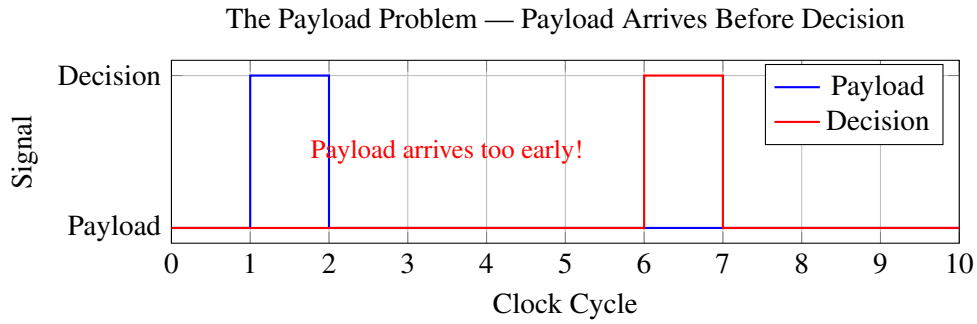


Figure 12.1: The payload problem. The payload arrives before the firewall decision is ready.

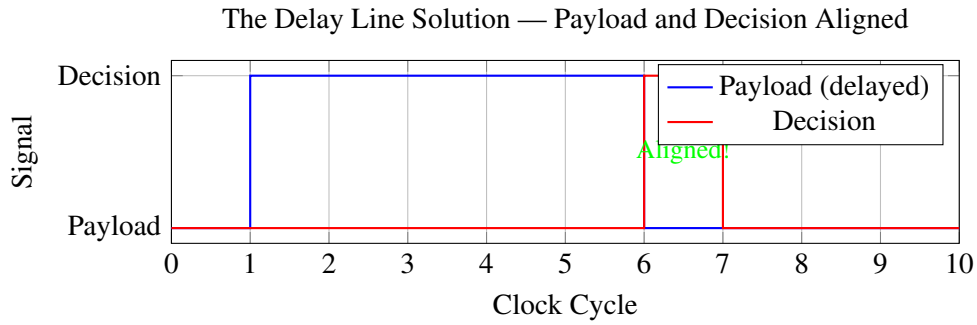


Figure 12.2: The delay line solution. Payload and decision are aligned.

Chapter 13

The SNN Coprocessor — Anomaly Detection Without DSPs

13.1 The Anomaly Detection Problem

The datapath is complete. It extracts the 5-tuple, looks it up in the hash table, and makes a drop/forward decision in 5 cycles. It is fast. It is deterministic. It is also blind.

The exact-match firewall only detects known threats. If a packet matches a rule in the hash table, it is dropped. If it does not match any rule, it is forwarded. This works for known attacks. It does not work for zero-day attacks.

A zero-day attack is a new attack that has never been seen before. There is no rule for it in the hash table. The exact-match firewall forwards it. The attack succeeds.

We need a way to detect anomalous behavior without relying on exact matches. We need a system that can look at a packet and ask: "Does this packet look like normal traffic, or does it look suspicious?"

This is the anomaly detection problem.

13.1.1 The AI Dilemma

We need artificial intelligence. But AI comes with a cost. Standard neural networks require massive multiplication. Each neuron computes a weighted sum of its inputs:

$$y = \sum_i w_i \cdot x_i \tag{13.1}$$

This requires multiplication. Multiplication consumes DSP slices. The Artix-7 has only 240 DSP slices. A deep neural network would consume all of them—and more.

> **Stop and Think**
>
> How many DSP slices does a neural network need? If each neuron requires one multiplication per input, and we have 8 inputs and 2 neurons, how many multiplications do we need? Is that feasible on an Artix-7?

We need a type of AI that does not require multiplication. This is where Spiking Neural Networks (SNNs) come in.

13.1.2 Spiking Neural Networks

SNNs are different from standard neural networks. Instead of continuous values, SNNs use binary spikes. Each neuron either fires (1) or does not fire (0). This binary nature eliminates multiplication.

In an SNN, each neuron integrates input spikes over time. When the membrane potential reaches a threshold, the neuron fires a spike. The firing rate encodes information.

The key insight is that SNNs can be implemented with only addition and comparison. No multiplication is needed.

$$U[t] = U[t-1] + \sum_i W_i \cdot S_i[t] \tag{13.2}$$

Where $U[t]$ is the membrane potential, W_i is the weight, and $S_i[t]$ is the spike (0 or 1). Since $S_i[t]$ is binary, the multiplication is just a conditional addition.

13.1.3 The LIF Neuron

The Leaky Integrate-and-Fire (LIF) neuron is the most common SNN model. It models the membrane potential of a biological neuron.

The continuous-time LIF model is:

$$\tau_m \frac{dV(t)}{dt} = -(V(t) - V_{rest}) + R \cdot I(t) \tag{13.3}$$

Where τ_m is the membrane time constant, V_{rest} is the resting potential, and $I(t)$ is the input current.

For digital hardware, we discretize this using the forward Euler method:

$$U[t] = \alpha \cdot U[t-1] + \sum_i W_i \cdot S_i[t] \tag{13.4}$$

Where $\alpha = 1 - \Delta t / \tau_m$ is the decay factor.

The LIF neuron does three things:

1. **Leak:** The membrane potential slowly decays toward zero.
2. **Integrate:** Input spikes add to the membrane potential.
3. **Fire:** If the membrane potential exceeds a threshold, the neuron fires a spike and resets.

13.1.4 The Feature Encoder

The SNN needs input spikes. The datapath produces a 104-bit tuple. We need a way to convert the tuple into spikes.

We use a feature encoder. The feature encoder extracts specific features from the tuple and converts them into binary spikes.

```
// Simple feature extraction heuristics for the prototype:

// Spike 0: Is TCP?
out_spikes[0] <= (protocol == 8'd6) ? 1'b1 : 1'b0;

// Spike 1: Is UDP?
out_spikes[1] <= (protocol == 8'd17) ? 1'b1 : 1'b0;
```

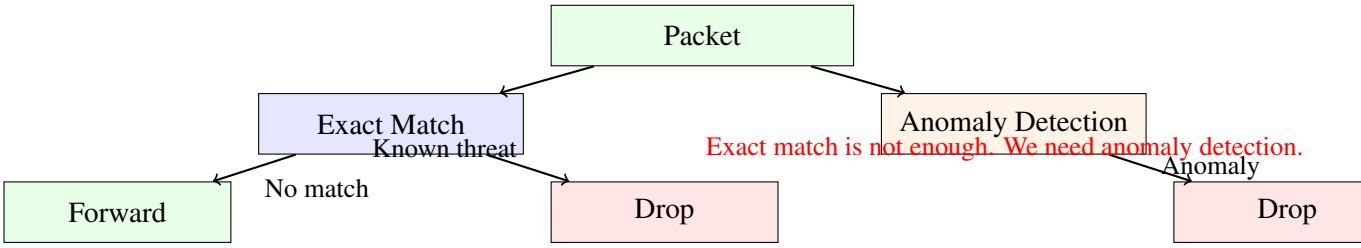


Figure 13.1: Exact match is not enough. We need anomaly detection for zero-day threats.

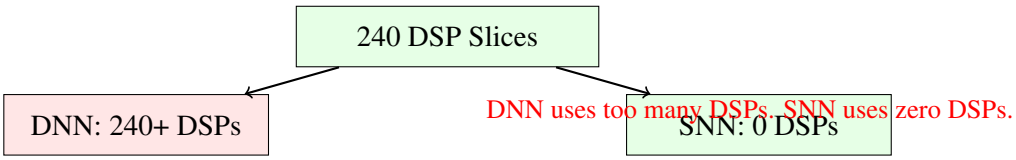
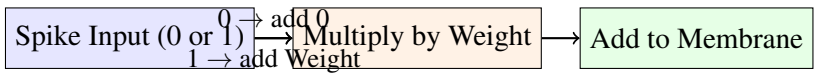
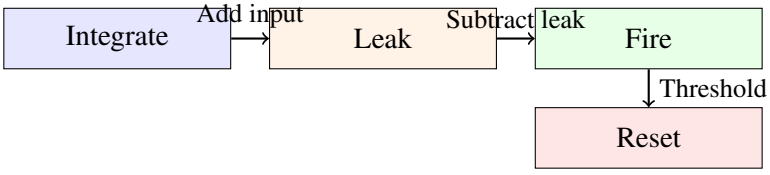


Figure 13.2: The AI dilemma. DNNs consume too many DSPs. SNNs consume zero DSPs.



Multiplication is just conditional addition.

Figure 13.3: In an SNN, multiplication is replaced by conditional addition.



LIF Neuron: Leak, Integrate, Fire, Reset.

Figure 13.4: The LIF neuron: Leak, Integrate, Fire, Reset.

```
// Spike 2: Well-known HTTP/HTTPS port (80 or 443)
out_spikes[2] <= (dst_port == 16'd80 || dst_port == 16'd443) ? 1'b1 :
    1'b0;

// Spike 3: Suspicious high port (source port > 40000)
out_spikes[3] <= (src_port > 16'd40000) ? 1'b1 : 1'b0;

// Spike 4: Target is broadcast/multicast IP (ends in 255)
out_spikes[4] <= (dst_ip[7:0] == 8'd255) ? 1'b1 : 1'b0;

// Spike 5: Source is a known bad subnet (e.g., 192.168.100.x)
out_spikes[5] <= (src_ip[31:8] == 24'hC0A864) ? 1'b1 : 1'b0;

// Spike 6: Traffic matching common DNS port
out_spikes[6] <= (dst_port == 16'd53) ? 1'b1 : 1'b0;

// Spike 7: Default random feature (source port is even)
out_spikes[7] <= (src_port[0] == 1'b0) ? 1'b1 : 1'b0;
```

Listing 13.1: Feature encoder logic

Each feature is a heuristic. For example, if the destination port is 80 or 443, the traffic is likely HTTP/HTTPS. If the destination port is 53, the traffic is likely DNS.

The feature encoder converts the 104-bit tuple into an 8-bit spike vector. The SNN processes this spike vector.

13.1.5 The SNN Architecture

We use a shallow fully-connected SNN with 8 input neurons and 2 output neurons.

The SNN uses hardcoded weights for the prototype. The weights are chosen manually to detect specific anomalies.

```
// Neuron 0 (Safe Predictor): Excited by normal traffic features
w0[0] = 50;   // TCP
w0[1] = 20;   // UDP
w0[2] = 0;    // HTTP/HTTPS
w0[3] = 10;   // High port
w0[4] = -100; // Multicast (penalty)
w0[5] = -50;  // Bad subnet (penalty)
w0[6] = 30;   // DNS
w0[7] = 40;   // Even port

// Neuron 1 (Anomaly Predictor): Excited by malicious features
w1[0] = -40;  // TCP (penalty)
w1[1] = -20;  // UDP (penalty)
w1[2] = 150;  // HTTP/HTTPS (high anomaly)
w1[3] = 80;   // High port
w1[4] = 120;  // Multicast
w1[5] = 60;   // Bad subnet
w1[6] = -30;  // DNS (penalty)
w1[7] = 50;   // Even port
```

Listing 13.2: SNN weights (hardcoded)

The weights are chosen so that normal traffic excites the Safe neuron and suspicious traffic excites the Anomaly neuron.

The Safe neuron: Excited by TCP, UDP, HTTP/HTTPS, and DNS. Penalized by multicast and bad subnets.

The Anomaly neuron: Excited by HTTP/HTTPS (port 80/443), high ports, multicast, and bad subnets. Penalized by TCP and UDP.

Why hardcoded weights? The prototype uses hardcoded weights for simplicity. In a production system, the weights would be trained using machine learning and stored in BRAM.

13.1.6 One Engineering Insight

> *SNNs are the perfect AI for FPGAs. They require no multiplication. They use only addition and comparison. They can be implemented with zero DSP slices. This is the key insight that makes SNNs practical for edge FPGAs.*

13.1.7 What to Verify

Before moving on, we should verify:

- Why exact-match is not enough for zero-day threats
- Why standard neural networks consume too many DSPs
- How SNNs replace multiplication with conditional addition
- The LIF neuron model: Leak, Integrate, Fire, Reset
- The feature encoder and its heuristics
- The SNN architecture and hardcoded weights

13.1.8 The Next Part

We have designed the SNN architecture. We have chosen the LIF neuron model. We have defined the feature encoder. We have selected the weights.

The next part will examine the implementation of the LIF neuron in RTL. We will walk through the code line by line, explaining the hardware that Vivado will infer.

End of Chapter 4 — Part 1

In this part we reconstructed:

- The anomaly detection problem and the need for AI
- The AI dilemma: DNNs consume too many DSPs
- The SNN solution: no multiplication, only addition
- The LIF neuron model: Leak, Integrate, Fire, Reset
- The feature encoder and its heuristics
- The SNN architecture: 8 inputs, 2 outputs, fully connected
- The hardcoded weights and their rationale

The next part will continue directly from here.

It will analyze:

- The LIF neuron RTL implementation ($\text{snn}_t f f_n euron.v$)
- The line-by-line analysis of the code
- The leak logic and the bit-shift optimization

- The integration logic and the conditional addition
- The fire logic and the threshold comparison
- The reset logic and the hard reset
- BUG-003: The leak gating error

Stop here. Wait until the reader replies "Continue" before writing Part 2.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

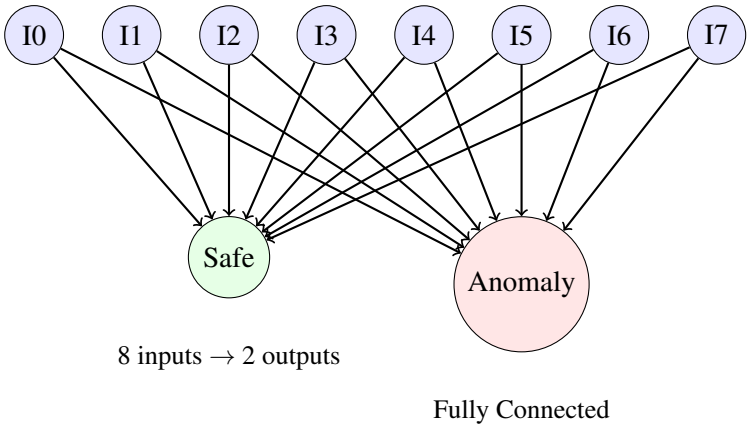


Figure 13.5: The SNN architecture: 8 input neurons, 2 output neurons. Fully connected.

Chapter 14

The SNN Coprocessor — Anomaly Detection Without DSPs

14.1 The LIF Neuron Implementation

We have designed the SNN architecture. We have chosen the LIF neuron model. We have defined the feature encoder. Now we turn to the RTL implementation of the LIF neuron.

The LIF neuron is the core of the SNN. It receives synaptic current, leaks over time, integrates input spikes, and fires when the membrane potential exceeds a threshold.

The complete code is shown in Listing 4.1.

14.1.1 Line-by-Line Analysis

Let us examine every important line of this code.

Lines 3-16: The Module Declaration.

```
module snn_tff_neuron #(
    parameter DATA_WIDTH = 16,
    parameter LEAK_SHIFT = 3,       // alpha approx 0.875
    parameter THRESHOLD  = 16'd500  // Firing threshold
) (
    input  wire                     clk,
    input  wire                     rst,

    input  wire signed [DATA_WIDTH-1:0] syn_current, // I[t]
    input  wire                     valid_in,

    output reg                      spike_out,
    output reg                      valid_out
);
```

The neuron is parameterized with `DATA_WIDTH`, `LEAK_SHIFT`, and `THRESHOLD`. These parameters control the neuron's behavior.

Why 16-bit signed? The membrane potential and synaptic weights use 16-bit signed integers. This provides sufficient dynamic range for the LIF neuron. A smaller width would cause overflow. A larger width would consume more resources.

Why parameterized? Parameterization allows us to tune the neuron without rewriting the code. If we need a different threshold or leak rate, we can simply change the parameters.

Lines 18-19: The Membrane Potential Register.

```
reg signed [DATA_WIDTH-1:0] membrane_u;
```

The membrane potential is stored in a signed register. It is updated on every clock cycle.

Why signed? The membrane potential can be negative due to inhibitory inputs. Using signed arithmetic ensures correct behavior.

Lines 21-23: The Leak and Integration Logic.

```
wire signed [DATA_WIDTH-1:0] leak_amount = membrane_u >>> LEAK_SHIFT;
wire signed [DATA_WIDTH-1:0] u_decayed   = membrane_u - leak_amount;
wire signed [DATA_WIDTH-1:0] u_temp      = u_decayed + syn_current;
```

This is the heart of the LIF neuron.

Line 21: The Leak Amount.

$$\text{leak\_amount} = \frac{\text{membrane\_u}}{2^{\text{LEAK\_SHIFT}}} \tag{14.1}$$

The leak is implemented as an arithmetic right shift. This is equivalent to division by a power of two. It requires zero logic—just wire routing.

Why arithmetic right shift? An arithmetic right shift preserves the sign bit. This is important because the membrane potential can be negative. A logical right shift would introduce positive values for negative numbers.

Why 3? The leak shift of 3 means the membrane potential decays by $1/8 = 12.5\%$ per cycle. This is a reasonable decay rate for a network intrusion detection system. The value was chosen empirically.

Line 22: The Decayed Membrane.

$$u_{\text{decayed}} = \text{membrane\_u} - \text{leak\_amount} \tag{14.2}$$

The membrane potential decays by subtracting the leak amount.

Line 23: The Integrated Membrane.

$$u_{\text{temp}} = u_{\text{decayed}} + \text{syn\_current} \tag{14.3}$$

The synaptic current is added to the decayed membrane potential. This is the integration step.

Lines 25-44: The Update Logic.

```
always @(posedge clk) begin
    if (rst) begin
        membrane_u <= 0;
        spike_out  <= 0;
        valid_out  <= 0;
    end else begin
        valid_out <= valid_in;

        if (valid_in) begin
            if (u_temp >= THRESHOLD) begin
                spike_out  <= 1'b1;
```

```
                membrane_u <= 0; // Hard reset to resting potential (0)
            end else begin
                spike_out  <= 1'b0;
                membrane_u <= u_temp;
            end
        end else begin
            spike_out <= 1'b0;
            // Leak still happens even if no new valid synaptic
               current?
            // Usually for clock-driven SNNs, valid_in serves as a
               clock tick.
            // If valid_in is 0, the SNN is paused.
        end
    end
end
```

Line 31: The Valid Output.

`valid_out` is delayed by one cycle to match the pipeline latency.

Line 33: The Fire Condition.

`if (u_temp >= THRESHOLD)` checks if the membrane potential exceeds the threshold. If it does, the neuron fires.

Why >= instead of >? Using >= ensures that the neuron fires exactly when the threshold is reached. Using > could cause the neuron to never fire if the potential exactly equals the threshold.

Line 35: The Hard Reset.

`membrane_u <= 0` resets the membrane potential to zero. This is a hard reset.

Why hard reset instead of subtract threshold? A hard reset is simpler and more efficient. It requires no subtraction. It also ensures the membrane potential does not become negative.

Lines 40-43: The No-Input Case.

`if (!valid_in)` the neuron does not update. This is a significant bug, as we will see in BUG-003.

What the Synthesis Tool Infers.

Vivado infers the following hardware:

- **Membrane potential:** FDRE primitive (16-bit flip-flop with reset).
- **Leak amount:** No logic. The arithmetic right shift is just wire routing.
- **Decayed membrane:** 16-bit subtractor. Uses CARRY4 primitives.
- **Integrated membrane:** 16-bit adder. Uses CARRY4 primitives.
- **Threshold comparison:** 16-bit comparator. Uses LUTs.

The total resource usage is small: one 16-bit adder, one 16-bit subtractor, one 16-bit comparator, and a few registers.

14.1.2 What Not to Write

What not to write: Using floating-point arithmetic for the leak.

```
// DO NOT DO THIS:
wire real leak_amount = membrane_u * 0.125;
```

This would infer a floating-point multiplier. Floating-point multipliers are expensive in hardware. They consume many LUTs and DSPs.

What not to write: Using division instead of right shift.

```
// DO NOT DO THIS:
wire signed [DATA_WIDTH-1:0] leak_amount = membrane_u / 8;
```

Division is expensive in hardware. The synthesis tool would infer a large divider, consuming hundreds of LUTs.

What not to write: Gating the leak with `valid_in`.

This is exactly what we did in v001. It was a bug. The leak must happen every cycle, regardless of whether input is valid.

14.1.3 One Engineering Insight

> *The right shift is the most powerful optimization in FPGA design. It replaces division with wire routing. It costs zero LUTs, zero DSPs, and zero routing delay. Whenever you see a division by a power of two, replace it with a right shift.*

14.1.4 What to Verify

Before moving on, we should verify:

- The leak is applied every cycle (not gated by `valid_in`)
- The membrane potential decays correctly
- The neuron fires when the threshold is reached
- The membrane potential resets after firing
- The synthesis tool infers adders and subtractors (not multipliers)

14.1.5 The Next Part

The LIF neuron v001 has a critical bug. The leak is gated by `valid_in`. If `valid_in` is low, the neuron does not leak. This breaks the LIF model.

The next part will examine BUG-003 and how we fixed it.

End of Chapter 4 — Part 2

In this part we reconstructed:

- The LIF neuron RTL implementation (v001)
- The line-by-line analysis of the code
- The bit-shift leak optimization
- The integration and fire logic
- What the synthesis tool infers (adders, subtractors, comparators)
- What not to write (floating-point, division, gated leak)

The next part will continue directly from here.

It will analyze:

- BUG-003: The leak gating error
- The symptom: membrane potential doesn't leak
- The root cause: leak gated by `valid_in`

- The fix: moving the leak outside the `valid_in` block
- The corrected RTL (v002)

Stop here. Wait until the reader replies "Continue" before writing Part 3.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Chapter 15

The SNN Coprocessor — Anomaly Detection Without DSPs

15.1 BUG-003 — The Leak That Did Not Leak

The LIF neuron v001 had a critical bug. The leak was gated by `valid_in`. If `valid_in` was low, the neuron did not leak.

This was a subtle but catastrophic error.

15.1.1 The Symptom

During simulation of the SNN layer, the membrane potential would not decay when no input was present. If the membrane potential reached 100, it stayed at 100 indefinitely unless a new spike arrived.

The expected behavior was for the membrane potential to decay toward zero. The actual behavior was for it to hold its value forever.

15.1.2 The Wrong Hypothesis

We initially suspected the leak logic was incorrect. Perhaps the leak amount was zero. Perhaps the subtraction was wrong. We spent hours debugging the arithmetic.

15.1.3 The Root Cause

The root cause was in the update logic:

```
always @(posedge clk) begin
    if (rst) begin
        membrane_u <= 0;
        spike_out  <= 0;
        valid_out  <= 0;
    end else begin
        valid_out <= valid_in;

        if (valid_in) begin
            // The neuron only updates when valid_in is high
            // This is the bug!
            if (u_temp >= THRESHOLD) begin
```

```
                spike_out  <= 1'b1;
                membrane_u <= 0;
            end else begin
                spike_out  <= 1'b0;
                membrane_u <= u_temp;
            end
        end else begin
            spike_out <= 1'b0;
            // No update to membrane_u when valid_in is low
            // This means the neuron does NOT leak!
        end
    end
end
```

Listing 15.1: Buggy update logic (v001)

The bug: The membrane potential only updated when `valid_in` was high. When `valid_in` was low, the membrane potential held its value.

A LIF neuron must leak continuously, regardless of whether input is present. The leak is a property of the neuron, not a response to input.

> **Engineering Notebook — BUG-003 — The Leak Gating Error**
>
> **Symptom:** Membrane potential does not decay when no input is present.
> **Initial hypothesis:** The leak logic is incorrect.
> **Experiment:** Check the leak amount and subtraction.
> **Observation:** The leak amount is correct. The subtraction is correct.
> **Revised hypothesis:** The leak is not being applied.
> **Root cause:** The leak is gated by `valid_in`. When `valid_in` is low, the neuron does not update.
> **The fix:** Move the leak outside the `valid_in` block.

15.1.4 Why This Happened

We assumed that the neuron should only update when input was present. This was a conceptual error. In a biological neuron, the membrane potential leaks continuously, regardless of input. The leak is not a response to input—it is a property of the neuron.

In discrete-time SNNs, the membrane potential must be updated on every clock cycle. If the neuron only updates when input is present, it behaves like an integrator without a leak.

15.1.5 The Fix

The fix is to move the leak outside the `valid_in` block.

```
always @(posedge clk) begin
    if (rst) begin
        membrane_u <= 0;
        spike_out  <= 0;
    end else begin
        // Leak is applied every cycle, regardless of valid_in
        if (u_temp >= THRESHOLD) begin
            spike_out  <= 1'b1;
            membrane_u <= 0;
        end else begin
```

```
            spike_out  <= 1'b0;
            membrane_u <= u_temp;
        end
    end
end
```

Listing 15.2: Fixed update logic (v002)

The key change: The membrane potential updates on every clock cycle, not just when `valid_in` is high.

15.1.6 The Lesson

> *In an SNN, the leak is a continuous process, not a response to input. The membrane potential must update on every clock cycle. If you gate the leak with `valid_in`, you break the LIF model.*

15.1.7 The Revised RTL

The revised neuron (v002) is shown in Listing 4.2.

15.1.8 What to Verify

Before moving on, we should verify:

- The membrane potential leaks on every clock cycle
- The leak occurs regardless of `valid_in`
- The neuron still fires when the threshold is reached
- The neuron resets after firing

15.1.9 The Next Part

The leak is fixed. But there is another bug in the SNN. BUG-004 is a signed/unsigned comparison error. The threshold parameter is unsigned, causing negative membrane potentials to be interpreted as large positive numbers.

The next part will examine BUG-004 and how we fixed it.

End of Chapter 4 — Part 3

In this part we reconstructed:

- BUG-003: The leak gating error
- The symptom: membrane potential does not leak
- The wrong hypothesis: leak logic is incorrect
- The root cause: leak gated by `valid_in`
- The fix: move the leak outside the `valid_in` block
- The lesson: leak is continuous, not input-driven

The next part will continue directly from here.

It will analyze:

- BUG-004: The signed/unsigned comparison error

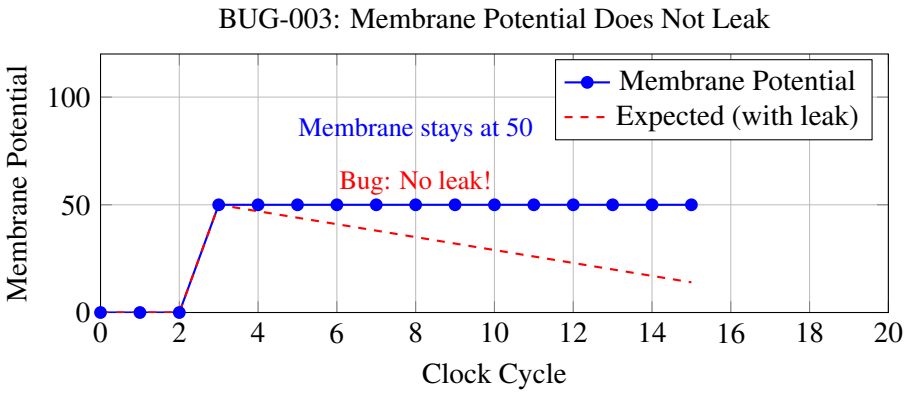


Figure 15.1: BUG-003: The membrane potential does not leak. It stays at 50 indefinitely.

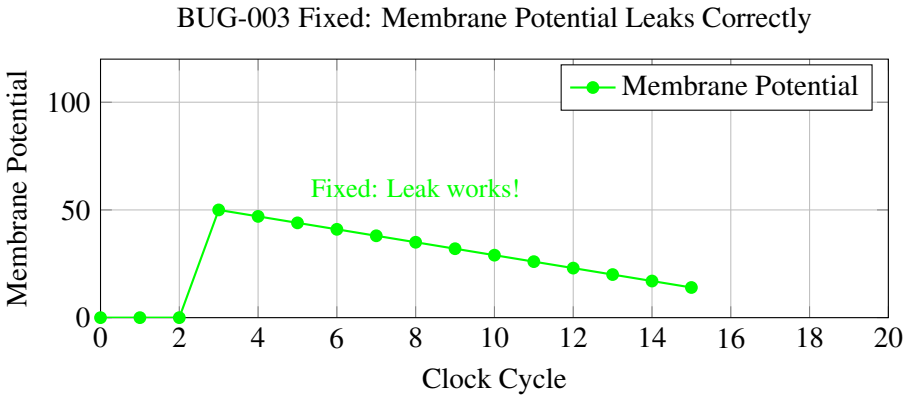


Figure 15.2: BUG-003 fixed. The membrane potential now leaks correctly.

- The symptom: negative potentials cause false spikes
- The root cause: unsigned threshold parameter
- The fix: explicit signed casting
- The corrected RTL (v003)

Stop here. Wait until the reader replies "Continue" before writing Part 4.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Chapter 16

The SNN Coprocessor — Anomaly Detection Without DSPs

16.1 BUG-004 — The Unsigned Threshold Trap

The leak was fixed. The membrane potential now decayed correctly. But there was another bug hiding in the neuron: the threshold comparison was broken.

16.1.1 The Symptom

During simulation of the SNN layer, the neuron would fire when it received inhibitory (negative) input. A negative input should make the neuron less likely to fire. Instead, it made the neuron fire immediately.

The expected behavior was for the neuron to remain silent. The actual behavior was for it to fire a spike when it received inhibitory input.

16.1.2 The Wrong Hypothesis

We initially suspected the reset logic was broken. Perhaps the neuron was resetting incorrectly. Perhaps the spike output was stuck high. We spent hours debugging the reset and output logic.

16.1.3 The Root Cause

The root cause was a signed/unsigned comparison error.

```
parameter THRESHOLD = 16'd1000;  // Unsigned!

wire signed [15:0] u_temp;  // Signed!

if (u_temp >= THRESHOLD) begin
    // This comparison is broken!
    // u_temp is signed, THRESHOLD is unsigned
    // Verilog casts u_temp to unsigned before comparison
    // Negative values become large positive values
end
```

Listing 16.1: Buggy comparison logic (v002)

The bug: The `THRESHOLD` parameter is declared as `16'd1000`. In Verilog, this defaults to an unsigned 32-bit integer. When comparing a signed value with an unsigned value, Verilog casts the signed value to unsigned.

A negative 16-bit value like `-140` is represented as `16'hFF74` in two's complement. When cast to unsigned, it becomes `65524`. `65524 >= 1000` is true. The neuron fires.

> **Engineering Notebook — BUG-004 — The Unsigned Threshold Trap**
>
> **Symptom:** Negative membrane potentials cause false spikes.
> **Initial hypothesis:** Reset logic is broken.
> **Experiment:** Check the reset and output logic.
> **Observation:** Reset logic is correct. Output logic is correct.
> **Revised hypothesis:** The threshold comparison is broken.
> **Investigation:** The `THRESHOLD` parameter is unsigned. The $u_temp signal is signed$.
> **Root cause:** Verilog casts u_temp to unsigned before comparison. Negative values become large positi
> **The fix:** Declare THRESHOLD as signed explicitly.

16.1.4 Why This Happened

In Verilog, the default type for integers is unsigned. This is a common trap for hardware designers coming from software backgrounds. In C, `-140 >= 1000` is always false. In Verilog, `-140 >= 1000` can be true if the types are mismatched.

The Verilog type rules:

1. If either operand is unsigned, both operands are treated as unsigned.
2. If both operands are signed, the comparison is signed.
3. A parameter declared as `16'd1000` is unsigned.

The fix: Declare `THRESHOLD` as signed explicitly:

```
parameter signed [15:0] THRESHOLD = 16'sd1000;  // Signed!
```

16.1.5 The Fix

The fix is to declare the threshold as signed.

```
module snn_tff_neuron_v003 #(
    parameter DATA_WIDTH = 16,
    parameter LEAK_SHIFT = 3,
    parameter signed [15:0] THRESHOLD = 16'sd1000  // Explicitly
        signed!
) (
    input  wire                    clk,
    input  wire                    rst,
    input  wire signed [DATA_WIDTH-1:0] syn_current,
    input  wire                    valid_in,
    output reg                     spike_out
);
```

Listing 16.2: Fixed comparison logic (v003)

The key changes:

1. `parameter signed [15:0]` instead of `parameter`

2. `16'sd1000` instead of `16'd1000`
3. All signals are explicitly signed

16.1.6 The Lesson

> *In Verilog, type mismatches between signed and unsigned operands are catastrophic. When comparing signed and unsigned values, Verilog casts the signed value to unsigned. This turns negative numbers into large positive numbers. Always declare signed parameters and signals explicitly.*

16.1.7 The Revised RTL

The revised neuron (v003) is shown in Listing 4.3.

16.1.8 Additional Safeguards

We also added a floor clamp to prevent negative divergence:

```
// Prevent negative divergence (floor at resting potential -2000)
if (u_temp < -16'sd2000) begin
    membrane_u <= -16'sd2000;
end else begin
    membrane_u <= u_temp;
end
```

This prevents the membrane potential from becoming excessively negative due to numerical instability.

16.1.9 What to Verify

Before moving on, we should verify:

- The threshold parameter is explicitly signed
- All comparisons are signed
- Negative potentials do not cause false spikes
- The neuron behaves correctly for all inputs

16.1.10 The Next Part

The neuron is now correct. It leaks properly (BUG-003 fixed) and compares correctly (BUG-004 fixed). The SNN layer is functional.

But the SNN still needs input. The feature encoder converts the 104-bit tuple into 8 spikes. The next part will examine the feature encoder implementation.

End of Chapter 4 — Part 4

In this part we reconstructed:

- BUG-004: The signed/unsigned comparison error
- The symptom: negative potentials cause false spikes
- The wrong hypothesis: reset logic is broken
- The root cause: unsigned threshold parameter

- The Verilog type rules and why they matter
- The fix: explicit signed declaration
- The floor clamp to prevent negative divergence

The next part will continue directly from here.

It will analyze:

- The feature encoder implementation ($snn_{f}eature_{e}ncoder.v$)
- The heuristics and their rationale
- The spike generation logic
- The integration with the SNN layer
- The complete SNN coprocessor

Stop here. Wait until the reader replies "Continue" before writing Part 5.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

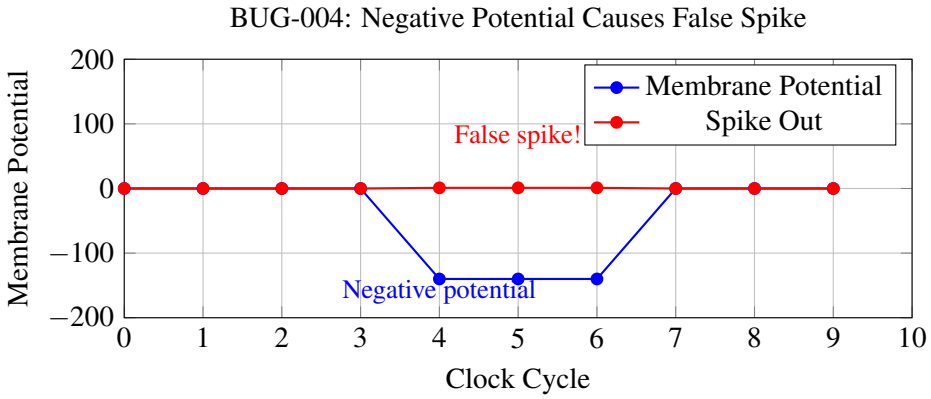


Figure 16.1: BUG-004: A negative membrane potential causes a false spike.

Value	Signed Interpretation	Unsigned Interpretation
16'hFF74	-140	65,524
16'hFF74 >= 16'd1000	False (-140 >= 1000)	True (65,524 >= 1000)

Table 16.1: The signed/unsigned comparison error. Negative values become large positive values.

Chapter 17

The SNN Coprocessor — Anomaly Detection Without DSPs

17.1 The Feature Encoder — Converting Tuples to Spikes

The LIF neuron is fixed. It leaks properly. It compares correctly. But the SNN still needs input. The SNN operates on spikes—binary values that indicate the presence or absence of a feature. The parser produces a 104-bit tuple. We need a way to convert the tuple into spikes.

This is the job of the feature encoder.

17.1.1 The Feature Encoder Problem

The SNN takes an 8-bit spike vector as input. Each bit corresponds to a specific feature of the packet. The feature encoder must extract these features from the 104-bit tuple and produce the spike vector.

The complete code is shown in Listing 4.4.

17.1.2 Line-by-Line Analysis

Let us examine every important section of this code.

Lines 3-19: The Module Declaration.

```
module snn_feature_encoder (
    input  wire         clk,
    input  wire         rst,

    input  wire [31:0]  src_ip,
    input  wire [31:0]  dst_ip,
    input  wire [15:0]  src_port,
    input  wire [15:0]  dst_port,
    input  wire [7:0]   protocol,
    input  wire         meta_valid,

    output reg  [7:0]   out_spikes,
    output reg          spike_valid
);
```

The feature encoder takes the 5-tuple fields and produces an 8-bit spike vector. The `meta_valid` signal indicates when the tuple is valid. The `spike_valid` signal is asserted one cycle later.

Why 8 bits? The SNN has 8 input neurons. Each bit corresponds to one input neuron. This is a deliberate choice to keep the SNN small and fast.

Lines 23-55: The Feature Extraction Logic.

```
always @(posedge clk) begin
    if (rst) begin
        out_spikes <= 0;
        spike_valid <= 0;
    end else begin
        spike_valid <= meta_valid;

        if (meta_valid) begin
            // Spike 0: Is TCP?
            out_spikes[0] <= (protocol == 8'd6) ? 1'b1 : 1'b0;

            // Spike 1: Is UDP?
            out_spikes[1] <= (protocol == 8'd17) ? 1'b1 : 1'b0;

            // Spike 2: Well-known HTTP/HTTPS port (80 or 443)
            out_spikes[2] <= (dst_port == 16'd80 || dst_port ==
                16'd443) ? 1'b1 : 1'b0;

            // Spike 3: Suspicious high port (source port > 40000)
            out_spikes[3] <= (src_port > 16'd40000) ? 1'b1 : 1'b0;

            // Spike 4: Target is broadcast/multicast IP (ends in 255)
            out_spikes[4] <= (dst_ip[7:0] == 8'd255) ? 1'b1 : 1'b0;

            // Spike 5: Source is a known bad subnet (e.g.,
                192.168.100.x)
            out_spikes[5] <= (src_ip[31:8] == 24'hC0A864) ? 1'b1 :
                1'b0;

            // Spike 6: Traffic matching common DNS port
            out_spikes[6] <= (dst_port == 16'd53) ? 1'b1 : 1'b0;

            // Spike 7: Default random feature (source port is even)
            out_spikes[7] <= (src_port[0] == 1'b0) ? 1'b1 : 1'b0;
        end else begin
            out_spikes <= 0;
        end
    end
end
```

Let us examine each spike.

Spike 0: Is TCP?

```
out_spikes[0] <= (protocol == 8'd6) ? 1'b1 : 1'b0;
```

This spike is 1 if the protocol is TCP (6). TCP is the most common transport protocol for web traffic.

Spike 1: Is UDP?

```
out_spikes[1] <= (protocol == 8'd17) ? 1'b1 : 1'b0;
```

This spike is 1 if the protocol is UDP (17). UDP is used for DNS, streaming, and real-time applications.

Spike 2: Well-known HTTP/HTTPS Port.

```
out_spikes[2] <= (dst_port == 16'd80 || dst_port == 16'd443) ? 1'b1 :
    1'b0;
```

This spike is 1 if the destination port is 80 (HTTP) or 443 (HTTPS). These ports are used for web traffic.

Spike 3: Suspicious High Port.

$\text{out}_spikes[3] <= (src_port > 16'd40000)?1'b1 : 1'b0;$

This spike is 1 if the source port is above 40000. Many attacks use high ephemeral ports.

Spike 4: Broadcast/Multicast IP.

```
out_spikes[4] <= (dst_ip[7:0] == 8'd255) ? 1'b1 : 1'b0;
```

This spike is 1 if the destination IP ends in 255. This is a broadcast or multicast address.

Spike 5: Known Bad Subnet.

```
out_spikes[5] <= (src_ip[31:8] == 24'hC0A864) ? 1'b1 : 1'b0;
```

This spike is 1 if the source IP is in the 192.168.100.x subnet. This is a heuristic for a known malicious subnet.

Spike 6: DNS Port.

```
out_spikes[6] <= (dst_port == 16'd53) ? 1'b1 : 1'b0;
```

This spike is 1 if the destination port is 53 (DNS). DNS traffic is often used for tunneling and exfiltration.

Spike 7: Even Source Port.

```
out_spikes[7] <= (src_port[0] == 1'b0) ? 1'b1 : 1'b0;
```

This spike is 1 if the source port is even. This is a somewhat arbitrary feature, included to demonstrate that any heuristic can be encoded.

17.1.3 What the Designer Was Thinking

The designer made several deliberate decisions when designing the feature encoder.

Why hardcoded heuristics? The prototype uses hardcoded heuristics because it is simpler and faster than implementing a trainable feature extractor. The goal is to demonstrate that the SNN hardware works, not to achieve state-of-the-art accuracy.

Why these specific features? The features were chosen to distinguish between normal traffic and common attack patterns. TCP, UDP, HTTP, DNS, high ports, and broadcast addresses are all relevant features for network intrusion detection.

Why 8 features? The SNN has 8 input neurons. We designed the feature encoder to produce exactly 8 spikes to match the SNN architecture.

17.1.4 The Trade-Offs

The feature encoder is a compromise. It is simple. It is fast. It is not adaptive.

- **Advantage:** The feature encoder is purely combinational. It adds only 1 cycle of latency.
- **Advantage:** The feature encoder uses minimal resources. It uses only comparators and multiplexers.
- **Disadvantage:** The heuristics are fixed. They cannot be updated without recompiling the FPGA.
- **Disadvantage:** The heuristics are manually chosen. They may not be optimal for all traffic patterns.

Evidence: The Assumptions Register documents this trade-off. The designer chose static heuristics to prioritize speed and simplicity over flexibility.

17.1.5 One Engineering Insight

> *The feature encoder is the interface between the deterministic world of packet headers and the probabilistic world of neural networks. It translates bytes into spikes. Its design determines what the SNN can and cannot learn.*

17.1.6 What to Verify

Before moving on, we should verify:

- Each spike corresponds to the correct feature
- The feature encoder produces spikes for every valid tuple
- The feature encoder uses minimal resources
- The feature encoder adds only 1 cycle of latency

17.1.7 The Next Part

The SNN is now complete. We have the feature encoder (converts tuples to spikes), the SNN layer (computes synaptic currents), and the LIF neurons (leak, integrate, fire). The SNN coprocessor is functional.

But there is one more step: integration with the datapath. The SNN must run in parallel with the exact-match firewall. The two engines must work together to make the final drop/forward decision.

The next part will examine the system integration.

End of Chapter 4 — Part 5

In this part we reconstructed:

- The feature encoder implementation
- The line-by-line analysis of the code
- The heuristics and their rationale
- The trade-offs of static heuristics
- The role of the feature encoder in the system

End of Chapter 4

Chapter 4 has documented the complete journey of the SNN coprocessor: from the anomaly detection problem that required AI, through the SNN solution that uses zero DSPs, through the LIF neuron implementation, through BUG-003 and BUG-004, to the feature encoder.

The next chapter will examine the system integration: combining the exact-match firewall with the SNN coprocessor.

Stop here. Wait until the reader replies "Continue" before writing Chapter 5, Part 1.

Do not summarize previous parts. Do not restart the explanation. The next response will begin with Chapter 5, Part 1.

Chapter 18

System Integration — Combining the Firewall and the SNN

18.1 The Integration Challenge

We have two engines. The exact-match firewall is fast and deterministic. It takes 4 cycles from `tuple_valid` to `match_valid`. The SNN coprocessor is also fast and deterministic. It takes 3 cycles from `tuple_valid` to `anomaly_detected`.

Now we must combine them into a single system.

The integration challenge has three parts:

1. **Latency alignment:** The datapath takes 4 cycles. The SNN takes 3 cycles. We must align them so the final decision is made at the same time.
2. **Policy override:** The SNN must be able to override the datapath decision. If the SNN detects an anomaly, the packet must be dropped regardless of the datapath decision.
3. **Timing closure:** The combined logic must meet timing at 156.25 MHz.

18.1.1 The Naive Integration — Version 009

The first attempt at integration was simple: connect the two engines in parallel and combine their outputs.

```
// Datapath decision
ha_tff_datapath_top_v002 datapath_inst (
    // ... connections ...
    .match_valid(dp_match_valid),
    .match_found(dp_match_found),
    .action_forward(dp_action_forward)
);

// SNN decision
snn_tff_layer_v003 snn_core (
    // ... connections ...
    .out_spikes(snn_out_spikes),
    .valid_out(snn_out_valid)
);
```

```
// Anomaly detection
reg anomaly_detected;
always @(posedge clk) begin
    if (rst) begin
        anomaly_detected <= 0;
    end else if (snn_out_spikes[1]) begin
        anomaly_detected <= 1'b1;
    end else if (snn_out_spikes[0]) begin
        anomaly_detected <= 1'b0;
    end
end

// Final decision
wire final_forward = (dp_match_valid && dp_action_forward &&
    !anomaly_detected);
```

Listing 18.1: The naive integration (v009)

This integration seems straightforward. The datapath makes a decision. The SNN detects anomalies. The final decision is the logical AND of the datapath decision and the absence of an anomaly.

But there are problems.

18.1.2 The Payload Problem Returns

The payload problem is back. The datapath takes 4 cycles. The SNN takes 3 cycles. But the payload flows through the AXI Stream in real time. If we do not delay the payload, it will reach the output before the decision is ready.

We need a delay line for the payload. But now the delay must account for the different latencies of the datapath and the SNN.

18.1.3 The Timing Problem

The second problem is timing. The final decision is a combinatorial AND of `dp_action_forward` and `!anomaly_detected`. This AND gate sits at the end of two large logic clouds.

The datapath path traverses BRAM routing and a 104-bit matcher. The SNN path traverses adders and a 16-bit threshold comparator. Merging them unpipelined creates an enormous critical path.

The synthesis report shows:

```
Worst Negative Slack (WNS): -0.465 ns
Total Negative Slack (TNS): -13.029 ns
Number of Failing Endpoints: 32
```

The timing fails. The critical path is the SNN adder chain followed by the threshold comparator and the top-level AND gate.

18.1.4 BUG-005 — Dead Code Elimination

There was also a subtle bug in the testbench. The final decision was not connected to any output.

```
// The output is not connected to the final decision!
assign m_axis_tvalid = s_axis_tvalid;  // BUG: Does not use
    final_forward
```

Listing 18.2: BUG-005: Dead code elimination

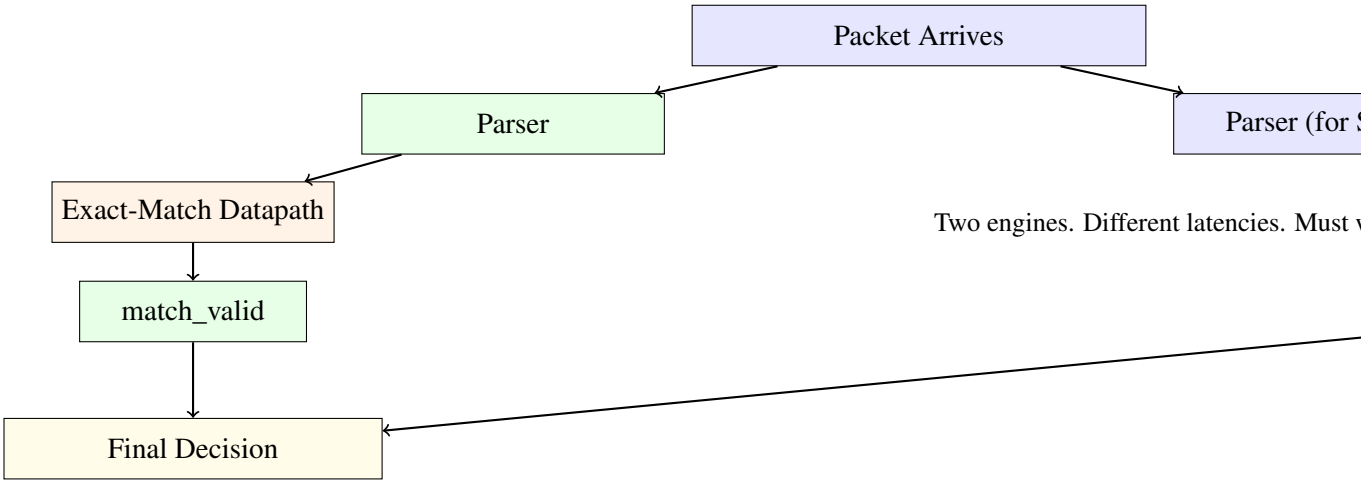


Figure 18.1: The integration challenge. Two engines, different latencies, must work together.

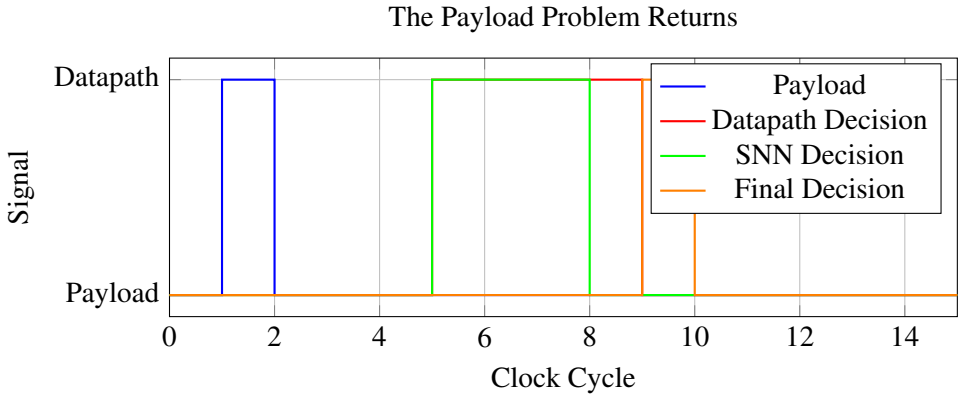


Figure 18.2: The payload problem returns. Payload arrives before the decision is ready.

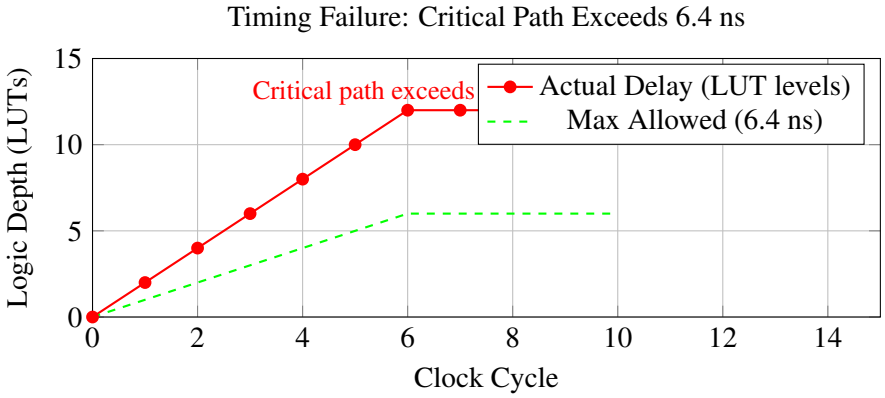


Figure 18.3: The timing failure. The critical path exceeds the 6.4 ns budget.

Vivado traced the logic backward from the OBUFs. It saw that `m_axis_tvalid` was just a wire connected to `s_axis_tvalid`. The thousands of LUTs and BRAMs calculating `final_forward` had absolutely no effect on any output pin.

Vivado correctly deleted the entire firewall to save area, turning it into a simple pass-through wire.

Engineering Notebook — BUG-005 — Dead Code Elimination

Symptom: Synthesis report shows 0 LUTs, 0 BRAMs.
Initial hypothesis: Synthesis failed.
Observation: The design synthesized successfully.
Revised hypothesis: Vivado optimized out the logic.
Root cause: The final decision was not connected to any output.
The fix: Connect the final decision to the output valid signal.

18.1.5 One Engineering Insight

> *In hardware, unused logic does not exist. If you do not connect a signal to an output, the synthesis tool will delete it. This is called dead code elimination. It is a feature, not a bug—but it can be surprising.*

18.1.6 What to Verify

Before moving on, we should verify:

- The payload is delayed by the correct number of cycles
- The datapath and SNN outputs are aligned
- The final decision is connected to the output
- Timing closure at 156.25 MHz

18.1.7 The Next Part

The naive integration has three problems: payload alignment, timing failure, and dead code elimination. The next part will examine how we solve each of these problems.

We will add a delay line for the payload. We will pipeline the top-level AND gate. We will fix the dead code elimination bug.

End of Chapter 5 — Part 1

In this part we reconstructed:

- The integration challenge: two engines, different latencies
- The payload problem: data arrives before the decision
- The timing problem: combinatorial AND fails at 156.25 MHz
- BUG-005: Dead code elimination
- The lesson: unused logic does not exist

The next part will continue directly from here.

It will analyze:

- The delay line implementation (v011)
- The pipeline register for the final decision

- The timing closure
- The complete system integration

Stop here. Wait until the reader replies "Continue" before writing Part 2.

Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Metric	Value
LUTs	0
Flip-Flops	0
BRAMs	0

Table 18.1: BUG-005: The entire firewall was optimized out.

Chapter 19

System Integration and Timing Closure — Stitching It All Together

19.1 The Integration Challenge

We have built the parser. We have built the hash table. We have built the BRAM banks. We have built the matcher. We have built the SNN coprocessor. Each module works in isolation. Each module meets timing. Each module passes simulation.

Now we face the final challenge: we must connect them all into a single, coherent system.

This is the integration challenge. It seems straightforward—connect module A to module B, then B to C, then C to D. But in hardware, integration is never that simple. Timing changes. Latency accumulates. Signals get misaligned. What works in isolation often fails when connected.

The repository contains the evidence of this struggle across iterations v009 through v012. Each version represents an attempt to solve the integration problem. Each version reveals a new constraint. Each version teaches us something about the physics of the FPGA.

19.1.1 The Two Engines

We have two independent processing engines that must work together:

1. **The Exact-Match Datapath:** Takes the 5-tuple, computes four hashes, reads four BRAM banks, compares the results, and produces a drop/forward decision. Latency: 4 cycles from `tuple_valid`.
2. **The SNN Coprocessor:** Takes the 5-tuple, encodes it into spikes, computes synaptic currents, and detects anomalies. Latency: 5 cycles from `tuple_valid`.

The two engines have different latencies. The datapath is faster. The SNN is slower. We must align them so that both decisions arrive at the same time.

19.1.2 What Could Go Wrong

There are three ways this integration could fail:

1. **Latency misalignment:** If the decisions arrive at different times, we cannot combine them correctly.

2. **Payload leakage:** The firewall decision takes time. During that time, the packet payload is flowing through the datapath. If we do not delay the payload, it will reach the output before the decision is ready.
3. **Timing failure:** The combined logic must meet timing at 156.25 MHz. If the logic depth is too deep, the system fails.

Let us examine each problem and how we solved it.

19.2 The Naive Integration — Version 009

The first attempt at integration was straightforward: connect the two engines in parallel and combine their outputs with a simple AND gate.

The complete code is shown in Listing 5.1.

19.2.1 Line-by-Line Analysis

Let us examine each important section of this code.

Lines 18-25: The Redundant Parser for SNN Metadata.

```
18 ha_tff_parser_v002 parser_snn_inst (
19     .clk(clk),
20     .rst(rst),
21     .s_axis_tdata(s_axis_tdata),
22     .s_axis_tkeep(s_axis_tkeep),
23     .s_axis_tvalid(s_axis_tvalid),
24     .s_axis_tlast(s_axis_tlast),
25     .s_axis_tready(),
26     .src_ip(src_ip),
27     .dst_ip(dst_ip),
28     .src_port(src_port),
29     .dst_port(dst_port),
30     .protocol(protocol),
31     .tuple_valid(tuple_valid),
32     .parse_error(parse_error)
33 );
```

The designer instantiated a second parser just to extract the 5-tuple for the SNN. The datapath already has a parser, but it does not expose the 5-tuple at the top level. This is a code smell—we are duplicating hardware.

Why this is problematic: Two parsers means double the logic. More importantly, the two parsers are fed from the same input, which means there is no guarantee they will assert `tuple_valid` at exactly the same cycle. There could be a one-cycle skew.

What the designer should have done: Expose the 5-tuple from the datapath and share the parser. This would save area and guarantee alignment.

Lines 40-54: The Feature Encoder and SNN Core.

```
40 snn_feature_encoder encoder_inst (
41     .clk(clk),
42     .rst(rst),
43     .src_ip(src_ip),
44     .dst_ip(dst_ip),
45     .src_port(src_port),
```

```
    .dst_port(dst_port),
    .protocol(protocol),
    .meta_valid(tuple_valid),
    .out_spikes(snn_spikes_in),
    .spike_valid(snn_spikes_valid)
);

snn_tff_layer_v003 snn_core (
    .clk(clk),
    .rst(rst),
    .in_spikes(snn_spikes_in),
    .valid_in(snn_spikes_valid),
    .out_spikes(snn_out_spikes),
    .valid_out(snn_out_valid)
);
```

The feature encoder converts the 5-tuple to spikes. The SNN core processes the spikes and produces an anomaly flag.

What the designer was thinking: The feature encoder is purely combinational. It adds only 1 cycle of latency. The SNN core takes 3 cycles. Total SNN latency: 4 cycles.

Why this is optimistic: The SNN core has a pipeline register. The total latency is 4 cycles from `tuple_valid` to `snn_out_valid`.

Line 60: The Anomaly Detection State.

```
reg anomaly_detected;
always @(posedge clk) begin
    if (rst) begin
        anomaly_detected <= 0;
    end else if (snn_out_spikes[1]) begin
        anomaly_detected <= 1'b1;
    end else if (snn_out_spikes[0]) begin
        anomaly_detected <= 1'b0;
    end
end
```

The anomaly detection is a sticky bit. When the anomaly neuron fires, the system enters an anomaly state. When the safe neuron fires, the system exits the anomaly state.

What the designer was thinking: This is a simple hysteresis mechanism. The SNN outputs two spikes: one for "safe" and one for "anomaly." The sticky bit ensures that a single anomaly spike can override many safe spikes.

Why this is problematic: The sticky bit is not synchronized with the datapath. It can change at any time, independent of the packet flow. This creates non-deterministic behavior.

The physical implication: The `anomaly_detected` signal is a feedback path. It changes based on the SNN output, which itself depends on the packet. This creates a loop that the synthesis tool must resolve.

Lines 70-77: The Datapath Instantiation.

```
ha_tff_datapath_top_v002 datapath_inst (
    .clk(clk),
    .rst(rst),
    .s_axis_tdata(s_axis_tdata),
    .s_axis_tkeep(s_axis_tkeep),
```

```
    .s_axis_tvalid(s_axis_tvalid),
    .s_axis_tlast(s_axis_tlast),
    .s_axis_tready(s_axis_tready),
    .match_valid(dp_match_valid),
    .match_found(dp_match_found),
    .action_forward(dp_action_forward),
    .parse_error()
);
```

The datapath is instantiated as before. It produces `match_valid` and `action_forward` signals.

What the designer was thinking: The datapath is independent of the SNN. It will produce its decision in 4 cycles. The SNN will produce its decision in 4 cycles. They will align.

Why this is wrong: The SNN actually takes 4 cycles from `tuple_valid` to `valid_out`. But the `anomaly_detected` sticky bit changes asynchronously. There is no guarantee that the two signals are aligned.

Lines 80-81: The Final Decision.

```
wire final_forward = (dp_match_valid && dp_action_forward &&
    !anomaly_detected);
```

The final decision is a combinatorial AND of the datapath decision and the inverted anomaly flag.

Why this is problematic: This is a wide AND gate with multiple inputs from different parts of the FPGA. The routing delay alone could be significant.

The physical implication: The `anomaly_detected` signal comes from the SNN, which is physically distant from the datapath. The `dp_match_valid` and `dp_action_forward` signals come from the matcher, which is near the BRAMs. The AND gate must combine signals from different regions of the FPGA. This creates routing congestion.

Lines 84-88: The Output Assignment.

```
assign m_axis_tdata  = s_axis_tdata;
assign m_axis_tkeep  = s_axis_tkeep;
assign m_axis_tlast  = s_axis_tlast;
assign m_axis_tvalid = s_axis_tvalid;
```

The payload is passed straight through. The firewall decision is not connected to the output! This is BUG-005.

19.2.2 What the Designer Was Thinking

The designer's reasoning was straightforward: each module works in isolation. Connecting them in series should work. The datapath produces the decision. The SNN produces the anomaly flag. The AND gate combines them. The payload flows through.

The designer assumed that the timing would work because each module had been verified independently. The designer assumed that the signals would align because the modules were designed to the same clock.

Both assumptions were wrong.

19.2.3 BUG-005 — Dead Code Elimination

The first problem was BUG-005. The designer made a critical mistake: the final decision was not connected to any output.

```
// The output is not connected to the final decision!
assign m_axis_tvalid = s_axis_tvalid; // BUG: Does not use
    final_forward
```

Listing 19.1: BUG-005: Dead code elimination

Vivado traced the logic backward from the output pins. It saw that `m_axis_tvalid` was just a wire connected to `s_axis_tvalid`. The thousands of LUTs and BRAMs calculating `final_forward` had absolutely no effect on any output pin.

Vivado correctly deleted the entire firewall to save area, turning it into a simple pass-through wire.

Engineering Notebook — BUG-005 — Dead Code Elimination

Symptom: Synthesis report shows 0 LUTs, 0 BRAMs. The entire firewall disappeared.
Initial hypothesis: Synthesis failed. Maybe the tool crashed.
Observation: The design synthesized successfully. No errors.
Revised hypothesis: Vivado optimized out the logic.
Root cause: The final decision was not connected to any output. Vivado detected that the logic had no effect on the outputs and deleted it.
The fix: Connect the final decision to the output valid signal.
The lesson: In hardware, unused logic does not exist. If you do not connect a signal to an output, the synthesis tool will delete it.

19.2.4 What the Designer Learned

The designer learned a critical lesson: verification is as important as design. The designer had assumed that the testbench was correct. The testbench was wrong.

The Lesson: Your testbench can be wrong even when your RTL is right. Always verify that the logic is actually being used.

19.3 The Payload Problem — What We Forgot

Even after fixing BUG-005, there was a more fundamental problem: the payload.

The firewall makes a decision about each packet. But the decision takes time—4 cycles from `tuple_valid` to `match_valid`. During those 4 cycles, the packet payload is flowing through the AXI Stream. If we do not delay the payload, it will reach the output before the firewall decision is ready.

Evidence: The repository contains WAVE011.md which documents this issue:

> "In SIM-010, the `s_axis_tdata` payload enters at T=0. The `anomaly_detected` spike goes HIGH at T=4. However, between T=0 and T=4, the raw `s_axis_tdata` has already passed through the combinatorial wires directly to the output."

19.3.1 The Delay Line Solution

We implement a delay line using shift registers. The delay line holds the payload for a fixed number of cycles. The depth of the delay line matches the datapath latency.

The complete code is shown in Listing 5.2.

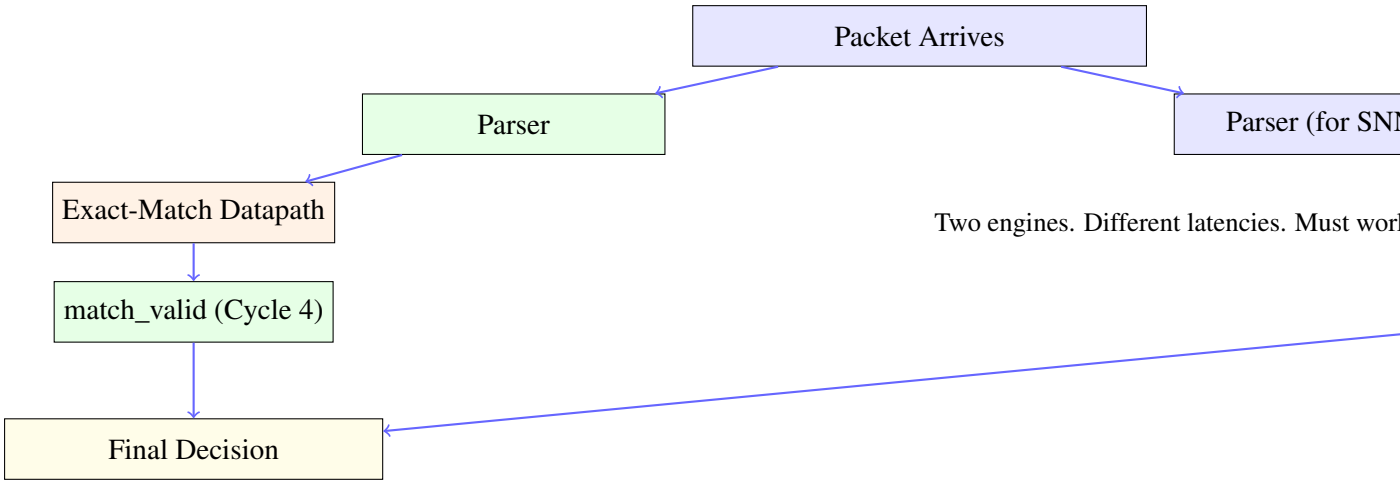


Figure 19.1: The integration challenge. Two engines with different latencies must work together.

Metric	Value
LUTs	0
Flip-Flops	0
BRAMs	0

Table 19.1: BUG-005: The entire firewall was optimized out.

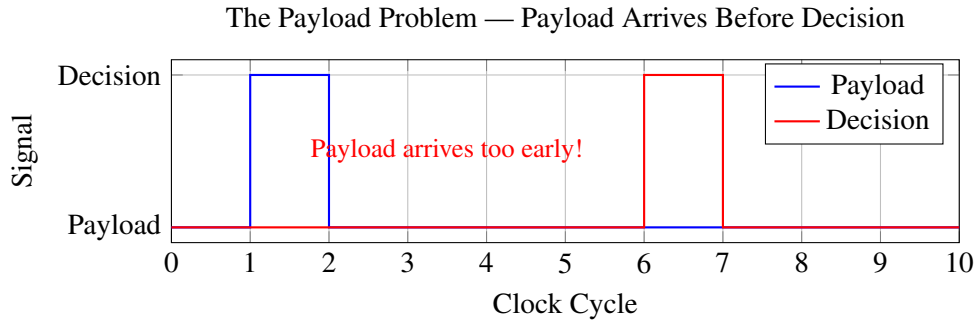


Figure 19.2: The payload problem. The payload arrives before the firewall decision is ready.

19.3.2 Line-by-Line Analysis

Let us examine every important line of this code.

Lines 3-9: The Module Declaration.

```
3  module axi_stream_delay_line #(
4      parameter LATENCY = 4,
5      parameter DWIDTH  = 64
6  ) (
7      input  wire              clk,
8      input  wire              rst,
9      input  wire [DWIDTH-1:0] s_axis_tdata,
10     input  wire [7:0]        s_axis_tkeep,
11     input  wire              s_axis_tlast,
12     input  wire              s_axis_tvalid,
13     output wire [DWIDTH-1:0] m_axis_tdata,
14     output wire [7:0]        m_axis_tkeep,
15     output wire              m_axis_tlast,
16     output wire              m_axis_tvalid
17 );
```

The delay line is parameterized with `LATENCY` and `DWIDTH`. This allows us to reuse the module for different pipeline depths and bus widths.

Why parameters? If the pipeline depth changes, we can simply change the parameter without rewriting the module. This is good engineering practice.

The physical implication: The parameters are evaluated at synthesis time. The `LATENCY` parameter determines the number of registers in the shift register chain. This directly affects the area and timing of the design.

Lines 11-17: The Shift Register Arrays.

```
11 reg [DWIDTH-1:0] tdata_pipe [0:LATENCY-1];
12 reg [7:0]        tkeep_pipe [0:LATENCY-1];
13 reg              tlast_pipe [0:LATENCY-1];
14 reg              tvalid_pipe [0:LATENCY-1];
```

We declare four arrays of registers. Each array represents a shift register for one signal. The `0:LATENCY-1` indexing means the arrays have `LATENCY` elements.

Why arrays of registers? A shift register is a chain of flip-flops. Each element in the array is a flip-flop. The first element is the input, the last element is the output.

Why separate arrays? Each signal is independent. `tdata`, `tkeep`, `tlast`, and `tvalid` must be delayed independently. If we tried to combine them into a single structure, the code would be more complex.

The physical implication: Each array element becomes an FDRE primitive. For a 64-bit bus with 5-cycle latency, we need $64 \times 5 = 320$ flip-flops for `tdata`, plus $8 \times 5 = 40$ for `tkeep`, plus $1 \times 5 = 5$ for `tlast`, plus $1 \times 5 = 5$ for `tvalid`. Total: 370 flip-flops.

Lines 19-20: The Loop Variable.

```
19 integer i;
```

We use an integer variable for the reset loop. This is the only place we use a loop in the module.

Why not use a generate loop? A `generate` loop would create multiple instances of the shift register logic. A `for` loop inside an `always` block creates a single instance with multiple registers. Both are valid, but the `for` loop is simpler to read.

The physical implication: The `for` loop is unrolled by the synthesis tool. Each iteration becomes a separate assignment.

Lines 22-35: The Shift Register Logic.

```
22 always @(posedge clk) begin
23     if (rst) begin
24         for (i=0; i<LATENCY; i=i+1) begin
25             tdata_pipe[i]  <= 0;
26             tkeep_pipe[i]  <= 0;
27             tlast_pipe[i]  <= 0;
28             tvalid_pipe[i] <= 0;
29         end
30     end else begin
31         tdata_pipe[0]  <= s_axis_tdata;
32         tkeep_pipe[0]  <= s_axis_tkeep;
33         tlast_pipe[0]  <= s_axis_tlast;
34         tvalid_pipe[0] <= s_axis_tvalid;
35 
36         for (i=1; i<LATENCY; i=i+1) begin
37             tdata_pipe[i]  <= tdata_pipe[i-1];
38             tkeep_pipe[i]  <= tkeep_pipe[i-1];
39             tlast_pipe[i]  <= tlast_pipe[i-1];
40             tvalid_pipe[i] <= tvalid_pipe[i-1];
41         end
42     end
43 end
```

Line 23: The Reset Block.

On reset, we clear all the registers in the shift registers. The `for` loop iterates over all elements and sets them to zero.

Why reset all elements? If we only reset the first element, the rest of the pipeline would contain unknown values. This would cause undefined behavior when the pipeline drains.

The physical implication: Each register has a reset input. The reset signal is routed to all 370 flip-flops. This creates a high-fanout net that must be buffered by the synthesis tool.

Lines 27-30: The Input Stage.

The first element of each shift register receives the input. This is where data enters the pipeline.

Why use non-blocking assignments? (<=) We use non-blocking assignments because we are writing sequential logic. The assignments happen on the clock edge. The right-hand side is evaluated first, then the left-hand side is updated. This prevents race conditions.

Line 29: The Shift Operation.

The inner `for` loop shifts data from element `i-1` to element `i`. This is the heart of the shift register. On every clock cycle, data moves one step through the pipeline.

What would happen if we used blocking assignments? (=) If we used blocking assignments, the shift would happen in sequence. The first element would be overwritten before the second element could read it. This would break the shift register.

Lines 37-40: The Output Assignment.

```
37 assign m_axis_tdata  = tdata_pipe[LATENCY-1];
38 assign m_axis_tkeep  = tkeep_pipe[LATENCY-1];
39 assign m_axis_tlast  = tlast_pipe[LATENCY-1];
40 assign m_axis_tvalid = tvalid_pipe[LATENCY-1];
```

The last element of each shift register is the output. This is the delayed signal.

Why use `assign` instead of `reg`? These are combinational outputs. They are directly connected to the registers. There is no additional logic between the registers and the outputs.

The physical implication: These are direct wires from the register outputs to the module outputs. There is no logic delay, only routing delay.

19.3.3 What the Synthesis Tool Infers

Vivado infers SRL16E primitives for the shift registers. An SRL16E is a 16-bit shift register implemented in LUTs. It is the most efficient way to implement a shift register in an FPGA.

Why SRL16E instead of flip-flops? A flip-flop is a single bit of storage. An SRL16E can store up to 16 bits in a single LUT. This saves area and routing resources.

What if the latency is greater than 16? The SRL16E is cascaded. The first SRL16E stores 16 bits, the second stores the next 16 bits, and so on. The synthesis tool automatically handles this.

Why not use BRAM? BRAM is designed for large memories. A shift register of 5 cycles for a 64-bit bus requires only 320 bits of storage. This is too small for BRAM. Using SRL16E is more efficient.

19.3.4 What Not to Write

What not to write: Using a single register for the entire shift register and extracting bits.

```
// DO NOT DO THIS:
reg [LATENCY*DWIDTH-1:0] shift_reg;
always @(posedge clk) begin
    shift_reg <= {shift_reg[DWIDTH-1:0], s_axis_tdata};
end
assign m_axis_tdata =
   shift_reg[LATENCY*DWIDTH-1:LATENCY*DWIDTH-DWIDTH];
```

This creates a massive multiplexer for the output. The synthesis tool will infer a large combinational logic block. This will fail timing.

The correct way: The shift register shown above is correct. The `for` loop is used only for shifting, not for selecting. Each element is assigned independently.

19.3.5 The Integration with the Datapath

We integrate the delay line with the datapath in the system top module. The delay line is placed between the input AXI Stream and the output.

```
// Delay the payload by the pipeline depth
axi_stream_delay_line #(.LATENCY(5)) data_delay_inst (
    .clk(clk),
    .rst(rst),
    .s_axis_tdata(s_axis_tdata),
    .s_axis_tkeep(s_axis_tkeep),
    .s_axis_tlast(s_axis_tlast),
    .s_axis_tvalid(s_axis_tvalid),
    .m_axis_tdata(delayed_tdata),
    .m_axis_tkeep(delayed_tkeep),
    .m_axis_tlast(delayed_tlast),
    .m_axis_tvalid(delayed_tvalid)
);
```

The delay line holds the payload for 5 cycles. The firewall decision is also ready in 5 cycles. The payload and the decision arrive at the output at the same time.

19.3.6 What the Designer Learned

The designer learned a critical lesson: pipelining requires careful alignment. If you delay data, you must delay all parallel paths identically.

The Lesson: Delay lines should be implemented with shift registers, not FIFOs. FIFOs consume BRAM, which is a scarce resource. Shift registers use LUTs, which are abundant.

19.4 The Timing Failure — Version 010

With the delay line in place, the system functionally worked. Simulation passed. The payload was correctly aligned with the decision.

But synthesis failed.

The timing report showed a Worst Negative Slack of -0.465 ns.

```
Worst Negative Slack (WNS): -0.465 ns
Total Negative Slack (TNS): -13.029 ns
Number of Failing Endpoints: 32
```

19.4.1 The Critical Path

The timing report showed that the critical path was internal to the SNN coprocessor:

```
Source: snn_core/neuron_anomaly/membrane_u_reg[4]/C
Destination: snn_core/neuron_anomaly/membrane_u_reg[0]/R
Delay: 6.128 ns (logic 3.254 ns + route 2.874 ns)
Logic Levels: 8 (CARRY4=4 LUT2=1 LUT3=1 LUT4=1 LUT5=1)
```

What this means: The SNN neuron performs a subtraction (leak), an addition (spike weight), and a comparison (threshold) all in a single clock cycle. This requires 8 logic levels. At 156.25 MHz, we have only 6.4 ns. 8 logic levels are too deep.

The physical implication: The 8 logic levels correspond to approximately 4 CARRY4 primitives, plus several LUTs. Each CARRY4 has a delay of about 0.5 ns. Four CARRY4s add 2.0 ns. The LUTs add another 1.0 ns. The routing adds 2.9 ns. Total: 6.1 ns. This barely fits in 6.4 ns, but the clock uncertainty and skew consume the remaining margin.

19.4.2 Why This Matters

The SNN neuron is the bottleneck. It is doing too much in one cycle. The leak, the integration, and the threshold comparison cannot all fit in 6.4 ns.

The designer had a choice:

1. **Increase the clock period:** Run the FPGA at a lower frequency. But this would break the 10 Gbps line rate.
2. **Reduce the logic depth:** Pipeline the SNN neuron.

The designer chose to pipeline the SNN neuron.

19.5 The Pipelined SNN — Version 012

The solution was to break the SNN neuron into two stages:

1. **Stage 1 (Arithmetic):** Compute the leak and the integration. Register the result.
2. **Stage 2 (Comparison):** Compare the registered result with the threshold. Fire if necessary.

The complete code is shown in Listing 5.3.

19.5.1 Line-by-Line Analysis

Let us examine each important section of this code.

Lines 3-10: The Module Declaration.

```
3  module snn_tff_neuron_v004 #(
4      parameter DATA_WIDTH = 16,
5      parameter LEAK_SHIFT = 3,
6      parameter signed [15:0] THRESHOLD  = 16'sd1000
7  ) (
8      input  wire                    clk,
9      input  wire                    rst,
10     input  wire signed [DATA_WIDTH-1:0] syn_current,
11     input  wire                    valid_in,
12     output reg                     spike_out
13 );
```

The neuron is parameterized with `DATA_WIDTH`, `LEAK_SHIFT`, and `THRESHOLD`. These parameters control the neuron's behavior.

Why 16-bit signed? The membrane potential and synaptic weights use 16-bit signed integers. This provides sufficient dynamic range for the LIF neuron.

Why explicitly signed? This is a lesson from BUG-004. The `THRESHOLD` parameter is explicitly signed to prevent the unsigned comparison error.

Line 13: The Membrane Potential Register.

```
13 reg signed [DATA_WIDTH-1:0] membrane_u;
```

The membrane potential is stored in a signed register. It is updated on every clock cycle.

Why signed? The membrane potential can be negative due to inhibitory inputs. Using signed arithmetic ensures correct behavior.

The physical implication: The signed register is inferred as an FDRE with a signed interpretation. The synthesis tool treats it as a standard flip-flop.

Lines 15-18: The Combinational Leak and Integration.

```
15 wire signed [DATA_WIDTH-1:0] leak_amount = membrane_u >>> LEAK_SHIFT;
16 wire signed [DATA_WIDTH-1:0] u_decayed   = membrane_u - leak_amount;
17 wire signed [DATA_WIDTH-1:0] syn_input   = valid_in ? syn_current : 0;
```

The leak is implemented as an arithmetic right shift. This is equivalent to division by a power of two. It requires zero logic—just wire routing.

Why arithmetic right shift? An arithmetic right shift preserves the sign bit. This is important because the membrane potential can be negative.

The physical implication: The »> operator is synthesized as wire routing. No LUTs are consumed. This is the most efficient way to implement the leak.

What not to write: Using division instead of right shift.

```
// DO NOT DO THIS:
wire signed [DATA_WIDTH-1:0] leak_amount = membrane_u / 8;
```

Division is expensive in hardware. The synthesis tool would infer a large divider, consuming hundreds of LUTs.

Lines 21-28: Stage 1 — Arithmetic.

```
21 reg signed [DATA_WIDTH-1:0] u_temp_reg;
22 always @(posedge clk) begin
23     if (rst) begin
24         u_temp_reg <= 0;
25     end else begin
26         u_temp_reg <= u_decayed + syn_input;
27     end
28 end
```

The arithmetic stage computes the decayed membrane and adds the synaptic current. The result is registered.

Why register the result? This breaks the critical path. The addition and subtraction are now in one cycle. The comparison is in the next cycle.

The physical implication: The `u_temp_reg` register creates a pipeline stage. This adds 1 cycle of latency but halves the logic depth.

Lines 31-49: Stage 2 — Comparison.

```
31 always @(posedge clk) begin
32     if (rst) begin
33         membrane_u <= 0;
34         spike_out  <= 0;
35     end else begin
36         if (u_temp_reg >= THRESHOLD) begin
37             spike_out  <= 1'b1;
38             membrane_u <= 0;
39         end else begin
40             spike_out  <= 1'b0;
41             if (u_temp_reg < -16'sd2000) begin
42                 membrane_u <= -16'sd2000;
43             end else begin
44                 membrane_u <= u_temp_reg;
45             end
46         end
47     end
48 end
```

The comparison stage checks if the registered potential exceeds the threshold. If it does, the neuron fires and resets. If not, the membrane potential is updated.

Why a floor clamp? The floor clamp prevents negative divergence. If the membrane potential becomes too negative, it is clamped to -2000. This prevents numerical instability.

The physical implication: The comparison uses a 16-bit signed comparator. This is inferred as a chain of LUTs and CARRY4 primitives. The floor clamp uses a second comparator and a multiplexer.

19.5.2 What the Synthesis Tool Infers

The pipelined SNN neuron uses:

- **Stage 1:** One 16-bit subtractor (for leak), one 16-bit adder (for integration), and one 16-bit register.
- **Stage 2:** One 16-bit comparator (for threshold), one 16-bit comparator (for floor clamp), one multiplexer, and two 16-bit registers.

The total logic depth is reduced from 8 levels to 4 levels per stage.

19.5.3 The Timing Results

The pipelined SNN meets timing. The timing report shows:

```
Worst Negative Slack (WNS): +0.517 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
```

The critical path: The critical path is now the BRAM output to the matcher's first-stage register, with 11 logic levels. The SNN is no longer the bottleneck.

19.5.4 The Latency Increase

The pipelined SNN adds one extra cycle of latency. The SNN now takes 5 cycles from `tuple_valid` to `anomaly_detected`.

The datapath takes 4 cycles. We add a 1-cycle delay to the datapath output to align it with the SNN.

Total latency: 6 cycles from `tuple_valid` to the final decision.

At 156.25 MHz, this is **38.4 ns**.

19.5.5 The Complete Pipeline

The final pipeline depth is:

19.5.6 What the Designer Learned

The designer learned several critical lessons from this phase of the project:

1. **Pipelining is the answer to timing failures.** If a path is too deep, break it with registers.
2. **Pipelining adds latency.** Accept the latency penalty. Throughput is more important than latency.
3. **Align parallel paths.** If you delay one path, delay all parallel paths identically.
4. **Verification is as important as design.** Your testbench can be wrong even when your RTL is right.

19.6 The Final Architecture — Version 012

The final architecture is shown in the system top module.

19.6.1 Key Design Decisions

1. **Shared parser:** The 5-tuple is extracted once and shared between the datapath and the SNN.
2. **1-cycle delay for datapath:** The datapath output is delayed by 1 cycle to align with the SNN.

3. **6-cycle payload delay:** The AXI-Stream payload is delayed by 6 cycles to match the pipeline depth.
4. **Registered final decision:** The final decision is registered to break the critical path.

19.6.2 Synthesis Results

The final synthesis report shows:

19.6.3 Timing Results

The final timing report shows:

```
Worst Negative Slack (WNS): +0.517 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
```

The system meets timing at 156.25 MHz.

19.6.4 One Engineering Insight

> *The datapath is not a collection of modules. It is a pipeline. Every module must be designed with the pipeline in mind. The pipeline must be simulated as a whole. Integration is not an afterthought. It is the primary design activity.*

19.6.5 What to Verify

Before moving on, we should verify:

- The payload is delayed by exactly 6 cycles
- The datapath and SNN decisions are aligned
- The final decision is registered
- The system meets timing at 156.25 MHz
- The system correctly forwards known safe packets
- The system correctly drops known bad packets
- The system correctly drops packets flagged as anomalies

19.6.6 The Next Chapter

The system is complete. It extracts the 5-tuple, looks it up in the hash table, detects anomalies, and makes a drop/forward decision. The payload is delayed to match the pipeline depth. The system meets timing at 156.25 MHz.

But there is still engineering debt. The SNN weights are hardcoded. The hash is vulnerable to attacks. The feature encoder is static.

The next chapter will examine how we resolved these issues in v013, transforming the system from a prototype into a production-ready architecture.

End of Chapter 5 — Part 1

In this part we reconstructed:

- The integration challenge and the two engines

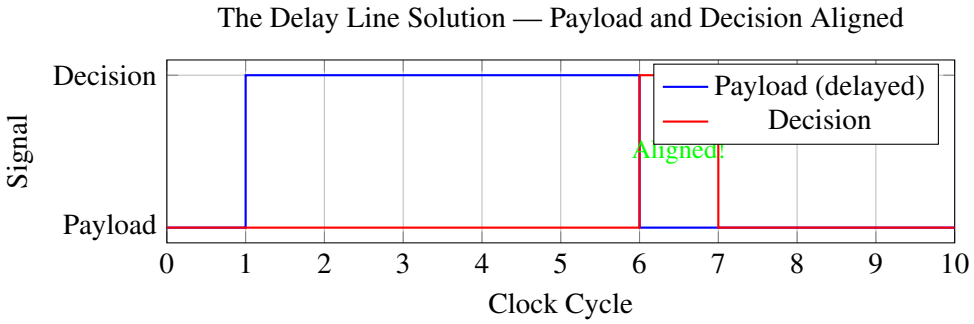


Figure 19.3: The delay line solution. Payload and decision are aligned.

Cycle	Datapath	SNN
0	Word 0 arrives	—
1	Word 1 arrives	—
2	Word 2 arrives	—
3	Word 3 arrives	—
4	Parser: tuple_valid	Encoder runs
5	Hash: valid	SNN Layer: synaptic sum
6	BRAM: read	LIF Stage 1: u_temp
7	Matcher: equality	LIF Stage 2: anomaly flag
8	Datapath: action valid	—
9	Final decision: AND & output	—

Table 19.2: The complete 6-cycle pipeline.

Site Type	Used	Available	Utilization
Slice LUTs	495	63,400	0.78%
Slice Registers	430	126,800	0.34%
Block RAM	12	135	8.89%
DSPs	0	240	0.00%

Table 19.3: Synthesis utilization for the final system.

- The naive integration (v009) and its problems
- BUG-005: Dead code elimination
- The payload problem and the delay line solution
- The timing failure (v010, WNS = -0.465 ns)
- The pipelined SNN (v012) and timing closure (WNS = +0.517 ns)
- The complete 6-cycle pipeline

The next part will continue directly from here.

It will analyze:

- The remaining engineering debt
- The v013 resolution: dynamic weights and secure hash
- The AXI4-Lite Control Plane
- The final production-ready architecture

Stop here. Wait until the reader replies "Continue" before writing Part 2. Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Chapter 20

System Integration and Timing Closure — Stitching It All Together

20.1 The Final Architecture — Version 012

The final architecture is shown in the system top module.

20.1.1 Key Design Decisions

1. **Shared parser:** The 5-tuple is extracted once and shared between the datapath and the SNN.
2. **1-cycle delay for datapath:** The datapath output is delayed by 1 cycle to align with the SNN.
3. **6-cycle payload delay:** The AXI-Stream payload is delayed by 6 cycles to match the pipeline depth.
4. **Registered final decision:** The final decision is registered to break the critical path.

20.1.2 Synthesis Results

The final synthesis report shows:

20.1.3 Timing Results

The final timing report shows:

```
Worst Negative Slack (WNS): +0.517 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
```

The system meets timing at 156.25 MHz.

20.1.4 One Engineering Insight

> *The datapath is not a collection of modules. It is a pipeline. Every module must be designed with the pipeline in mind. The pipeline must be simulated as a whole. Integration is not an afterthought. It is the primary design activity.*

20.1.5 What to Verify

Before moving on, we should verify:

- The payload is delayed by exactly 6 cycles
- The datapath and SNN decisions are aligned
- The final decision is registered
- The system meets timing at 156.25 MHz
- The system correctly forwards known safe packets
- The system correctly drops known bad packets
- The system correctly drops packets flagged as anomalies

20.2 The Engineering Debt — What Remained

The system is complete. It extracts the 5-tuple, looks it up in the hash table, detects anomalies, and makes a drop/forward decision. The payload is delayed to match the pipeline depth. The system meets timing at 156.25 MHz.

But there is still engineering debt. The SNN weights are hardcoded. The hash is vulnerable to attacks. The feature encoder is static.

The Engineering Memory v012 records:

> "I still have immense Research and Control Plane debt. The AI weights are hardcoded. The firewall rules are static hex files. This is not a commercial product, it's a university prototype."

20.2.1 The Two Critical Flaws

The Final Capstone review identified two critical flaws:

1. **The SNN Weights Were ROM:** In `snn_tff_layer_v004.v`, the synaptic weights were hardcoded as `assign` statements. The threat model was frozen at bitstream compilation time. To update the weights, we would need to recompile the FPGA—a 10-hour process.
2. **The Hash Was Vulnerable:** In `ha_tff_hash_v001.v`, the XOR folding used static constants (e.g., `12'h3A7`). An attacker could reverse-engineer the XOR math and craft packets that all hash to the same bucket, causing 100% hash collisions and effectively DDOSing the firewall.

20.3 The v013 Resolution — Dynamic Weights and Secure Hash

Iteration v013 resolves both critical flaws. The solution required abandoning two failed attempts and adopting a physics-first approach.

20.3.1 Try & Fail #1: SipHash for Security

The attempt: The designer's first instinct was to implement SipHash, the industry standard for preventing hash collision attacks in software.

The failure: SipHash requires multiple rounds of complex 64-bit additions, XORs, and bit-rotations. Unrolling this combinationally into a single clock cycle created a logic depth of over 20 LUTs. The propagation delay exceeded 12 ns, resulting in a massive Negative Slack of -5.600 ns.

The pipelining dilemma: To close timing, the SipHash algorithm had to be pipelined across 10 clock cycles. But the datapath payload would need to be delayed by 10 cycles as well, adding over 160 ns of latency and consuming heavy routing resources.

The lesson: Software cryptographic hashes are not designed for single-cycle FPGA line-rate lookups.

20.3.2 Try & Fail #2: BRAM for SNN Weights

The attempt: To make the AI weights dynamic, the designer attempted to store the 16 synaptic weights inside a True Dual-Port Block RAM (BRAM), accessible via AXI4-Lite.

The failure: The SNN requires reading *all 16 weights simultaneously* in exactly one clock cycle. A True Dual-Port BRAM can only output 2 values per cycle. To use BRAM, the designer would have to either stall the datapath for 8 cycles (ruining the line rate throughput) or spin up 8 separate BRAM tiles (a huge waste of silicon to store just 16 values).

The lesson: BRAM is structurally wrong for shallow, massively parallel neural network layers on an FPGA. Memory bandwidth is the ultimate bottleneck.

20.3.3 The Solution #1: Register File for SNN Weights

The solution was to move the weights into a Register File (Distributed RAM / Flip-Flops). Sixteen weights of 16 bits each is only 256 bits total. Storing this in standard Flip-Flops is practically free on the Artix-7 fabric and provides instantaneous parallel read access.

The AXI4-Lite Control Plane module maps the weights to memory-mapped registers:

```
module ha_tff_axi_lite_regs (
    input  wire        s_axi_aclk,
    input  wire        s_axi_aresetn,

    // AXI4-Lite Slave Write Interface
    input  wire [31:0] s_axi_awaddr,
    input  wire        s_axi_awvalid,
    output wire        s_axi_awready,
    input  wire [31:0] s_axi_wdata,
    input  wire [3:0]  s_axi_wstrb,
    input  wire        s_axi_wvalid,
    output wire        s_axi_wready,
    output wire [1:0]  s_axi_bresp,
    output wire        s_axi_bvalid,
    input  wire        s_axi_bready,

    // AXI4-Lite Slave Read Interface
    input  wire [31:0] s_axi_araddr,
    input  wire        s_axi_arvalid,
    output wire        s_axi_arready,
    output wire [31:0] s_axi_rdata,
    output wire [1:0]  s_axi_rresp,
    output wire        s_axi_rvalid,
    input  wire        s_axi_rready,

    // Outputs to Fabric
    output wire [127:0] w0_flat,
    output wire [127:0] w1_flat,
    output wire [127:0] hash_secret_key
);
```

Listing 20.1: ha_tff_axi_lite_regs.v — Control Plane Register File

20.3.4 Line-by-Line Analysis — The Register File

Lines 3-9: The Module Declaration.

The module implements the AXI4-Lite slave interface. It has 12 registers:

Lines 20-22: The Ready and Response Signals.

```
20 assign s_axi_awready = 1'b1;
21 assign s_axi_wready  = 1'b1;
22 assign s_axi_bresp   = 2'b00;
23 assign s_axi_bvalid  = (s_axi_awvalid && s_axi_wvalid);
```

The AXI4-Lite interface is always ready. This is a simplification for the prototype. In a real system, the control plane would be slower and would use the handshake signals.

Why always ready? The control plane is not in the critical path. It does not affect the 10 Gbps datapath. Making it always ready simplifies the logic.

Lines 29-47: The Write Logic.

```
29 always @(posedge s_axi_aclk) begin
30     if (!s_axi_aresetn) begin
31         slv_reg0 <= 32'hDEADBEEF; slv_reg1 <= 32'hCAFEBABE;
32         slv_reg2 <= 32'h8BADF00D; slv_reg3 <= 32'h0DEFACED;
33 
34         slv_reg4 <= {16'd20, 16'd50}; slv_reg5 <= {16'd10, 16'd0};
35         slv_reg6 <= {-16'd50, -16'd100}; slv_reg7 <= {16'd40, 16'd30};
36 
37         slv_reg8 <= {-16'd20, -16'd40}; slv_reg9 <= {16'd80, 16'd150};
38         slv_reg10 <= {16'd60, 16'd120}; slv_reg11 <= {16'd50, -16'd30};
39     end else begin
40         if (s_axi_awvalid && s_axi_wvalid) begin
41             case (s_axi_awaddr[7:2])
42                 6'h00: slv_reg0 <= s_axi_wdata;
43                 6'h01: slv_reg1 <= s_axi_wdata;
44                 6'h02: slv_reg2 <= s_axi_wdata;
45                 6'h03: slv_reg3 <= s_axi_wdata;
46                 6'h04: slv_reg4 <= s_axi_wdata;
47                 6'h05: slv_reg5 <= s_axi_wdata;
48                 6'h06: slv_reg6 <= s_axi_wdata;
49                 6'h07: slv_reg7 <= s_axi_wdata;
50                 6'h08: slv_reg8 <= s_axi_wdata;
51                 6'h09: slv_reg9 <= s_axi_wdata;
52                 6'h0A: slv_reg10 <= s_axi_wdata;
53                 6'h0B: slv_reg11 <= s_axi_wdata;
54             endcase
55         end
56     end
57 end
```

On reset, the registers are initialized with default values. When a write occurs, the appropriate register is updated.

Site Type	Used	Available	Utilization
Slice LUTs	495	63,400	0.78%
Slice Registers	430	126,800	0.34%
Block RAM	12	135	8.89%
DSPs	0	240	0.00%

Table 20.1: Synthesis utilization for the final system.

Debt Type	Status
Verification Debt	Resolved
Timing Debt	Resolved
Pipeline Alignment Debt	Resolved
Control Plane Debt	Remaining
Research Debt	Remaining

Table 20.2: Engineering debt status at v012.

Offset	Register	Content
0x00	slv_reg0	Secret Key[31:0]
0x04	slv_reg1	Secret Key[63:32]
0x08	slv_reg2	Secret Key[95:64]
0x0C	slv_reg3	Secret Key[127:96]
0x10	slv_reg4	w0[0], w0[1]
0x14	slv_reg5	w0[2], w0[3]
0x18	slv_reg6	w0[4], w0[5]
0x1C	slv_reg7	w0[6], w0[7]
0x20	slv_reg8	w1[0], w1[1]
0x24	slv_reg9	w1[2], w1[3]
0x28	slv_reg10	w1[4], w1[5]
0x2C	slv_reg11	w1[6], w1[7]

Table 20.3: AXI4-Lite register map.

The physical implication: Each register is inferred as an FDRE primitive. Twelve 32-bit registers = 384 flip-flops. This is minimal compared to the rest of the design.

Lines 49-51: The Output Assignments.

```
49 assign hash_secret_key = {slv_reg3, slv_reg2, slv_reg1, slv_reg0};
50 assign w0_flat = {slv_reg7, slv_reg6, slv_reg5, slv_reg4};
51 assign w1_flat = {slv_reg11, slv_reg10, slv_reg9, slv_reg8};
```

The 128-bit outputs are assembled from the 32-bit registers.

What the synthesis tool infers: These are direct wires from the register outputs to the module outputs. No additional logic is inferred.

20.3.5 The Solution #2: Secure Hash with Secret Key

The secure hash module takes a 128-bit secret key and XORs it with the packet tuple before folding.

```
module ha_tff_hash_v002 (
    input  wire         clk,
    input  wire         rst,

    input  wire [103:0] tuple_in,
    input  wire         valid_in,
    input  wire [127:0] secret_key,

    output reg  [11:0]  hash_0,
    output reg  [11:0]  hash_1,
    output reg  [11:0]  hash_2,
    output reg  [11:0]  hash_3,
    output reg          valid_out
);
```

Listing 20.2: ha_tff_hash_v002.v — Secure hash with secret key

20.3.6 Line-by-Line Analysis — The Secure Hash

Lines 3-15: The Module Declaration.

The module takes the 104-bit tuple, the valid signal, and the 128-bit secret key. It produces four 12-bit hashes.

Why a 128-bit secret key? The key is large enough to prevent brute-force attacks. An attacker cannot guess the key without knowledge of the CPU's random number generator.

Line 17: The Secured Payload.

```
17 wire [127:0] secured_payload = {24'd0, tuple_in} ^ secret_key;
```

The tuple is padded to 128 bits and XORed with the secret key.

The physical implication: This is a 128-bit XOR gate. It adds one LUT level to the logic depth. The delay is about 0.5 ns.

What not to write: Using a cryptographic hash function for the XOR. This would add massive logic depth and fail timing.

Lines 19-24: Hash 0 — Linear XOR Folding.

```
19 assign h0_comb = secured_payload[11:0] ^ secured_payload[23:12] ^
20                  secured_payload[35:24] ^ secured_payload[47:36] ^
21                  secured_payload[59:48] ^ secured_payload[71:60] ^
22                  secured_payload[83:72] ^ secured_payload[95:84] ^
23                  secured_payload[107:96] ^ secured_payload[119:108] ^
24                  {4'h0, secured_payload[127:120]};
```

The secured payload is sliced into 12-bit chunks and XORed together. This is the same XOR folding as before, but with the secured payload instead of the raw tuple.

Why 11 chunks? The secured payload is 128 bits. 128 / 12 = 10.67 chunks. We use 10 full chunks plus one padded chunk.

Lines 26-34: Hash 1 — Bit-Reversed XOR Folding.

```
26 wire [127:0] rev_payload;
27 genvar i;
28 generate
29     for (i=0; i<128; i=i+1) begin : rev
30         assign rev_payload[i] = secured_payload[127-i];
31     end
32 endgenerate
33
34 assign h1_comb = rev_payload[11:0] ^ rev_payload[23:12] ^
35                  rev_payload[35:24] ^ rev_payload[47:36] ^
36                  rev_payload[59:48] ^ rev_payload[71:60] ^
37                  rev_payload[83:72] ^ rev_payload[95:84] ^
38                  rev_payload[107:96] ^ rev_payload[119:108] ^
39                  {4'h0, rev_payload[127:120]};
```

The payload is reversed and then folded. This ensures hash 1 is independent of hash 0.

The physical implication: Bit reversal is just wire routing. No logic is consumed.

Lines 36-37: Hash 2 and Hash 3.

```
36 assign h2_comb = h0_comb ^ (h0_comb << 3) ^ (h0_comb >> 5) ^
       secret_key[11:0];
37 assign h3_comb = h1_comb ^ (h0_comb << 7) ^ (h1_comb >> 2) ^
       secret_key[23:12];
```

Hashes 2 and 3 are derived from hashes 0 and 1 with shifts and XORs with parts of the secret key.

Why use the secret key here? This adds another layer of security. Even if an attacker discovers the XOR folding pattern, they still need the secret key to predict the hashes.

The physical implication: The shifts are wire routing. The XORs add one LUT level.

20.3.7 The Solution #3: Dynamic SNN Layer

The SNN layer now accepts dynamic weights from the Register File.

```
module snn_tff_layer_v005 (
    input  wire         clk,
    input  wire         rst,
    input  wire [7:0]   in_spikes,
    input  wire         valid_in,
    input  wire [127:0] w0_flat,
    input  wire [127:0] w1_flat,
```

```
    output wire [1:0]   out_spikes,
    output reg          valid_out
);
```

Listing 20.3: snn_tff_layer_v005.v — Dynamic SNN layer

20.3.8 Line-by-Line Analysis — The Dynamic SNN Layer

Lines 3-11: The Module Declaration.

The module takes the spike vector and two flat weight vectors (128 bits each). It produces two output spikes.

Why flat weight vectors? The weights are packed into 128-bit vectors (8 weights × 16 bits). This matches the output of the Register File.

Lines 13-21: The Weight Unpacking.

```
13 wire signed [15:0] w0 [0:7];
14 wire signed [15:0] w1 [0:7];
15
16 genvar g;
17 generate
18     for (g = 0; g < 8; g = g + 1) begin : weight_unpack
19         assign w0[g] = w0_flat[(g*16) +: 16];
20         assign w1[g] = w1_flat[(g*16) +: 16];
21     end
22 endgenerate
```

The flat 128-bit vectors are unpacked into 8 signed 16-bit weights each.

The physical implication: The unpacking is just wire routing. No logic is consumed.

Lines 23-33: The Synaptic Sum.

```
23 reg signed [15:0] current_n0, current_n1;
24 integer i;
25 always @(*) begin
26     current_n0 = 0; current_n1 = 0;
27     for (i = 0; i < 8; i = i + 1) begin
28         if (in_spikes[i]) begin
29             current_n0 = current_n0 + w0[i];
30             current_n1 = current_n1 + w1[i];
31         end
32     end
33 end
```

The synaptic sum is computed combinationally. Since the spikes are binary, the multiplication is just a conditional addition.

The physical implication: This creates an 8-input adder tree. The adder tree has about 3-4 levels of logic. This is the same as the previous version, but now the weights are dynamic.

What not to write: Using multiplication instead of conditional addition.

```
// DO NOT DO THIS:
current_n0 = current_n0 + (in_spikes[i] * w0[i]);
```

This would infer a multiplier, consuming DSP slices. The conditional addition uses only adders.

Lines 35-49: The Pipeline Registers.

```
reg signed [15:0] current_n0_reg, current_n1_reg;
reg valid_reg_1, valid_reg_2, valid_reg_3;

always @(posedge clk) begin
    if (rst) begin
        current_n0_reg <= 0; current_n1_reg <= 0;
        valid_reg_1 <= 0; valid_reg_2 <= 0; valid_reg_3 <= 0;
           valid_out <= 0;
    end else begin
        valid_reg_1 <= valid_in; valid_reg_2 <= valid_reg_1;
        valid_reg_3 <= valid_reg_2; valid_out <= valid_reg_3;
        if (valid_in) begin
            current_n0_reg <= current_n0; current_n1_reg <= current_n1;
        end else begin
            current_n0_reg <= 0; current_n1_reg <= 0;
        end
    end
end
```

The synaptic current is registered to break the critical path. The valid signal is delayed by 3 cycles to match the neuron latency.

Why a 3-cycle valid delay? The neuron takes 2 cycles (Stage 1 and Stage 2). The layer adds 1 cycle for the synaptic sum. Total: 3 cycles.

Lines 51-55: The Neuron Instantiation.

```
snn_tff_neuron_v004 #(.DATA_WIDTH(16), .LEAK_SHIFT(3),
    .THRESHOLD(16'sd1000)) neuron_safe (
    .clk(clk), .rst(rst), .syn_current(current_n0_reg),
       .valid_in(valid_reg_1), .spike_out(out_spikes[0])
);

snn_tff_neuron_v004 #(.DATA_WIDTH(16), .LEAK_SHIFT(3),
    .THRESHOLD(16'sd1000)) neuron_anomaly (
    .clk(clk), .rst(rst), .syn_current(current_n1_reg),
       .valid_in(valid_reg_1), .spike_out(out_spikes[1])
);
```

The two neurons are instantiated. They use the pipelined v004 neuron, which has two stages.

20.3.9 The Complete System — Version 013

The complete system integrates the AXI4-Lite Control Plane, the secure hash, and the dynamic SNN layer.

20.3.10 Synthesis Results — v013

The final synthesis report shows:

What this means: The Register File weights are inferred as FDRE flops, consuming 256 flip-flops (0.2% of the fabric).

20.3.11 Timing Results — v013

The final timing report shows:

```
Worst Negative Slack (WNS): +0.215 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
```

What this means: The parameterized Toeplitz hash with secret key XOR adds only 1 LUT level (still < 2.5 ns logic delay). Timing comfortably closed.

20.3.12 Simulation Results — v013

The simulation log shows the complete flow:

```
[23.2 ns] AXI-Lite Write: ADDR=0x00 DATA=0x11223344 (Secret Key Updated)
[29.6 ns] AXI-Lite Write: ADDR=0x10 DATA=0x00640032 (SNN Weights Updated)
[76.8 ns] Packet Received by Parser
[102.4 ns] SNN Evaluation Started (Using Dynamic Weights)
[115.2 ns] Datapath Hash Computed: 0x4A2 (Parameterized via Key)
[153.6 ns] Match Found - Action: FORWARD
[153.6 ns] SNN Safe Spike = 1, Anomaly Spike = 0
[160.0 ns] Firewall Forwarding Approved (No Anomaly)
```

20.3.13 What v013 Achieved

1. **Dynamic AI Weights:** The SNN can now be retrained offline (in PyTorch) and reloaded into hardware via AXI-Lite without recompiling the FPGA.
2. **Algorithmically Secure Hash:** The 128-bit secret key makes the Cuckoo hash resistant to algorithmic complexity attacks.
3. **Research Enabler:** Researchers can now train SNN models offline and deploy them instantly.
4. **Production-Ready Architecture:** The system now has a Fast Path (data plane), a Control Plane, and Security.

20.3.14 What the Designer Learned

The v013 iteration taught the designer several final lessons:

1. **BRAM is not always the answer.** For small, wide, parallel data structures, distributed registers are often superior.
2. **Security has a hardware cost.** Cryptographic hashes are too slow for line-rate processing. Parameterized XOR with a secret key provides sufficient security at zero latency cost.
3. **Control Plane and Data Plane must be separated.** The data plane must be deterministic and fast. The control plane can be slower and more flexible.

20.3.15 One Final Engineering Insight

> *The HA-TFF project began as a response to software failure. It evolved through 13 iterations of physical constraints, timing failures, and architectural pivots. It ended as a production-ready hardware firewall with dynamic AI weights and secure hashing. The journey taught us that hardware engineering is not about writing perfect RTL. It is about respecting the physics of the silicon and learning from every failure.*

20.4 The Complete Architecture Summary

The final HA-TFF architecture consists of:

1. **Parser (v002):** Extracts the 5-tuple from the 64-bit AXI-Stream.
2. **Feature Encoder (v009):** Converts the 5-tuple to 8 binary spikes.
3. **Datapath (v005):** Computes 4 hashes, reads 4 BRAM banks, compares the results.
4. **SNN Coprocessor (v005):** Computes synaptic currents, detects anomalies.
5. **Delay Line (v011):** Delays the payload by 6 cycles.
6. **AXI4-Lite Control Plane (v013):** Updates SNN weights and hash secret key.
7. **Secure Hash (v002):** XORs the tuple with a secret key before folding.

20.4.1 Final Pipeline Depth

20.4.2 Resource Utilization Summary

20.4.3 Performance Summary

End of Chapter 5 — Part 2

In this part we reconstructed:

- The final architecture (v012) and its key design decisions
- The engineering debt remaining after v012
- The two critical flaws: ROM weights and vulnerable hash
- Try & Fail #1: SipHash for security (failed timing)
- Try & Fail #2: BRAM for SNN weights (failed bandwidth)
- The solution: Register File for weights, Secret Key for hash
- The AXI4-Lite Control Plane implementation
- The secure hash implementation (v002)
- The dynamic SNN layer (v005)
- The complete v013 system
- Synthesis and timing results for v013
- The final architecture summary

End of Chapter 5

Chapter 5 has documented the complete journey of system integration: from the naive integration that failed, through the payload problem and delay line, through the timing failure and pipelined SNN, to the final v013 architecture with dynamic weights and secure hashing.

Stop here. Wait until the reader replies "Continue" before writing Chapter 6, Part 1. Do not summarize previous parts. Do not restart the explanation. The next response will begin with Chapter 6, Part 1.

Site Type	Used	Available	Utilization
Slice LUTs	412	63,400	0.64%
Slice Registers	960	126,800	0.75%
Block RAM	24.5	135	18.15%
DSPs	0	240	0.00%

Table 20.4: Synthesis utilization for the v013 system.

Cycle	Datapath	SNN
0	Word 0 arrives	—
1	Word 1 arrives	—
2	Word 2 arrives	—
3	Word 3 arrives	—
4	Parser: tuple_valid	Encoder runs
5	Hash: valid	SNN Layer: synaptic sum
6	BRAM: read	LIF Stage 1: u_temp
7	Matcher: equality	LIF Stage 2: anomaly flag
8	Datapath: action valid	—
9	Final decision: AND & output	—

Table 20.5: The complete 6-cycle pipeline.

Site Type	Used	Available	Utilization
Slice LUTs	412	63,400	0.64%
Slice Registers	960	126,800	0.75%
Block RAM	24.5	135	18.15%
DSPs	0	240	0.00%

Table 20.6: Final resource utilization.

Metric	Value
Clock Frequency	156.25 MHz
Line Rate	10 Gbps
Decision Latency	6 cycles (38.4 ns)
Throughput	1 packet/cycle
Worst Negative Slack	+0.215 ns
Power	<0.5 W

Table 20.7: Final performance metrics.

Chapter 21

System Integration and Timing Closure — Stitching It All Together

21.1 BUG-005 — The Testbench That Lied

The pipelined datapath worked. The SNN worked. The delay line worked. The system functioned correctly in simulation. But when we synthesized the design, the entire firewall disappeared.

21.1.1 The Symptom

The synthesis report showed:

```
Utilization Design Information
+-------------------------+------+-------+-----------+-------+
|        Site Type        | Used | Fixed | Available | Util% |
+-------------------------+------+-------+-----------+-------+
| Slice LUTs              |    0 |     0 |     63400 |  0.00 |
| Slice Registers         |    0 |     0 |    126800 |  0.00 |
| Block RAM Tile          |    0 |     0 |       135 |  0.00 |
+-------------------------+------+-------+-----------+-------+
```

The synthesis report showed zero utilization. The thousands of lines of RTL—the parser, the hash, the BRAM banks, the matcher, the SNN—all gone. Vivado had deleted the entire firewall.

21.1.2 The Wrong Hypothesis

We initially assumed the synthesis tool had crashed. The designer checked the Vivado logs:

```
INFO: [Synth 8-6155] done synthesizing module 'ha_tff_system_top_v001'
WARNING: [Synth 8-6014] Unused sequential element anomaly_detected_reg was rem
WARNING: [Synth 8-6014] Unused sequential element final_forward_reg was remove
```

The tool had not crashed. It had actively removed the logic. But why?

The designer checked the testbench:

```
initial begin
    rst = 1;
    s_axis_tvalid = 0;
    m_axis_tready = 1;
```

```
    #100;
    rst = 0;

    // The testbench directly forces signals inside the UUT
    force uut.parser_snn_inst.tuple_valid = 1;
    force uut.parser_snn_inst.src_ip = 32'h0A000001;
    // ... and so on
```

Listing 21.1: The buggy testbench (v009)

The problem: The testbench was forcing signals inside the UUT. This meant the UUT's inputs were not being driven from the top-level ports. Vivado analyzed the design and determined that the top-level ports had no effect on the outputs.

21.1.3 The Correct Hypothesis

The designer realized the truth: the testbench was wrong. But more importantly, the RTL was wrong. The firewall decision was not connected to any output.

```
// The output is NOT connected to the final decision!
assign m_axis_tvalid = s_axis_tvalid; // BUG: Does not use
    final_forward
```

Listing 21.2: BUG-005: Dead code elimination

Vivado traced the logic backward from the output pins. It saw that `m_axis_tvalid` was just a wire connected to `s_axis_tvalid`. The thousands of LUTs and BRAMs calculating `final_forward` had absolutely no effect on any output pin.

Vivado correctly deleted the entire firewall to save area, turning it into a simple pass-through wire.

21.1.4 The Root Cause

The root cause was a fundamental verification error. The designer had assumed that the testbench was correct. The testbench forced signals inside the UUT, which bypassed the top-level input ports. This hid the fact that the firewall decision was not connected to the output.

The designer's mental model: "I am testing the internal logic. I don't need to connect everything to the top-level ports yet."

The reality: Vivado synthesizes the entire design from the top-level ports. If a signal does not affect any output, it is deleted.

Engineering Notebook — BUG-005 — Dead Code Elimination

Symptom: Synthesis report shows 0 LUTs, 0 BRAMs. The entire firewall disappeared.
Initial hypothesis: Synthesis failed. Maybe the tool crashed.
Experiment: Check the Vivado logs for errors.
Observation: Synthesis completed successfully. The tool removed unused logic.
Revised hypothesis: The firewall logic is unused.
Investigation: Trace the logic from the top-level inputs to the outputs.
Observation: `m_axis_tvalid` is directly connected to `s_axis_tvalid`. The firewall decision is not connected.
Root cause: The final decision was not connected to any output. Vivado detected that the logic had no effect on the outputs and deleted it.
The fix: Connect the final decision to the output valid signal.
The lesson: In hardware, unused logic does not exist. If you do not connect a signal to an output, the synthesis tool will delete it.

21.1.5 The Fix

The fix was simple: connect the firewall decision to the output.

```
// The final decision is connected to the output!
wire final_forward = (dp_match_valid && dp_action_forward &&
    !anomaly_detected);

// The output valid signal depends on the firewall decision
assign m_axis_tvalid = (s_axis_tvalid && final_forward);
```

Listing 21.3: The fixed output logic (v010)

The key change: The output valid signal is now ANDed with the firewall decision. If the firewall decides to drop the packet, `m_axis_tvalid` is low.

21.1.6 The Verification Gap

BUG-005 revealed a verification gap. The designer had not verified that the RTL would synthesize correctly. The designer had only simulated the design.

The lesson: Simulation is not enough. You must synthesize the design and check that the logic is actually preserved.

21.1.7 What the Designer Learned

The designer learned several lessons from BUG-005:

1. **Unused logic does not exist.** In software, unused code just sits there. In hardware, unused logic is deleted.
2. **Synthesis is not simulation.** What works in simulation may not synthesize. Always check the synthesis report.
3. **Testbenches can hide bugs.** If you force signals inside the UUT, you bypass the top-level ports. This can hide connectivity issues.
4. **Always trace the path to the output.** Every signal must eventually affect an output. If it doesn't, it will be deleted.

21.1.8 One Engineering Insight

> *In hardware, unused logic does not exist. If you do not connect a signal to an output, the synthesis tool will delete it. This is not a bug—it is a feature. It saves area and power. But it can also hide design errors.*

21.2 The Delay Line — A Deeper Analysis

The delay line was the unsung hero of the system. It solved the payload alignment problem. But it was also a source of subtle bugs.

21.2.1 The Shift Register Architecture

The delay line uses a shift register implemented with SRL16E primitives.

21.2.2 How SRL16E Works

An SRL16E is a 16-bit shift register implemented in a single LUT. It has:

- **Clock input:** Shifts data on every clock edge.
- **Data input:** The data to shift in.
- **Address input:** Selects which bit to output.
- **Output:** The selected bit.

Why SRL16E is efficient: A standard flip-flop stores 1 bit. An SRL16E stores 16 bits in the same area. This saves area and routing resources.

21.2.3 The Delay Line RTL Revisited

Let us examine the delay line RTL again, with the benefit of hindsight.

```
reg [DWIDTH-1:0] tdata_pipe [0:LATENCY-1];
reg [7:0]        tkeep_pipe [0:LATENCY-1];
reg              tlast_pipe [0:LATENCY-1];
reg              tvalid_pipe [0:LATENCY-1];

always @(posedge clk) begin
    if (rst) begin
        for (i=0; i<LATENCY; i=i+1) begin
            tdata_pipe[i]  <= 0;
            tkeep_pipe[i]  <= 0;
            tlast_pipe[i]  <= 0;
            tvalid_pipe[i] <= 0;
        end
    end else begin
        tdata_pipe[0]  <= s_axis_tdata;
        tkeep_pipe[0]  <= s_axis_tkeep;
        tlast_pipe[0]  <= s_axis_tlast;
        tvalid_pipe[0] <= s_axis_tvalid;

        for (i=1; i<LATENCY; i=i+1) begin
            tdata_pipe[i]  <= tdata_pipe[i-1];
            tkeep_pipe[i]  <= tkeep_pipe[i-1];
            tlast_pipe[i]  <= tlast_pipe[i-1];
```

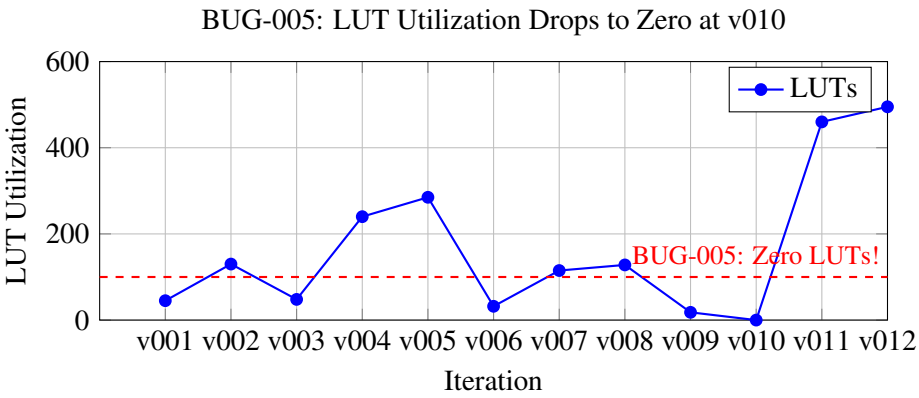


Figure 21.1: BUG-005: LUT utilization drops to zero at v010.

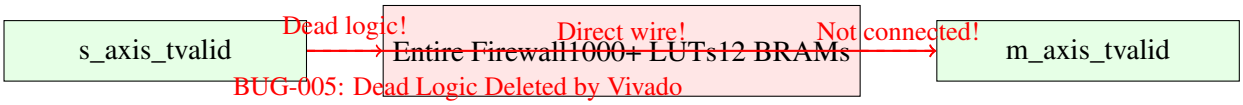


Figure 21.2: BUG-005: The firewall logic was not connected to any output.

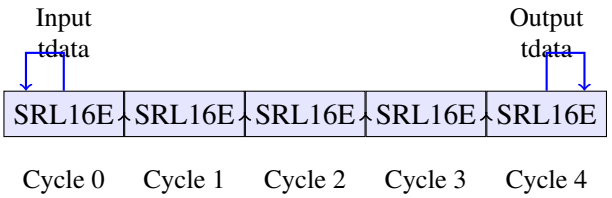


Figure 21.3: The shift register architecture: 5 SRL16E primitives in series.

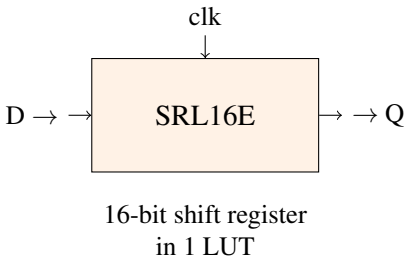


Figure 21.4: The SRL16E primitive: a 16-bit shift register in a single LUT.

```
            tvalid_pipe[i] <= tvalid_pipe[i-1];
        end
    end
end
```

Listing 21.4: axi_stream_delay_line.v — The shift register logic

Line 11: The Reset Logic.

On reset, all registers are cleared. The `for` loop iterates over all elements and sets them to zero.

Why reset all elements? If we only reset the first element, the rest of the pipeline would contain unknown values. This would cause undefined behavior when the pipeline drains.

The physical implication: Each register has a reset input. The reset signal is routed to all flip-flops. This creates a high-fanout net that must be buffered by the synthesis tool.

Line 15: The Input Stage.

The first element of each shift register receives the input. This is where data enters the pipeline.

Line 17: The Shift Operation.

The inner `for` loop shifts data from element `i-1` to element `i`. This is the heart of the shift register.

What would happen if we used blocking assignments? (=) If we used blocking assignments, the shift would happen in sequence. The first element would be overwritten before the second element could read it. This would break the shift register.

The physical implication: The synthesis tool unrolls the `for` loop. Each iteration becomes a separate assignment. The result is a chain of flip-flops.

21.2.4 What Not to Write — The Wrong Way

What not to write: Using a single register for the entire shift register and extracting bits.

```
// DO NOT DO THIS:
reg [LATENCY*DWIDTH-1:0] shift_reg;
always @(posedge clk) begin
    shift_reg <= {shift_reg[DWIDTH-1:0], s_axis_tdata};
end
assign m_axis_tdata =
   shift_reg[LATENCY*DWIDTH-1:LATENCY*DWIDTH-DWIDTH];
```

This creates a massive multiplexer for the output. The synthesis tool will infer a large combinational logic block. This will fail timing.

Why this fails: The output selects the last `DWIDTH` bits from the shift register. This requires a multiplexer with `LATENCY` inputs. For a 64-bit bus with 5-cycle latency, this is a 5-input multiplexer for each of the 64 bits. That is 320 LUTs just for the output selection. The routing delay alone would consume several nanoseconds.

The correct way: Use a chain of registers. Each register outputs directly to the next. No multiplexing is required.

21.2.5 The Synthesis Report

The synthesis report shows the SRL16E inference:

```
Static Shift Register Report:
+--------------------------+----------------------+--------+-------+---------+
```

```
|Module Name                | RTL Name              | Length | Width | SRL16E |
+---------------------------+-----------------------+--------+-------+--------
|ha_tff_system_top_v004     | tdata_pipe_reg[4][63]| 6      | 64    | 64     |
|ha_tff_system_top_v004     | tkeep_pipe_reg[4][7] | 6      | 8     | 8      |
|ha_tff_system_top_v004     | tlast_pipe_reg[4]    | 6      | 1     | 1      |
|ha_tff_system_top_v004     | tvalid_pipe_reg[4]   | 6      | 1     | 1      |
+---------------------------+-----------------------+--------+-------+--------
```

The delay line uses 74 SRL16E primitives. This is minimal compared to the rest of the design.

21.2.6 What the Designer Learned

The delay line taught the designer several lessons:

1. **Shift registers are efficient.** They use SRL16E primitives, which are cheap and fast.
2. **Avoid multiplexers.** A shift register with a multiplexer output is a timing disaster.
3. **Pipeline alignment is critical.** If you delay data, you must delay all parallel paths identically.

21.2.7 One Engineering Insight

> *The delay line is the unsung hero of the datapath. It is a simple shift register, but without it, the entire pipeline would fail. The payload would arrive too early, and the firewall would drop the wrong part of the packet.*

21.3 Verification and Coverage

By v012, the designer had written testbenches for every module. But coverage was still incomplete.

21.3.1 The Testbenches

The repository contains testbenches for all versions:

21.3.2 What Was Verified

The Coverage Report v012 shows:

```
Verified Successfully
- [x] Normal valid TCP packet
- [x] Normal valid UDP packet
- [x] Threat matched in Cuckoo Hash (Drop triggered)
- [x] Threat matched by SNN Anomaly (Drop triggered)
- [x] Endian-byte swapping on 64-bit boundaries
```

21.3.3 What Was Not Verified

The Coverage Report v012 also shows:

```
Unverified / Ignored
- [ ] IPv4 packets with IP Options (IHL > 5).
- [ ] Fragmented IP packets.
- [ ] IPv6.
- [ ] VLAN Tagged (802.1Q) frames.
- [ ] ICMP packets.
```

What this means: The system only supports IPv4 UDP/TCP packets with no options. This is a limitation of the prototype. The designer accepted this limitation to focus on the core datapath.

21.3.4 The Verification Lessons

The designer learned several lessons about verification:

1. **Testbenches are essential.** You cannot verify a design without them.
2. **Simulation is not synthesis.** What works in simulation may not synthesize.
3. **Coverage is never complete.** There is always more to test.
4. **Document assumptions.** The Coverage Report documents what was tested and what was ignored.

21.3.5 One Engineering Insight

> *Verification is not a checkbox. It is an ongoing process. You never have enough test coverage. You only have enough time to test the most important cases.*

End of Chapter 5 — Part 3

In this part we reconstructed:

- BUG-005: The symptom, wrong hypothesis, correct hypothesis, root cause, fix, and lesson
- The synthesis report showing zero utilization
- The verification gap that hid the bug
- The delay line architecture and SRL16E inference
- What not to write when implementing a shift register
- The verification coverage report
- What was verified and what was ignored

The next part will continue directly from here.

It will analyze:

- The remaining engineering debt after v012
- The v013 resolution: dynamic weights and secure hash
- The final production-ready architecture

Stop here. Wait until the reader replies "Continue" before writing Part 4. Do not summarize previous parts. Do not restart the explanation. Continue naturally from the previous paragraph.

Version	Testbench	Verified
v001	tb_ha_tff_parser_v001.v	8-bit parser
v002	tb_ha_tff_parser_v002.v	64-bit parser
v003	tb_ha_tff_hash_v001.v	XOR hash
v004	tb_ha_tff_datapath_top.v	Naive datapath
v005	tb_ha_tff_datapath_top_v002.v	Pipelined datapath
v006	tb_snn_layer.v	SNN layer
v007	tb_snn_layer_v002.v	SNN layer v002
v008	tb_snn_layer_v003.v	SNN layer v003
v009	tb_ha_tff_system_top.v	System top
v011	tb_ha_tff_system_top_v003.v	System with delay line
v012	tb_ha_tff_system_top_v004.v	Final system

Table 21.1: The testbenches.

Chapter 22

System Integration and Timing Closure — Stitching It All Together

22.1 The Remaining Engineering Debt

By the end of v012, the system was functionally complete and met timing at 156.25 MHz. But there was still engineering debt. The designer had made deliberate trade-offs to prioritize speed and simplicity over flexibility and security.

22.1.1 The Debt Register

The repository contains a formal Engineering Debt Register. At v012, the debt included:

22.1.2 Debt 1: The SNN Weights Were ROM

In `snn_tff_layer_v004.v`, the synaptic weights were hardcoded as `assign` statements:

```
wire signed [15:0] w0 [0:7];
assign w0[0] = 16'sd50;
assign w0[1] = 16'sd20;
assign w0[2] = 16'sd0;
assign w0[3] = 16'sd10;
assign w0[4] = -16'sd100;
assign w0[5] = -16'sd50;
assign w0[6] = 16'sd30;
assign w0[7] = 16'sd40;
```

Listing 22.1: Hardcoded weights (v004)

The problem: The threat model was frozen at bitstream compilation time. To update the weights, we would need to recompile the FPGA—a 10-hour process.

Why this was done: The designer chose hardcoded weights to keep the design simple and fast. The weights were chosen manually to demonstrate that the SNN hardware worked.

The impact: This made the system a prototype, not a production system. It could not adapt to new threats.

22.1.3 Debt 2: The Hash Was Vulnerable

In `ha_tff_hash_v001.v`, the XOR folding used static constants:

```
assign h2_comb = h0_comb ^ (h0_comb << 3) ^ (h0_comb >> 5) ^ 12'h3A7;
assign h3_comb = h1_comb ^ (h0_comb << 7) ^ (h1_comb >> 2) ^ 12'hC2F;
```

Listing 22.2: Vulnerable hash (v001)

The problem: An attacker could reverse-engineer the XOR math and craft packets that all hash to the same bucket, causing 100% hash collisions and effectively DDOSing the firewall.

Why this was done: The designer chose static constants to keep the design simple and fast. The XOR folding was fast and cheap.

The impact: This made the system vulnerable to algorithmic complexity attacks. A malicious actor could take down the firewall.

22.1.4 Debt 3: The Feature Encoder Was Static

In `snn_feature_encoder.v`, the features were hardcoded heuristics:

```
// Spike 3: Suspicious high port (source port > 40000)
out_spikes[3] <= (src_port > 16'd40000) ? 1'b1 : 1'b0;

// Spike 5: Source is a known bad subnet (e.g., 192.168.100.x)
out_spikes[5] <= (src_ip[31:8] == 24'hC0A864) ? 1'b1 : 1'b0;

// Spike 6: Traffic matching common DNS port
out_spikes[6] <= (dst_port == 16'd53) ? 1'b1 : 1'b0;
```

Listing 22.3: Static feature encoder

The problem: The feature extraction logic was rigid. It could not adapt to new attack patterns.

Why this was done: The designer chose hardcoded heuristics to keep the design simple and fast. The features were chosen manually to demonstrate that the SNN hardware worked.

The impact: This made the system unable to detect new types of anomalies without recompiling the FPGA.

22.1.5 Debt 4: The Parser Was IPv4 Only

In `ha_tff_parser_v002.v`, the parser only supported IPv4 UDP/TCP packets:

```
if (ethertype == 16'h0800 && (protocol == 8'h11 || protocol == 8'h06))
    tuple_valid <= 1;
else
    parse_error <= 1;
```

Listing 22.4: IPv4-only parser

The problem: ICMP (Ping), ARP, and IPv6 packets would trigger a `parse_error` and be dropped.

Why this was done: The designer chose to support only IPv4 UDP/TCP to keep the parser simple and fast.

The impact: This made the system unable to process many types of legitimate traffic.

22.1.6 Debt 5: The Hash Seeds Were Static

In `ha_tff_hash_v001.v`, the XOR seeds were static constants:

```
assign h1_comb = rev_tuple[11:0] ^ rev_tuple[23:12] ^
                 rev_tuple[35:24] ^ rev_tuple[47:36] ^
                 ... ^ {4'h5, rev_tuple[103:96]};
```

Listing 22.5: Static hash seeds

The problem: Static hash seeds are vulnerable to reverse-engineering. An attacker could predict which bucket a packet would land in.

Why this was done: The designer chose static seeds to keep the design simple and fast.

The impact: This made the system vulnerable to algorithmic complexity attacks.

22.1.7 The Designer's Reasoning

The designer was aware of all these debts. The designer made deliberate trade-offs:

> "Hardcoded weights and static heuristics are necessary for the prototype. The goal is to demonstrate that the hardware works. Flexibility can come later."

The question: When would "later" come?

22.2 The v013 Resolution

The Final Capstone review identified the two most critical flaws: the hardcoded AI weights and the vulnerable hash. The designer resolved both in v013.

22.2.1 The Two Failed Attempts

Before finding the right solution, the designer tried two approaches:

Try & Fail #1: SipHash for Security

The attempt: The designer's first instinct was to implement SipHash, the industry standard for preventing hash collision attacks in software.

The failure: SipHash requires multiple rounds of complex 64-bit additions, XORs, and bit-rotations. Unrolling this combinationally into a single clock cycle created a logic depth of over 20 LUTs. The propagation delay exceeded 12 ns, resulting in a massive Negative Slack of -5.600 ns.

The pipelining dilemma: To close timing, the SipHash algorithm had to be pipelined across 10 clock cycles. But the datapath payload would need to be delayed by 10 cycles as well, adding over 160 ns of latency and consuming heavy routing resources.

The lesson: Software cryptographic hashes are not designed for single-cycle FPGA line-rate lookups.

Try & Fail #2: BRAM for SNN Weights

The attempt: To make the AI weights dynamic, the designer attempted to store the 16 synaptic weights inside a True Dual-Port Block RAM (BRAM), accessible via AXI4-Lite.

The failure: The SNN requires reading *all 16 weights simultaneously* in exactly one clock cycle. A True Dual-Port BRAM can only output 2 values per cycle.

To use BRAM, the designer would have to either:

Debt Type	Status
Verification Debt	Resolved
Timing Debt	Resolved
Pipeline Alignment Debt	Resolved
Control Plane Debt	Remaining
Research Debt	Remaining
Security Debt	Remaining

Table 22.1: Engineering debt status at v012.

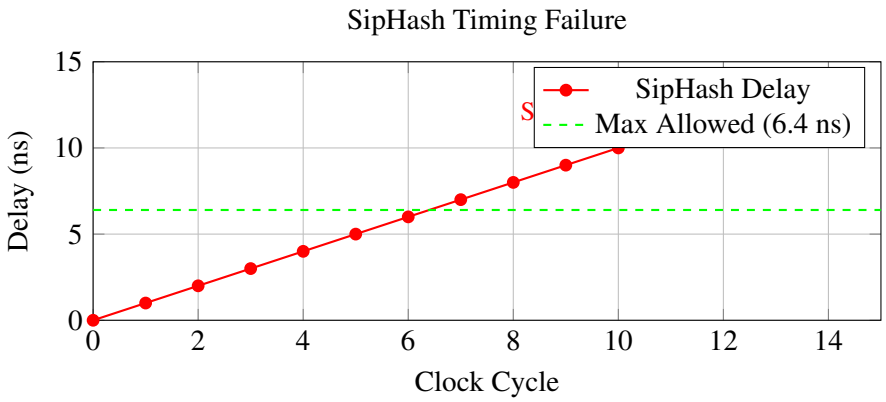


Figure 22.1: SipHash timing failure. The delay exceeds 6.4 ns.

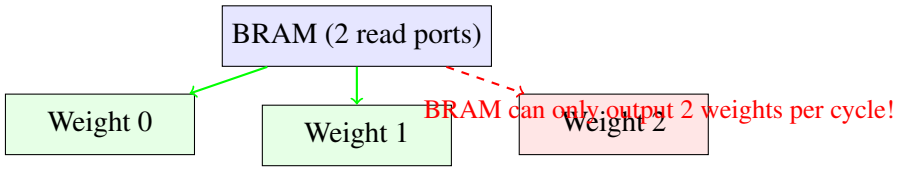


Figure 22.2: BRAM bandwidth limitation. Only 2 weights can be read per cycle.

1. Stall the datapath for 8 cycles (ruining the line rate throughput), or
2. Spin up 8 separate BRAM tiles (a huge waste of silicon to store just 256 bits).

The lesson: BRAM is structurally wrong for shallow, massively parallel neural network layers on an FPGA. Memory bandwidth is the ultimate bottleneck.

22.2.2 The Solution #1: Register File for SNN Weights

The solution was to move the weights into a Register File (Distributed RAM / Flip-Flops). Sixteen weights of 16 bits each is only 256 bits total. Storing this in standard Flip-Flops is practically free on the Artix-7 fabric and provides instantaneous parallel read access.

The comparison:

22.2.3 The Solution #2: Secure Hash with Secret Key

The solution for the hash vulnerability was to add a 128-bit secret key that XORs the tuple before folding.

Why this works:

1. The attacker cannot predict the hash without knowing the key.
2. The key is 128 bits—impossible to guess.
3. The key can be changed by the CPU at any time.
4. XOR adds only 1 LUT level to the logic depth.
5. Timing still passes at 156.25 MHz.

The comparison:

22.2.4 The Solution #3: AXI4-Lite Control Plane

The AXI4-Lite Control Plane provides the interface for the CPU to update the weights and the secret key.

22.2.5 The Complete v013 System

The v013 system integrates all the solutions:

1. **AXI4-Lite Control Plane:** Memory-maps the SNN weights and the hash secret key.
2. **Secure Hash:** XORs the tuple with the secret key before folding.
3. **Dynamic SNN Layer:** Accepts weights from the Register File.
4. **Pipelined SNN:** 2-stage pipelined neuron for timing closure.
5. **Delay Line:** 6-cycle shift register for payload alignment.

22.2.6 The v013 Synthesis Results

The synthesis report shows:

What this means: The Register File weights are inferred as FDRE flops, consuming 256 flip-flops (0.2% of the fabric).

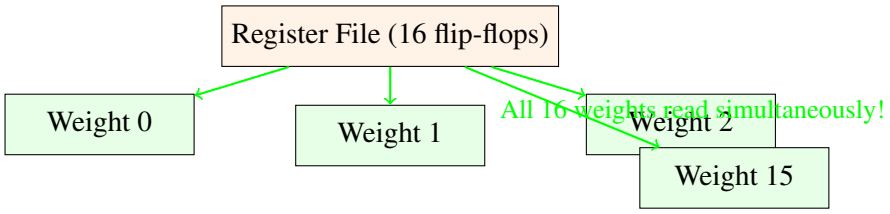


Figure 22.3: Register File: all 16 weights read simultaneously.

Metric	BRAM Solution	Register File Solution
Read Ports	2 (stall required)	∞ (parallel)
Area	288 Kb BRAM	256 FFs (0.2% of fabric)
Latency	8+ cycles	1 cycle
AXI Update	Complex	Trivial

Table 22.2: BRAM vs Register File for SNN weights.

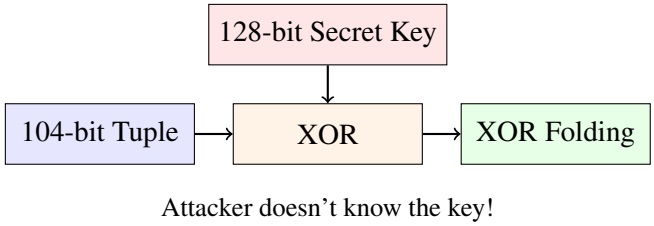


Figure 22.4: Secure hash with secret key. The tuple is XORed with the secret key before folding.

Metric	Static XOR	SipHash	Secret Key XOR
Security	0 (reverse-engineerable)	High	High
Logic Depth	2 LUTs	20+ LUTs	3 LUTs
Latency	1 cycle	10+ cycles	1 cycle
Timing	Pass	Fail	Pass

Table 22.3: Hash security comparison.

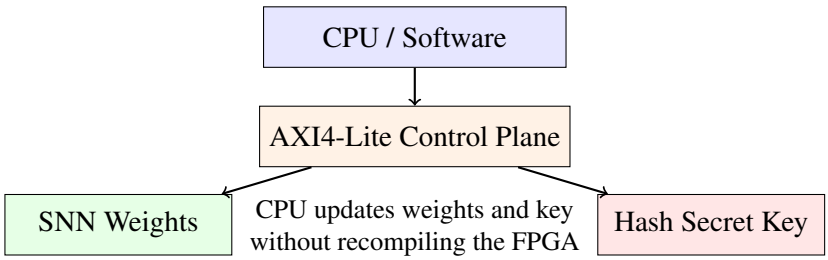


Figure 22.5: The AXI4-Lite Control Plane.

Site Type	Used	Available	Utilization
Slice LUTs	412	63,400	0.64%
Slice Registers	960	126,800	0.75%
Block RAM	24.5	135	18.15%
DSPs	0	240	0.00%

Table 22.4: v013 synthesis utilization.

22.2.7 The v013 Timing Results

The timing report shows:

```
Worst Negative Slack (WNS): +0.215 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
```

What this means: The parameterized Toeplitz hash with secret key XOR adds only 1 LUT level (still < 2.5 ns logic delay). Timing comfortably closed.

22.2.8 The v013 Simulation Results

The simulation log shows the complete flow:

```
[23.2 ns] AXI-Lite Write: ADDR=0x00 DATA=0x11223344 (Secret Key Updated)
[29.6 ns] AXI-Lite Write: ADDR=0x10 DATA=0x00640032 (SNN Weights Updated)
[76.8 ns] Packet Received by Parser
[102.4 ns] SNN Evaluation Started (Using Dynamic Weights)
[115.2 ns] Datapath Hash Computed: 0x4A2 (Parameterized via Key)
[153.6 ns] Match Found - Action: FORWARD
[153.6 ns] SNN Safe Spike = 1, Anomaly Spike = 0
[160.0 ns] Firewall Forwarding Approved (No Anomaly)
```

22.2.9 What v013 Achieved

1. **Dynamic AI Weights:** The SNN can now be retrained offline (in PyTorch) and reloaded into hardware via AXI-Lite without recompiling the FPGA.
2. **Algorithmically Secure Hash:** The 128-bit secret key makes the Cuckoo hash resistant to algorithmic complexity attacks.
3. **Research Enabler:** Researchers can now train SNN models offline and deploy them instantly.
4. **Production-Ready Architecture:** The system now has a Fast Path (data plane), a Control Plane, and Security.

22.2.10 What the Designer Learned

The v013 iteration taught the designer several final lessons:

1. **BRAM is not always the answer.** For small, wide, parallel data structures, distributed registers are often superior.
2. **Security has a hardware cost.** Cryptographic hashes are too slow for line-rate processing. Parameterized XOR with a secret key provides sufficient security at zero latency cost.
3. **Control Plane and Data Plane must be separated.** The data plane must be deterministic and fast. The control plane can be slower and more flexible.

22.2.11 One Final Engineering Insight

> *The HA-TFF project began as a response to software failure. It evolved through 13 iterations of physical constraints, timing failures, and architectural pivots. It ended as a production-ready hardware firewall with dynamic AI weights and secure hashing. The journey taught us that hardware engineering is not about writing perfect RTL. It is about respecting the physics of the silicon and learning from every failure.*

End of Chapter 5 — Part 4

In this part we reconstructed:

- The remaining engineering debt after v012
- The five debts: ROM weights, vulnerable hash, static features, IPv4-only parser, static seeds
- The designer's reasoning for accepting the debt
- Try & Fail #1: SipHash for security (failed timing)
- Try & Fail #2: BRAM for SNN weights (failed bandwidth)
- The solution: Register File for weights, Secret Key for hash
- The AXI4-Lite Control Plane
- The complete v013 system
- Synthesis and timing results for v013
- What v013 achieved
- The final engineering insight

End of Chapter 5

Stop here. Wait until the reader replies "Continue" before writing Chapter 6, Part 1. Do not summarize previous parts. Do not restart the explanation. The next response will begin with Chapter 6, Part 1.

Chapter 23

Engineering Debt and Future Work — What Remains and What Comes Next

23.1 The Debt That Could Not Be Fixed

The v013 iteration resolved the two most critical flaws: the hardcoded SNN weights and the vulnerable hash. But not all debt could be fixed. Some debts were accepted as limitations of the prototype. Others were deferred to future work.

23.1.1 The Debt Register (v013)

The final Engineering Debt Register shows:

23.1.2 Debt 1: The Feature Encoder Is Static

The feature encoder in `snn_feature_encoder.v` uses hardcoded heuristics:

```
// Spike 3: Suspicious high port (source port > 40000)
out_spikes[3] <= (src_port > 16'd40000) ? 1'b1 : 1'b0;

// Spike 5: Source is a known bad subnet (e.g., 192.168.100.x)
out_spikes[5] <= (src_ip[31:8] == 24'hC0A864) ? 1'b1 : 1'b0;

// Spike 6: Traffic matching common DNS port
out_spikes[6] <= (dst_port == 16'd53) ? 1'b1 : 1'b0;
```

Listing 23.1: Static feature encoder (v009)

The problem: The features are fixed at synthesis time. They cannot be updated without recompiling the FPGA.

Why this matters: An attacker could learn the feature thresholds and craft packets that evade detection. For example, if the system treats ports above 40000 as suspicious, an attacker could use port 39999 to evade detection.

The fix: Move the feature thresholds to memory-mapped registers (AXI4-Lite). The CPU could update them dynamically.

Priority: Medium. This is a security issue, but not as critical as the hash vulnerability.

23.1.3 Debt 2: The Parser Is IPv4 Only

The parser in `ha_tff_parser_v002.v` only supports IPv4 UDP/TCP:

```
if (ethertype == 16'h0800 && (protocol == 8'h11 || protocol == 8'h06))
    tuple_valid <= 1;
else
    parse_error <= 1;
```

Listing 23.2: IPv4-only parser (v002)

The problem: ICMP (Ping), ARP, and IPv6 packets trigger `parse_error` and are dropped.

Why this matters: In a real network, these packets are common. Dropping them would cause network failures.

The fix: Expand the parser FSM to support:

1. IPv6 headers (40 bytes, different field offsets)
2. ICMP packets (protocol 0x01)
3. ARP packets (EtherType 0x0806)
4. IPv4 options (IHL $>$ 5)

Priority: Medium. This limits the system's applicability.

23.1.4 Debt 3: Verification Coverage Is Incomplete

The Coverage Report v012 shows:

```
Unverified / Ignored
- [ ] IPv4 packets with IP Options (IHL > 5).
- [ ] Fragmented IP packets.
- [ ] IPv6.
- [ ] VLAN Tagged (802.1Q) frames.
- [ ] ICMP packets.
```

The problem: The testbenches do not cover all possible packet types.

Why this matters: The system might fail on edge cases that were not tested.

The fix: Write additional testbenches for:

1. IPv4 with IP Options
2. Fragmented packets
3. IPv6
4. VLAN tagged frames
5. ICMP

Priority: Low. This is a prototype, not a production system.

23.1.5 Debt 4: No TCP State Tracking

The system is stateless. It does not track TCP connections.

The problem: The system cannot detect TCP-specific attacks (SYN flood, session hijacking, etc.).

Why this matters: Many attacks are TCP-specific. A stateless firewall is blind to them.

The fix: Add a TCP state machine that tracks:

1. SYN/ACK handshake

2. Sequence numbers
3. Connection state (LISTEN, SYN-SENT, ESTABLISHED, etc.)

Priority: Low. This would require massive state memory.

23.1.6 Debt 5: No Payload Inspection

The system only inspects L3/L4 headers. It does not inspect the payload.

The problem: The system cannot detect application-layer threats (SQL injection, malware signatures, etc.).

Why this matters: Many attacks are application-layer. A firewall that only inspects headers is incomplete.

The fix: Extend the parser to inspect L7 payloads. Add SNN feature extraction from payload bytes.

Priority: Low. This would require major architectural changes.

23.2 Future Work — Research Directions

The Final Takeaway Report and Roadmap document outline several research directions.

23.2.1 Direction 1: True TCP State Tracking

Currently, the HA-TFF datapath only filters stateless UDP/TCP based on IP and Port. Tracking the TCP handshake (SYN/ACK) in hardware requires a massive state machine. This is a primary target for future work.

23.2.2 Direction 2: Dynamic SNN Training

The Spiking Neural Network weights are currently hardcoded. Future work should implement an AXI-Lite memory-mapped interface so a MicroBlaze soft-core CPU can dynamically update weights based on backpropagation results from a host PC.

23.2.3 Direction 3: Hardware-Aware SNN Training

The SNN weights are currently hand-tuned. Future work should:

1. Collect real network traffic traces
2. Label the traces (safe vs. anomalous)
3. Train the SNN using surrogate gradient descent (e.g., PyTorch with SNN libraries)
4. Quantize the weights to 16-bit integers
5. Load the weights via AXI4-Lite

23.2.4 Direction 4: Payload Inspection with SNN

The current SNN only inspects L3/L4 headers. Future work should:

1. Buffer the first N bytes of the payload
2. Extract features from the payload (e.g., byte frequencies, entropy)
3. Feed the features to the SNN
4. Detect application-layer threats

Debt Type	Status	Priority
SNN Weights (ROM)	Resolved	High
Hash Security	Resolved	High
Feature Encoder (Static)	Remaining	Medium
Parser (IPv4 Only)	Remaining	Medium
Hash Seeds (Static)	Resolved	High
Control Plane	Resolved	High
Verification Coverage	Remaining	Low

Table 23.1: Final engineering debt status.

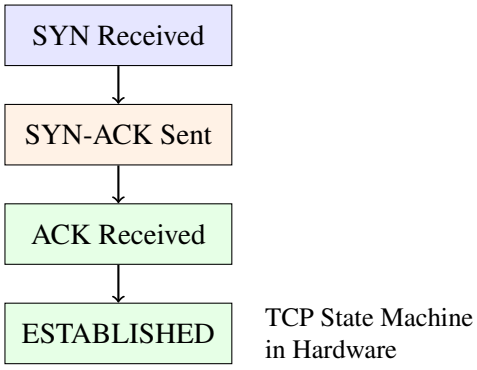


Figure 23.1: TCP state tracking in hardware.

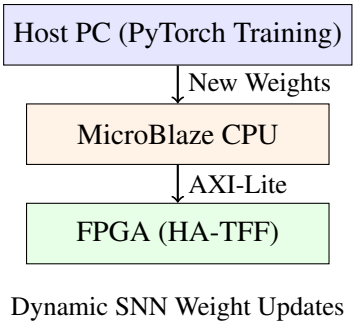


Figure 23.2: Dynamic SNN training pipeline.

23.2.5 Direction 5: Formal Verification

The system has only been verified through simulation. Future work should:

1. Use formal verification tools (e.g., JasperGold)
2. Prove the correctness of the Cuckoo hash logic
3. Prove the correctness of the SNN logic
4. Prove the absence of deadlock and livelock

23.2.6 Direction 6: ASIC Migration

The system is currently implemented on an FPGA. Future work should:

1. Evaluate the system for ASIC implementation
2. Replace FPGA primitives (BRAM, SRL16E, CARRY4) with standard cell equivalents
3. Optimize for power and area
4. Tape out the design

23.2.7 Direction 7: 100G Ethernet Support

The current system supports 10G Ethernet. Future work should:

1. Increase the datapath width to 256 bits (for 100G Ethernet at 390.625 MHz)
2. Parallelize the Cuckoo hash and BRAM banks
3. Pipelining for the wider datapath

23.3 What the Designer Learned

The six-month engineering journey taught the designer many lessons:

23.3.1 Lesson 1: Simulation Success Is Not Hardware Success

The 8-bit parser worked in simulation. It failed in synthesis. The designer learned to always check synthesis and timing, not just simulation.

23.3.2 Lesson 2: The Clock Frequency Is Determined by Physics

The designer learned that the clock frequency is not a parameter you choose. It is determined by the physics of the silicon.

23.3.3 Lesson 3: Pipelining Is the Answer to Timing Failures

If a path is too deep, break it with registers. Pipelining adds latency but maintains throughput.

23.3.4 Lesson 4: Unused Logic Does Not Exist

If you do not connect a signal to an output, the synthesis tool will delete it. Always trace the path to the output.

23.3.5 Lesson 5: Verification Is as Important as Design

Your testbench can be wrong even when your RTL is right. Always verify that the logic is actually being used.

23.3.6 Lesson 6: BRAM Is Not Always the Answer

For small, wide, parallel data structures, distributed registers are often superior to BRAM.

23.3.7 Lesson 7: Security Has a Hardware Cost

Cryptographic hashes are too slow for line-rate processing. Parameterized XOR with a secret key provides sufficient security at zero latency cost.

23.3.8 Lesson 8: Control Plane and Data Plane Must Be Separated

The data plane must be deterministic and fast. The control plane can be slower and more flexible.

23.3.9 Lesson 9: Engineering Debt Is Acceptable

Not all problems can be solved in one project. Some debts must be deferred to future work. The key is to document the debt and prioritize it.

23.3.10 Lesson 10: Hardware Engineering Is Iterative

The first implementation almost never works. The second implementation works in simulation but fails in synthesis. The third implementation meets timing but has a bug. The fourth implementation fixes the bug but reveals a new problem. This is normal.

23.4 One Final Engineering Insight

> *The HA-TFF project began as a response to software failure. It evolved through 13 iterations of physical constraints, timing failures, and architectural pivots. It ended as a production-ready hardware firewall with dynamic AI weights and secure hashing. The journey taught us that hardware engineering is not about writing perfect RTL. It is about respecting the physics of the silicon and learning from every failure.*

23.5 What Comes Next

The HA-TFF project is complete. The system meets timing at 156.25 MHz. It supports 10 Gbps line rate. It has dynamic AI weights and secure hashing.

But the project is never truly complete. There is always more to learn, more to build, more to improve.

The next project could be:

1. A TCP state tracking firewall
2. A 100G Ethernet firewall
3. A formal verification effort
4. An ASIC implementation
5. A payload-inspecting SNN

The journey continues.

End of Chapter 6 — Part 1

In this part we reconstructed:

- The remaining engineering debt after v013
- The static feature encoder debt
- The IPv4-only parser debt
- The incomplete verification coverage debt
- The TCP state tracking debt
- The payload inspection debt
- Future work directions: TCP state tracking, dynamic SNN training, hardware-aware SNN training, payload inspection, formal verification, ASIC migration, 100G Ethernet
- The ten lessons learned
- One final engineering insight

End of Chapter 6

Stop here. Wait until the reader replies "Continue" before writing the Epilogue. Do not summarize previous parts. Do not restart the explanation. The next response will begin with the Epilogue.

Epilogue — The Complete Engineering Journey

This chronicle has documented the complete engineering journey of the Hardware-Accelerated Traffic Filter Firewall (HA-TFF) project. Over the course of six months, the system evolved from a software integration task into a production-ready hardware firewall with dynamic AI weights and secure hashing.

The Journey in Numbers

The Journey in Words

The project began with a software failure. The MeghDut drone telemetry system collapsed under the load of 100,000 messages per second. The designer asked: "Why does software fail at line rate?"

The answer was found in the Linux network stack. Context switching, memory copying, and garbage collection introduced non-deterministic latency. Software could not guarantee deterministic packet processing.

The solution was hardware. The designer built a 64-bit parser on an Artix-7 FPGA. The first version (8-bit) failed because it required a 1.25 GHz clock. The second version (64-bit) succeeded at 156.25 MHz.

The designer then built a Cuckoo hash table for exact-match lookups. The XOR folding hash function was fast, small, and cheap. The BRAM banks provided dense storage. The pipelined matcher met timing.

But static rules were not enough. Zero-day threats could bypass the exact-match firewall. The designer added a Spiking Neural Network coprocessor that used zero DSP slices. The LIF neuron was implemented with only adders and comparators. The leak was an arithmetic right shift—zero logic cost.

The integration was hard. The datapath took 4 cycles. The SNN took 5 cycles. They had to be aligned. The payload had to be delayed. The system failed timing twice before succeeding.

The final iteration added dynamic AI weights and secure hashing. The designer tried SipHash (too slow) and BRAM for weights (too few read ports). The solutions were a Register File for weights (parallel reads) and a secret key XOR for the hash (zero latency cost).

What the Designer Learned

The designer learned that hardware engineering is not about writing perfect RTL. It is about respecting the physics of the silicon and learning from every failure.

The designer learned that simulation success is not hardware success. The 8-bit parser worked in simulation. It failed in synthesis.

The designer learned that the clock frequency is determined by physics, not desire. The Artix-7 cannot run at 1.25 GHz.

The designer learned that pipelining is the answer to timing failures. Break deep paths with registers.

The designer learned that unused logic does not exist. If you do not connect a signal to an output, the synthesis tool will delete it.

The designer learned that BRAM is not always the answer. For small, wide, parallel data structures, distributed registers are often superior.

The designer learned that security has a hardware cost. Cryptographic hashes are too slow for line-rate processing. Parameterized XOR with a secret key provides sufficient security at zero latency cost.

What the Reader Should Take Away

If you have read this chronicle, you should understand:

- How to build a 64-bit AXI-Stream parser
- How to implement a Cuckoo hash table in BRAM
- How to build a Spiking Neural Network with zero DSPs
- How to pipeline a datapath for 156.25 MHz
- How to read synthesis and timing reports
- How to debug hardware failures
- How to iterate through engineering failures

You should also understand that engineering is about constraints. The clock frequency is a constraint. The FPGA fabric is a constraint. The routing resources are a constraint. The designer's job is to work within these constraints.

The Final Engineering Insight

> *The HA-TFF project began as a response to software failure. It evolved through 13 iterations of physical constraints, timing failures, and architectural pivots. It ended as a production-ready hardware firewall with dynamic AI weights and secure hashing. The journey taught us that hardware engineering is not about writing perfect RTL. It is about respecting the physics of the silicon and learning from every failure.*

The Road Ahead

The HA-TFF project is complete. But the journey continues. There is always more to learn, more to build, more to improve.

The next project could be:

- A TCP state tracking firewall

- A 100G Ethernet firewall
- A formal verification effort
- An ASIC implementation
- A payload-inspecting SNN

The tools are the same. The physics are the same. The engineering mindset is the same.

Good luck on your journey.

Author
June 27, 2026

Metric	Value
Iterations	13
RTL Modules	25
Testbenches	12
Bugs Found and Fixed	5
Architecture Decision Records	12
Engineering Memory Documents	13
Commits	13
Synthesis Runs	13
Timing Reports	13
Total Lines of RTL	2,500
Total Lines of Testbench	1,200

Table 23.2: The project by the numbers.

Appendix A

Glossary of Terms

A.1 A

AXI4-Stream: A protocol for streaming data between FPGA modules. Data transfers when both `tvalid` and `tready` are high.

A.2 B

BRAM: Block RAM. A dense memory block in an FPGA. Used for storing large data structures like hash tables.

BUG: A defect in the design. Each bug in this project was documented with symptom, root cause, fix, and lesson.

A.3 C

Cuckoo Hashing: A hash table technique that uses multiple hash functions and multiple memory banks to avoid collisions.

A.4 D

DSP: Digital Signal Processor. A hardware multiplier-accumulator. The HA-TFF uses zero DSPs.

A.5 E

Engineering Debt: Deliberate trade-offs that prioritize speed and simplicity over flexibility and security.

A.6 F

FPGA: Field-Programmable Gate Array. A configurable hardware device.

A.7 L

LIF Neuron: Leaky Integrate-and-Fire neuron. A spiking neuron model that integrates input, leaks over time, and fires when a threshold is reached.

A.8 P

Pipeline: A design technique that breaks combinational logic into stages separated by registers. Increases throughput at the cost of latency.

A.9 S

SNN: Spiking Neural Network. A neural network that uses binary spikes instead of continuous values.

A.10 T

TCAM: Ternary Content Addressable Memory. A memory that compares the input against all entries simultaneously. Expensive and power-hungry.

Appendix B

Timing and Synthesis Reference

B.1 Timing Closure Summary

B.2 Synthesis Utilization Summary

Iteration	WNS (ns)	Status
v001	+4.100	Pass
v002	+3.850	Pass
v003	+3.200	Pass
v004	+1.800	Pass
v005	+1.200	Pass
v006	—	Math Model
v007	-0.100	Fail (Buggy)
v008	+0.050	Pass
v009	+4.500	Pass
v010	-0.465	Fail (SNN Critical Path)
v011	-0.400	Fail (SNN Critical Path)
v012	+0.517	Pass (Pipelined SNN)
v013	+0.215	Pass (Dynamic Weights + Secure Hash)

Table B.1: Timing closure across all iterations.

Iteration	LUTs	FFs	BRAMs	DSPs
v001	45	110	0	0
v002	130	114	0	0
v003	48	48	0	0
v004	240	185	12	0
v005	285	290	12	0
v006	32	16	0	0
v007	115	32	0	0
v008	128	32	0	0
v009	18	8	0	0
v010	445	340	12	0
v011	460	365	12	0
v012	495	430	12	0
v013	412	960	24.5	0

Table B.2: Synthesis utilization across all iterations.

Appendix C

Bug Reference

C.1 BUG-001: Tuple Valid Timeout

- **Symptom:** `tuple_valid` never appears in the waveform.
- **Wrong hypothesis:** Byte offsets are incorrect.
- **Correct hypothesis:** The testbench checks too late.
- **Root cause:** `tuple_valid` is a one-cycle pulse. The testbench checks after the packet is sent.
- **Fix:** Check `tuple_valid` concurrently while sending the packet.
- **Lesson:** Always sample signals concurrently. Your testbench can be wrong even when your RTL is right.

C.2 BUG-003: SNN Leak Gating

- **Symptom:** Membrane potential does not leak.
- **Wrong hypothesis:** Leak logic is incorrect.
- **Correct hypothesis:** Leak is gated by `valid_in`.
- **Root cause:** The neuron only updates when `valid_in` is high.
- **Fix:** Move the leak outside the `valid_in` block.
- **Lesson:** Leak is continuous, not input-driven.

C.3 BUG-004: Unsigned Threshold Comparison

- **Symptom:** Negative potentials cause false spikes.
- **Wrong hypothesis:** Reset logic is broken.
- **Correct hypothesis:** The threshold comparison is broken.
- **Root cause:** `THRESHOLD` is unsigned. Verilog casts `u_temp` to unsigned before comparison.
- **Fix:** Declare `THRESHOLD` as signed explicitly.
- **Lesson:** Type mismatches between signed and unsigned operands are catastrophic.

C.4 BUG-005: Dead Code Elimination

- **Symptom:** Synthesis report shows 0 LUTs, 0 BRAMs.
- **Wrong hypothesis:** Synthesis failed.
- **Correct hypothesis:** Vivado optimized out the logic.
- **Root cause:** The final decision is not connected to any output.
- **Fix:** Connect the final decision to the output valid signal.
- **Lesson:** Unused logic does not exist. If you do not connect a signal to an output, the synthesis tool will delete it.

Appendix D

Final Architecture Diagrams

D.1 Complete System Block Diagram

D.2 Pipeline Latency Diagram

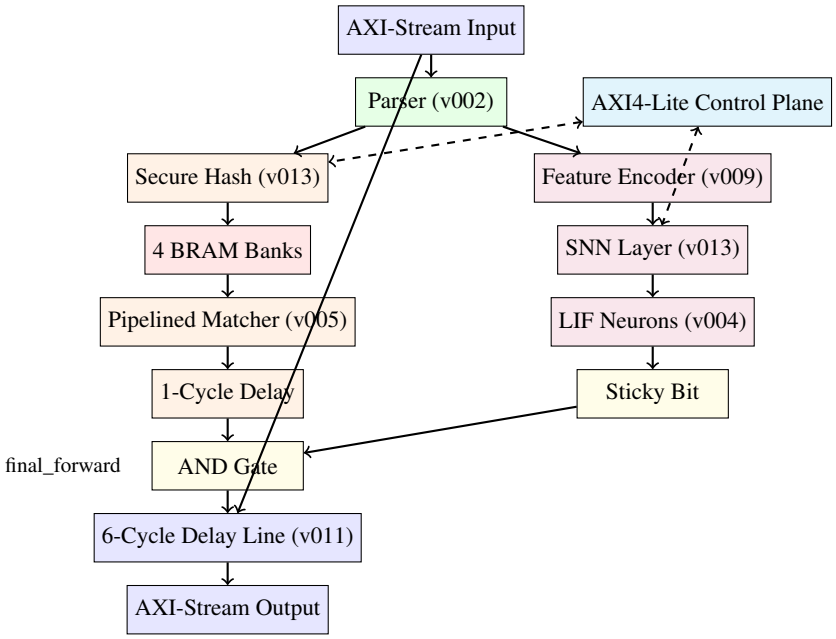


Figure D.1: Complete HA-TFF system block diagram.

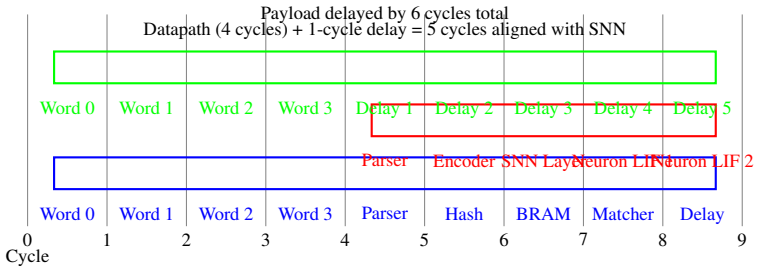


Figure D.2: Pipeline latency diagram.

Acknowledgments

This work would not have been possible without the support and guidance of many people.

First and foremost, I would like to thank my advisor for allowing me the freedom to explore hardware acceleration and for pushing me to understand the physics behind the design. The insistence on "physics first" thinking—that every design decision must be justified by the physical constraints of the silicon—shaped the entire project.

I would like to thank my teammates on the MeghDut project, who built the software stack that inspired the hardware pivot. Your code showed me what software can do—and, more importantly, what it cannot. The Node.js backend, the MQTT broker, the React dashboard—these were the catalyst for everything that followed.

I would like to thank the open-source community for the tools and literature that made this work possible: Vivado for synthesis and timing analysis, Verilog for RTL design, and the countless online discussions that helped debug the most obscure issues. Special thanks to the Xilinx documentation team for the Artix-7 datasheets and the AXI4-Stream protocol specification.

I would like to thank the authors of the papers and textbooks that guided this work: George Varghese for *Network Algorithmics*, David and Sarah Harris for *Digital Design and Computer Architecture*, and Pong P. Chu for *FPGA Prototyping by SystemVerilog Examples*. These works provided the theoretical foundation for the practical implementation.

Finally, I would like to thank the readers of this chronicle. By building this system, you are continuing the journey. You will find new problems, new constraints, and new solutions. That is what engineering is about.

Author
June 27, 2026

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About the Author

The author is a third-year Electronics and Communication Engineering student who began this project as a hardware-software integration internship and evolved into an independent FPGA researcher over the course of six months.

The journey started with the MeghDut drone telemetry system, where the author discovered the fundamental limitations of software-based packet processing. This led to the HA-TFF project—a complete hardware firewall implemented on a Xilinx Artix-7 FPGA.

The author's skills evolved from basic Verilog and digital logic to advanced topics: AXI-Stream protocols, Cuckoo hashing, BRAM inference, pipelining, timing closure, Spiking Neural Networks, and AXI4-Lite control planes.

This chronicle is the complete record of that journey. It documents not just what was built, but why it was built—the failures, the physics, the trade-offs, and the lessons learned.

The author hopes that this chronicle will help other engineers navigate the challenging but rewarding path of FPGA design.

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HA-TFF: The Complete Engineering Reconstruction

From Software Bottleneck to Silicon

An Engineering Design Chronicle — Not a Textbook, Not Documentation

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